# Engineering Mechanics

**Static & Dynamics**

# Strength of Materials

**Flexural Formula**

# Theory of Structure

**Statically Determinate**

**Statically Indeterminate**

# Engineering Mechanics

## Static

Problem Determine the x and y components of the forces shown below in Fig P-001.

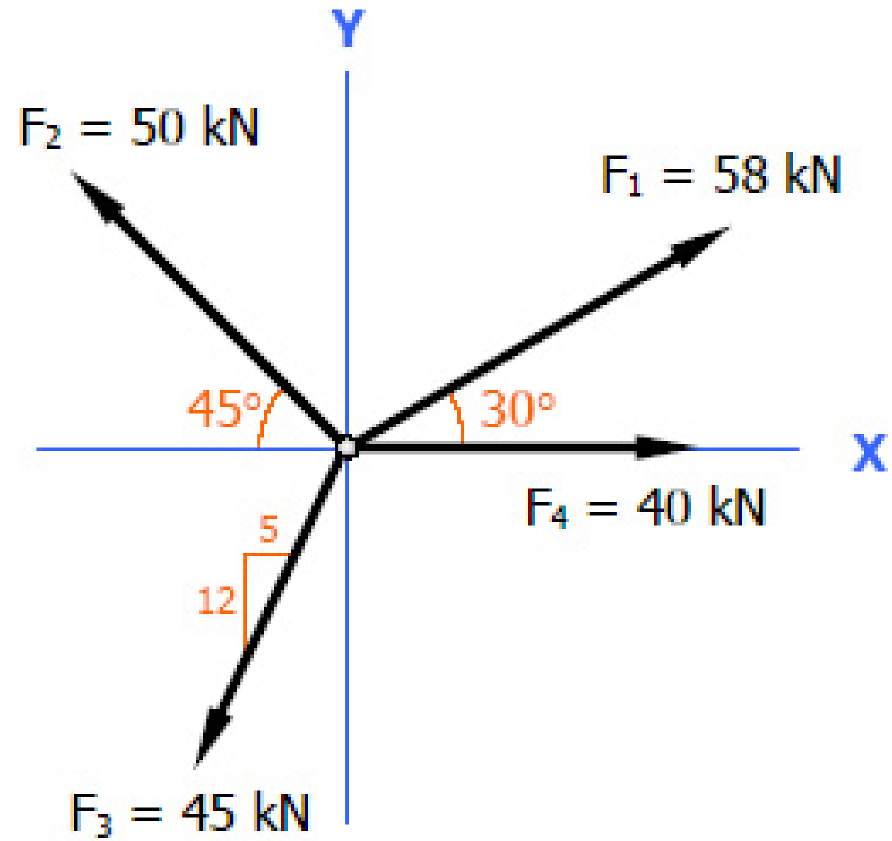



**Figure P-001**

Which of the following correctly defines the 500 N force that passes from A(4, 0, 3) to B(0, 6, 0)?

A. 256**i** - 384**j** + 192**k** N

B. -256**i** + 384**j** - 192**k** N

C. -384**i** + 192**j** - 256**k** N

D. 384**i** - 192**j** + 256**k** N

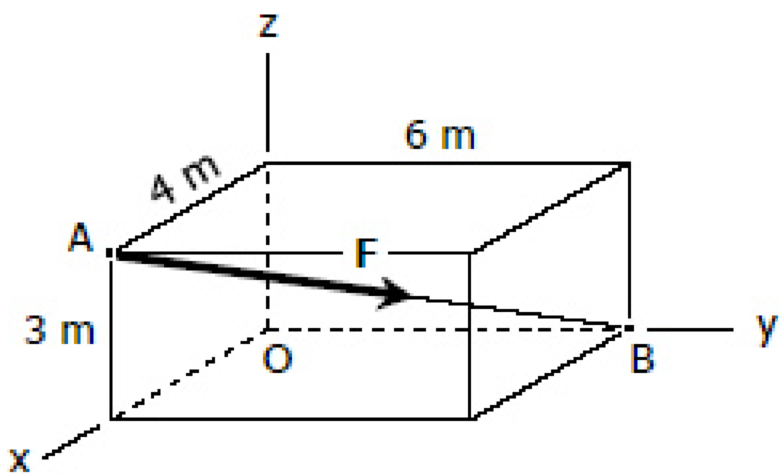



**Figure 1.3**

Referring to Fig. 1.4, determine the angle between vector A and the y-axis.

A. 65.7°

B. 73.1°

C. 67.5°

D. 71.3°

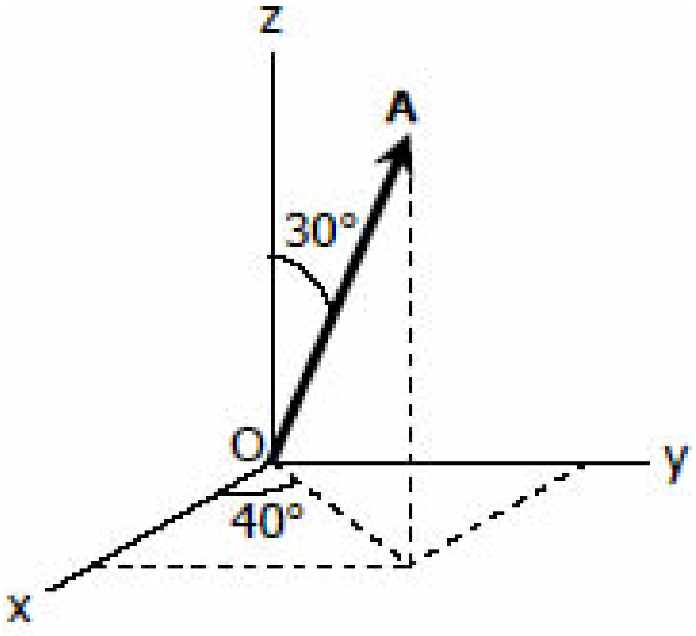



**Figure 1.4**

Find the components in the x, y, u and v directions of the force P = 10 kN shown in Fig. P-005.

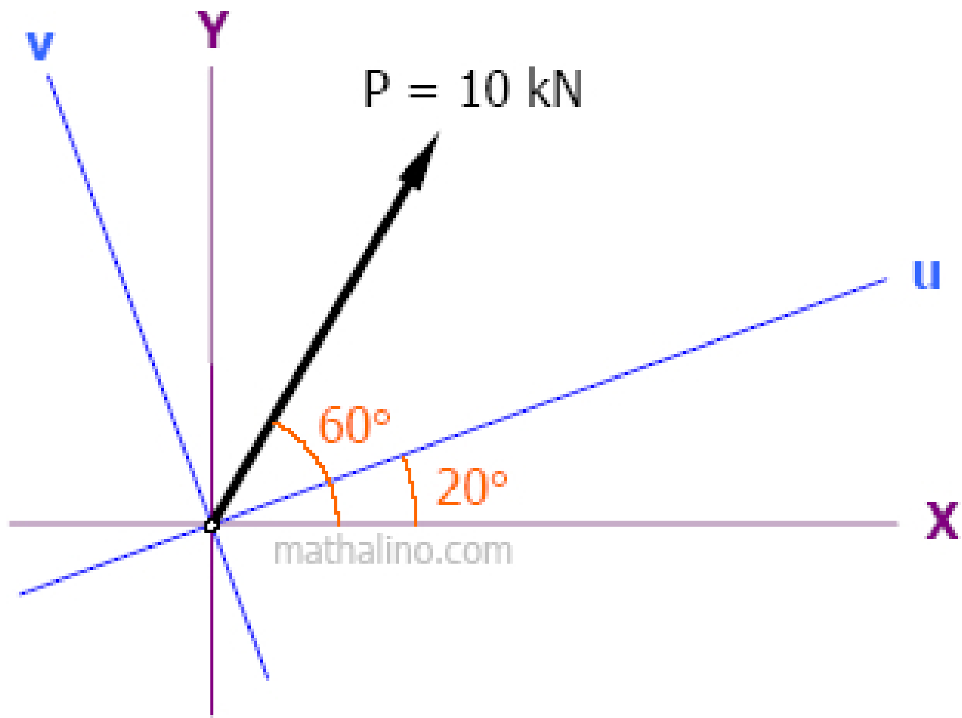



**Figure P-005**

The force P of magnitude 50 kN is acting at 215° from the x-axis. Find the components of P in u 157° from x, and v negative 69° from x.

A block is resting on an incline of slope 5:12 as shown in Fig. P-007. It is subjected to a force F = 500 N on a slope of 3:4. Determine the components of F parallel and perpendicular to the incline.

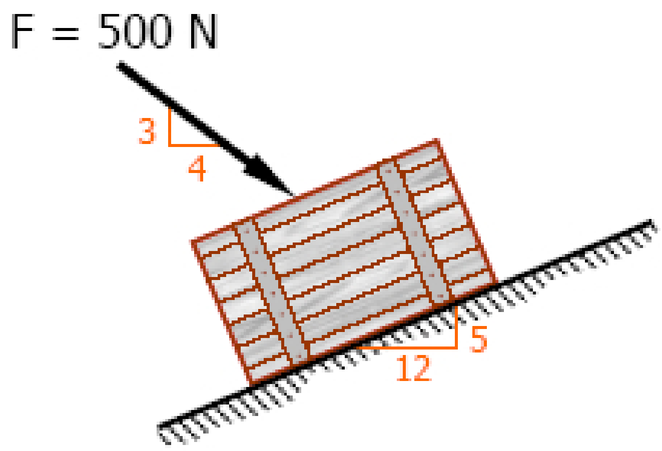



**Figure P-007**

A force P = 800 N is shown in Fig. P-008.

a. Find the y-component of P with respect to x and y axis.
b. Find the y'-component of P with respect to x' and y' axis.
c. Find the y-component of P with respect to x' and y axis.
d. Find the y'-component of P with respect to x and y' axis.

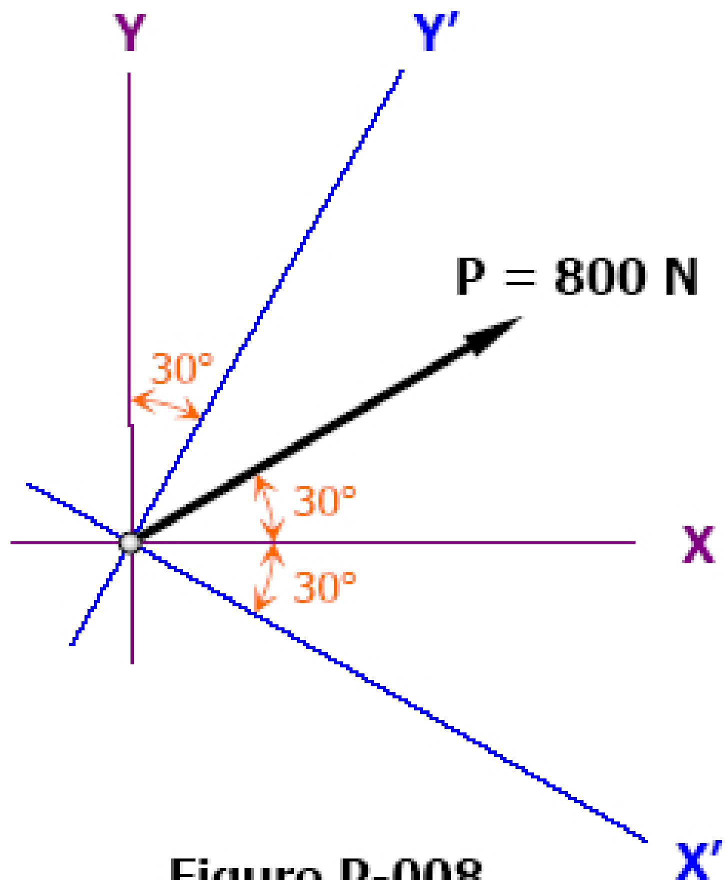



Figure P-008

The body on the 30° incline in Fig. P-009 is acted upon by a force P inclined at 20° with the horizontal. If P is resolved into components parallel and perpendicular to incline and the value of the parallel component is 1800 N, compute the value of the perpendicular component and that of P.

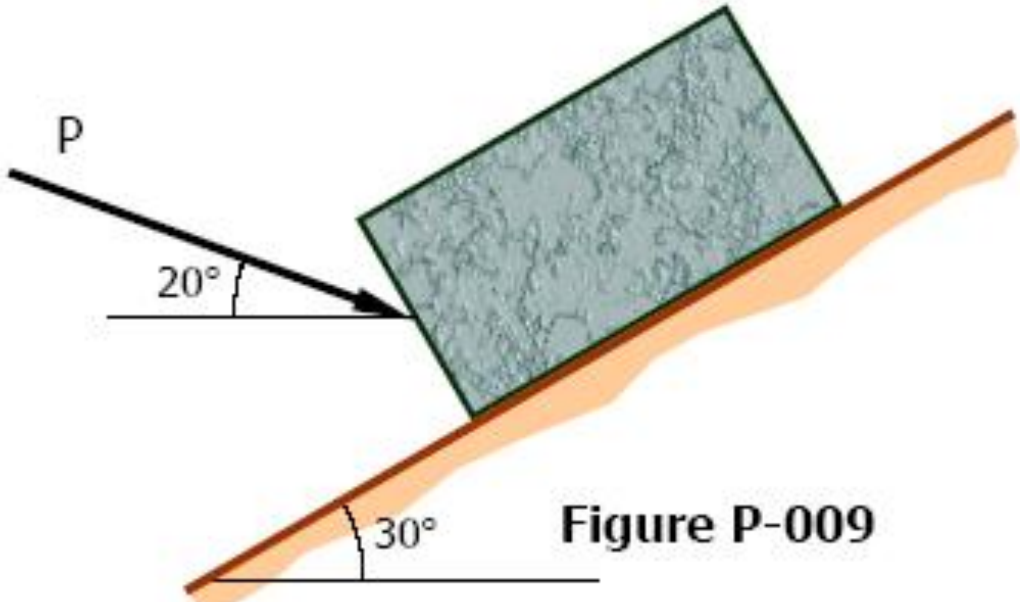
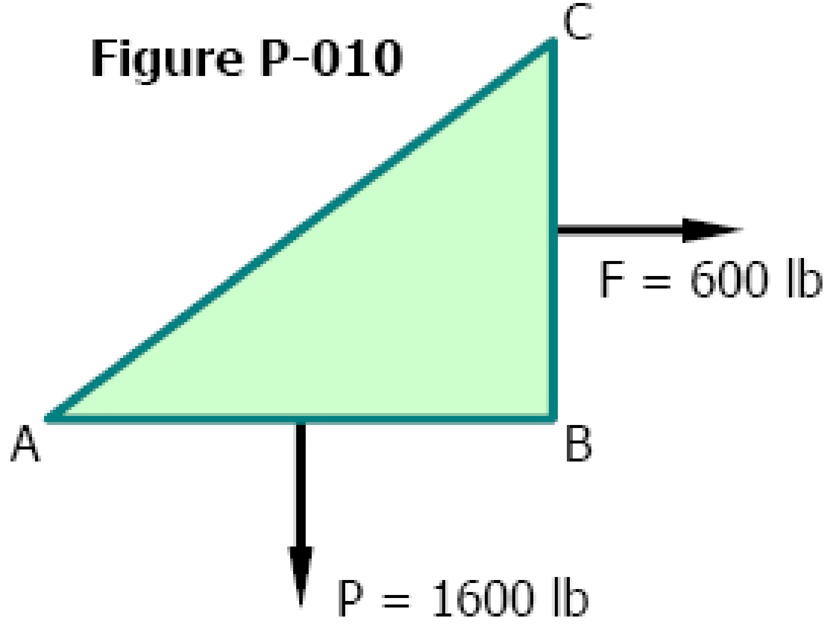



Figure P-009

The triangular block shown in Fig. P-010 is subjected to the loads P = 1600 lb and F = 600 lb. If AB = 8 in. and BC = 6 in., resolve each load into components normal and tangential to AC.

**Figure P-010**



The magnitude of vertical force F shown in Fig. P-016 is 8000 N. Resolve F into components parallel to the bars AB and AC.

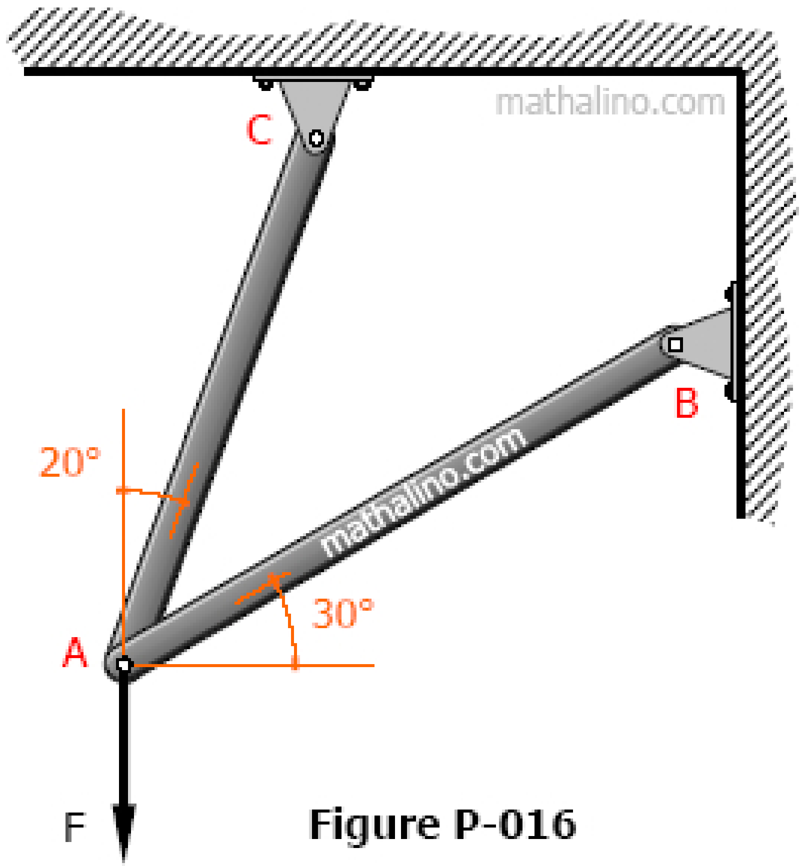



**Figure P-016**

If the force F shown in Fig. P-017 is resolved into components parallel to the bars AB and BC, the magnitude of the component parallel to bar BC is 4 kN. What are the magnitudes of F and its component parallel to AB?

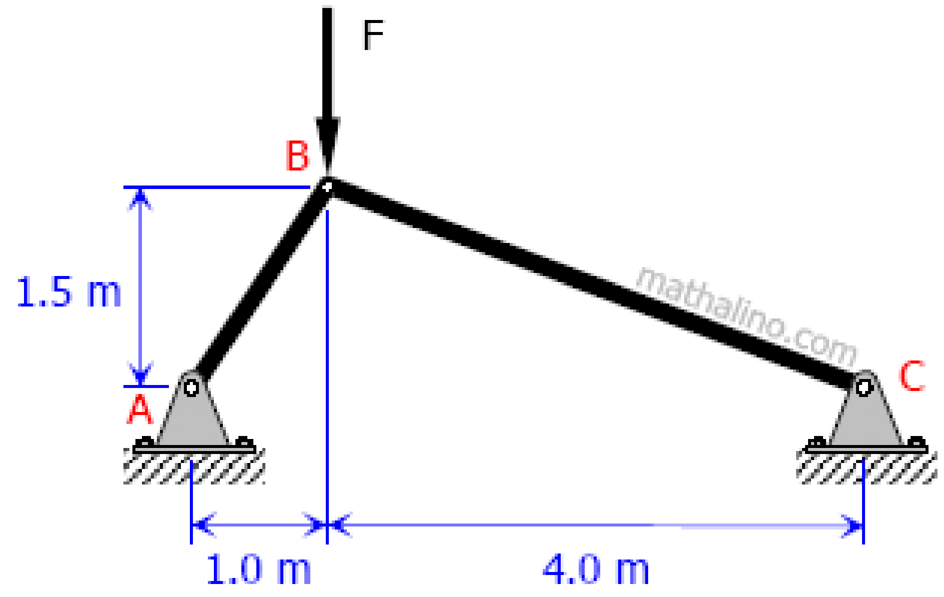



Figure P-017

Without computing the magnitude of the resultant, compute where the resultant of the forces shown in Fig. P-228 intersects the x and y axes.

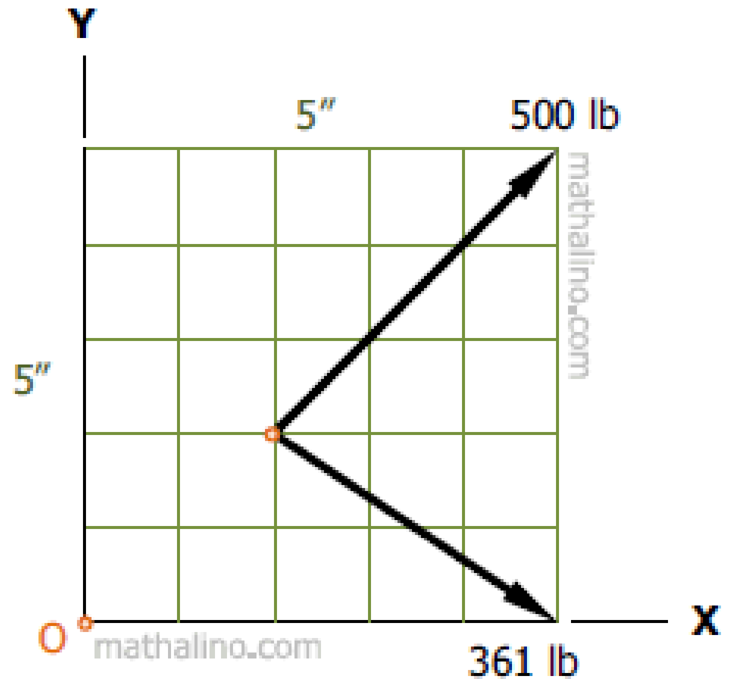



**Figure P-228**

Determine the resultant moment about point A of the system of forces shown in Fig. P-246. Each square is 1 ft on a side.

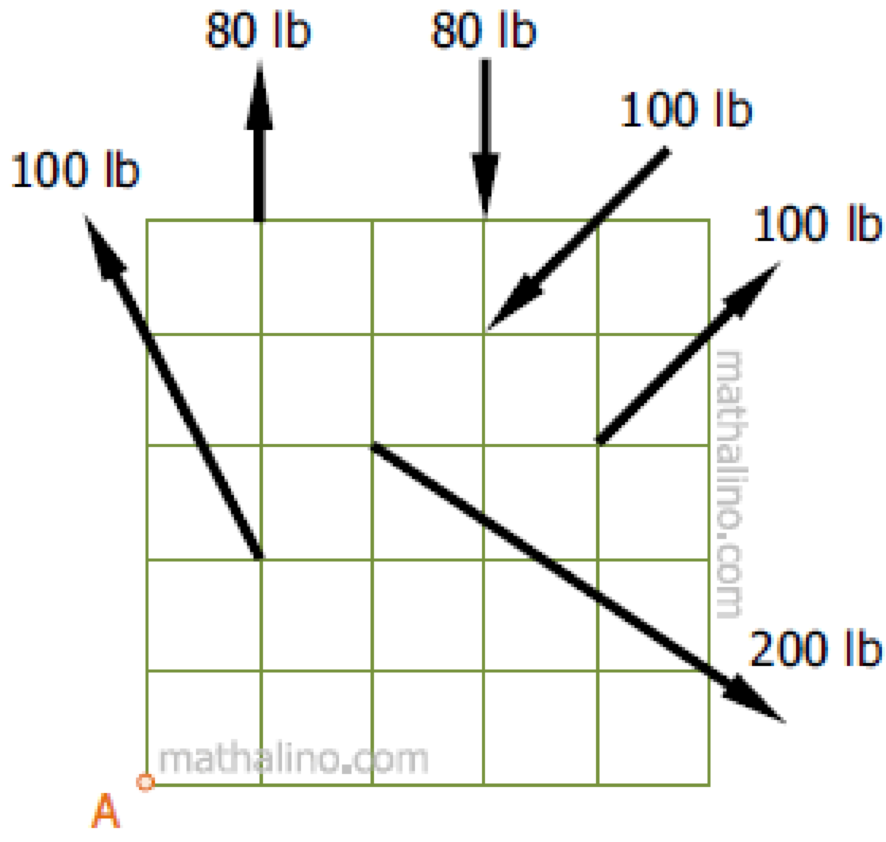



**Figure P-246**

The three-step pulley shown in Fig. P-247 is subjected to the given couples. Compute the value of the resultant couple. Also determine the forces acting at the rim of the middle pulley that are required to balance the given system.

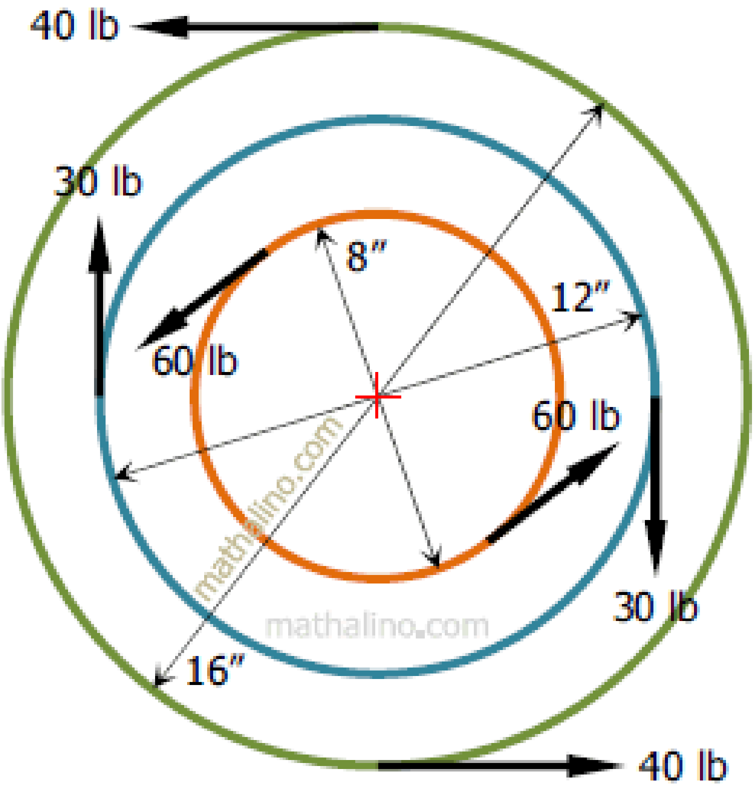



**Figure P-247**

To close a gate valve it is necessary to exert two forces of 60 lb at opposite sides of a handwheel 3 ft in diameter. Through an accident the wheel is broken and the valve must be closed by a thrusting bar through a slot in the valve stem and exerting a force 4 ft out from the center. Determine the force required and draw a free-body diagram of the bar.

A force system consists of a clockwise couple of 480 N·m plus a 240 N force directed up to the right through the origin of X and Y axes at $\theta_x = 30°$. Replace the given system by an equivalent single force and compute the intercepts its line of action with the X and Y axes.

Fig. P-249 represents the top view of a speed reducer which is geared for a four to one reduction in speed. The torque input at the horizontal shaft C is 100 lb·ft. The torque output at the horizontal shaft D, because of the speed reduction, is 400 lb·ft. Compute the torque reaction at the mounting bolts A and B holding the reducer to the floor. Hint: The torque reaction is caused by the unbalanced torque, which is a couple.

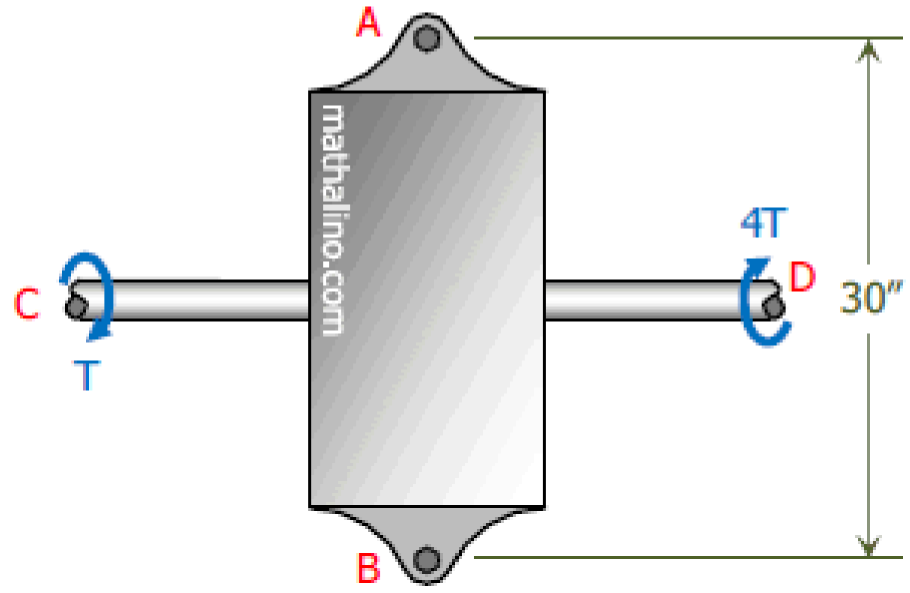



**Figure P-249**

Replace the system of forces acting on the frame in Fig. P-257 by a resultant R at A and a couple acting horizontally through B and C.

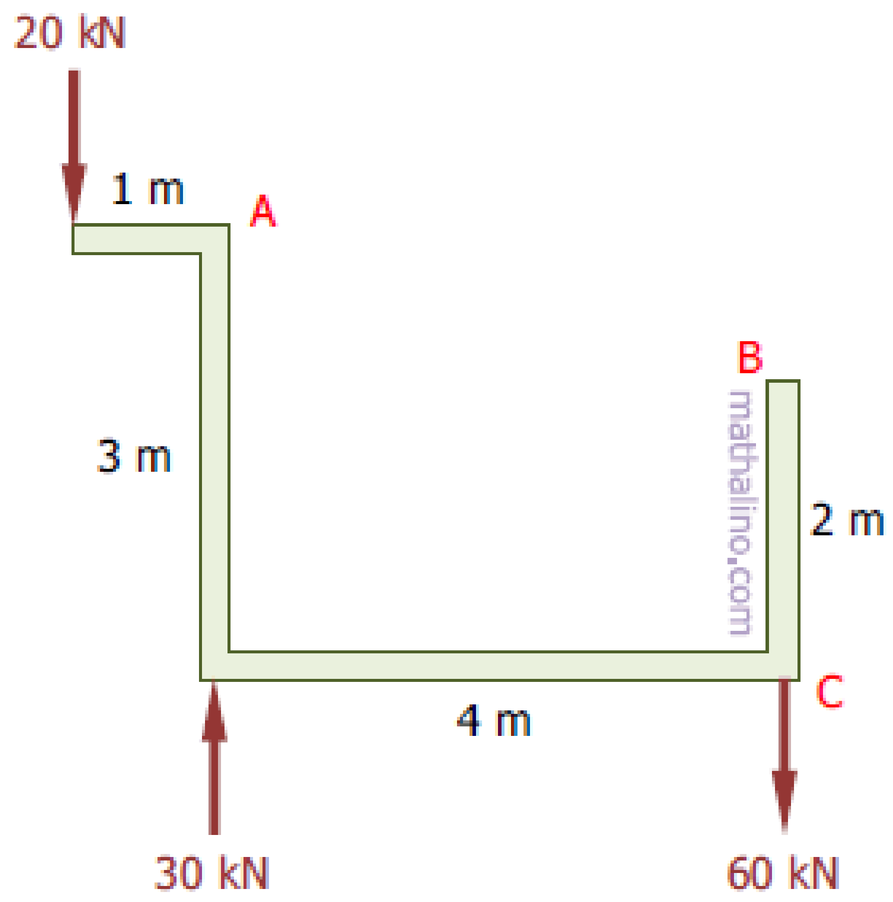



**Figure P-257**

Three ropes are tied to a small metal ring. At the end of each rope three students are pulling, each trying to move the ring in their direction. If we look down from above, the forces and directions they are applying are shown in Fig. P-011. Find the net force on the ring due to the three applied forces.

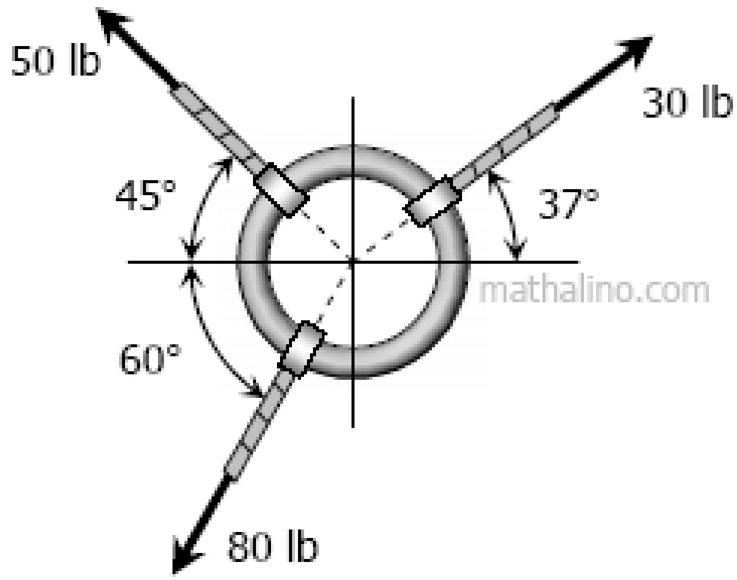



**Figure 011**

Forces F, P, and T are concurrent and acting in the direction as shown in Fig. P-015.

a. Find the value of F and $\alpha$ if T = 450 N, P = 250 N, $\beta$ = 30Â°, and the resultant is 300 N acting up along the y-axis.
b. Find the value of F and $\alpha$ if T = 450 N, P = 250 N, $\beta$ = 30Â° and the resultant is zero.
c. Find the value of $\alpha$ and $\beta$ if T = 450 N, P = 250 N, F = 350 N, and the resultant is zero.

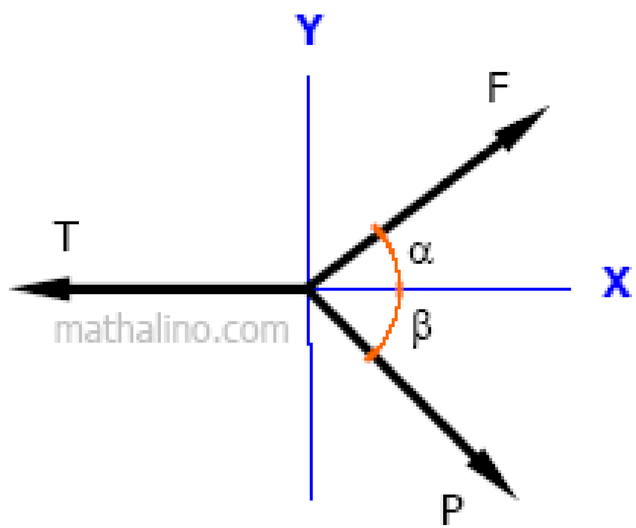



**Figure P-015**

The beam AB in Fig. P-238 supports a load which varies an intensity of 220 N/m to 890 N/m. Calculate the magnitude and position of the resultant load.

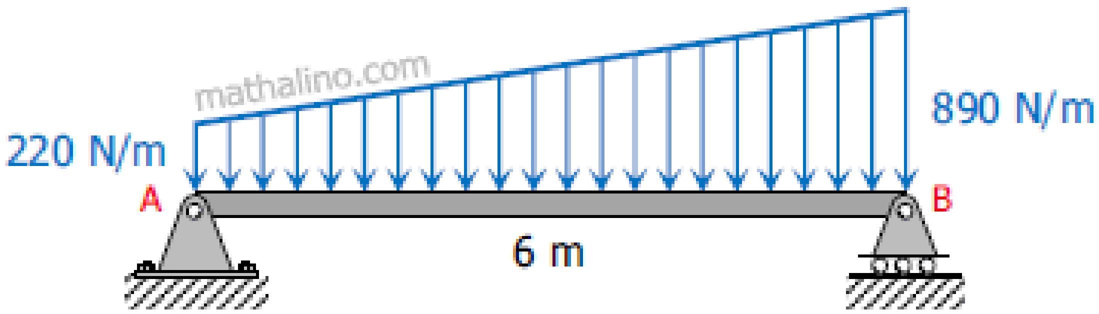



**Figure P-238**

Determine completely the resultant of the forces acting on the step pulley shown in Fig. P-262.

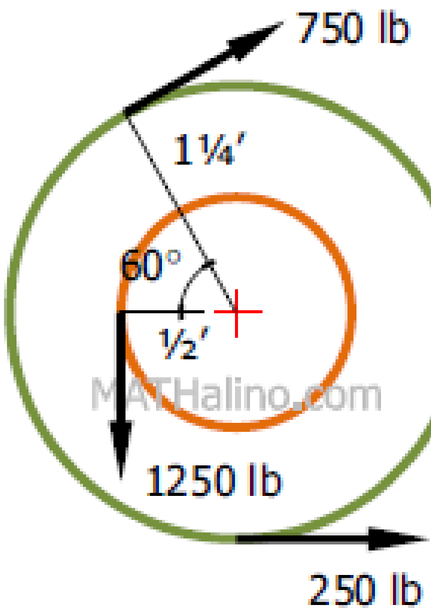



**Figure P-262**

Determine the resultant of the three forces acting on the dam shown in Fig. P-266 and locate its intersection with the base AB. For good design, this intersection should occur within the middle third of the base. Does it?

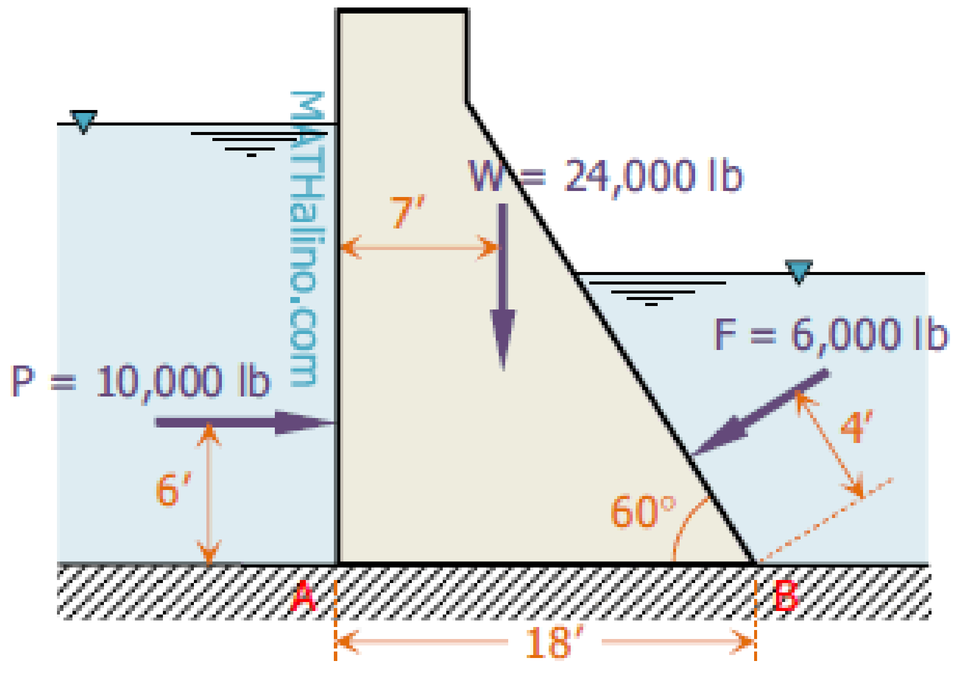



**Figure P-266**

The three forces in Fig. P-270 create a vertical resultant acting through point A. If T is known to be 361 lb, compute the values of F and P.

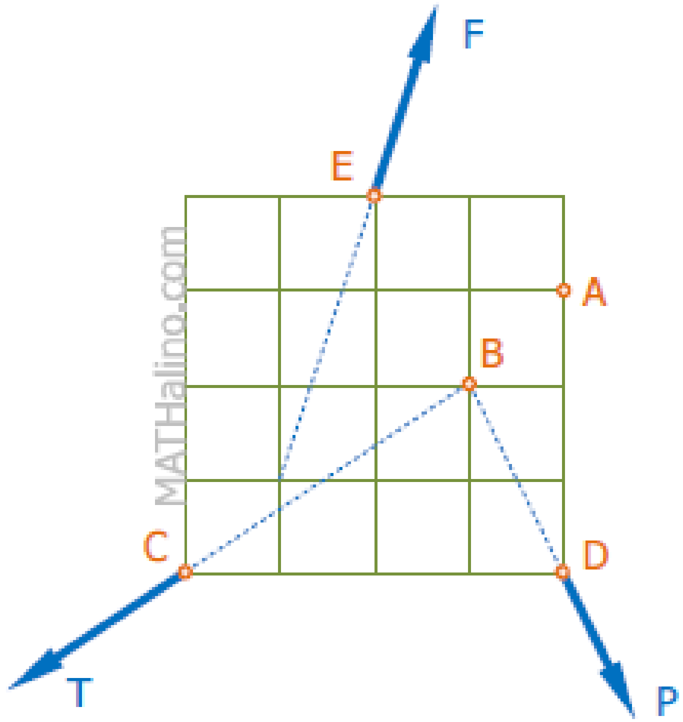



**Figure P-270 and P-271**

The system of knotted cords shown in Fig. P-317 support the indicated weights. Compute the tensile force in each cord.

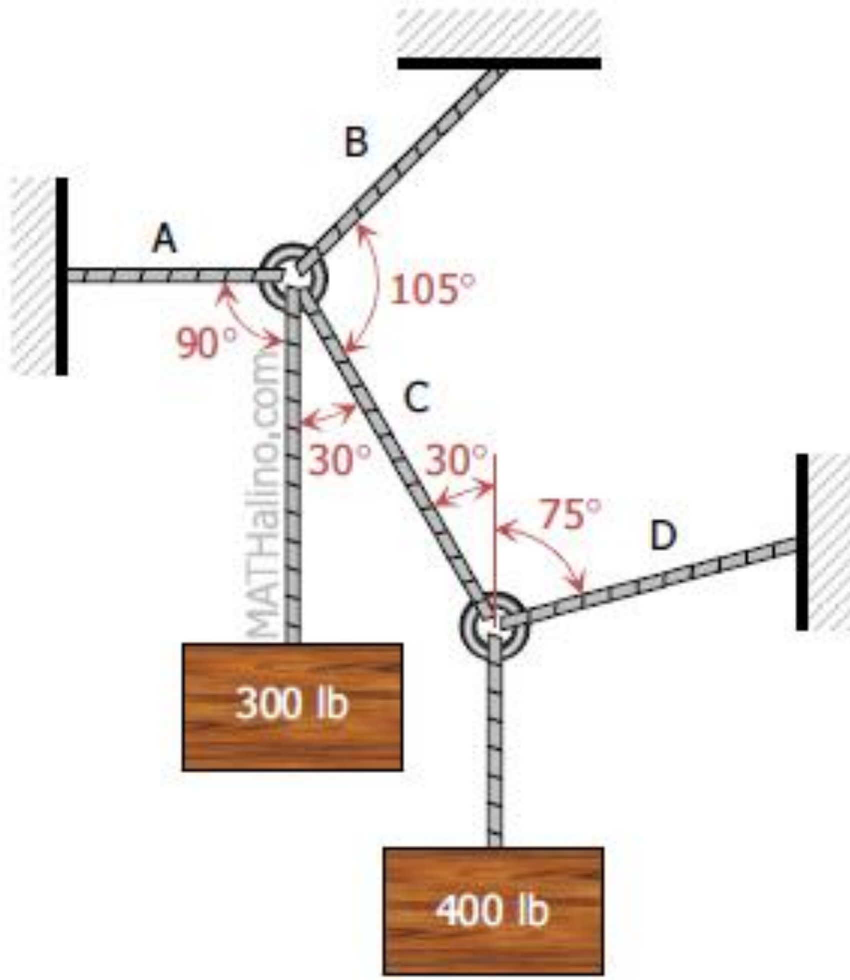



**Figure P-317**

Three bars, hinged at A and D and pinned at B and C as shown in Fig. P-318, form a four-link mechanism. Determine the value of P that will prevent motion.

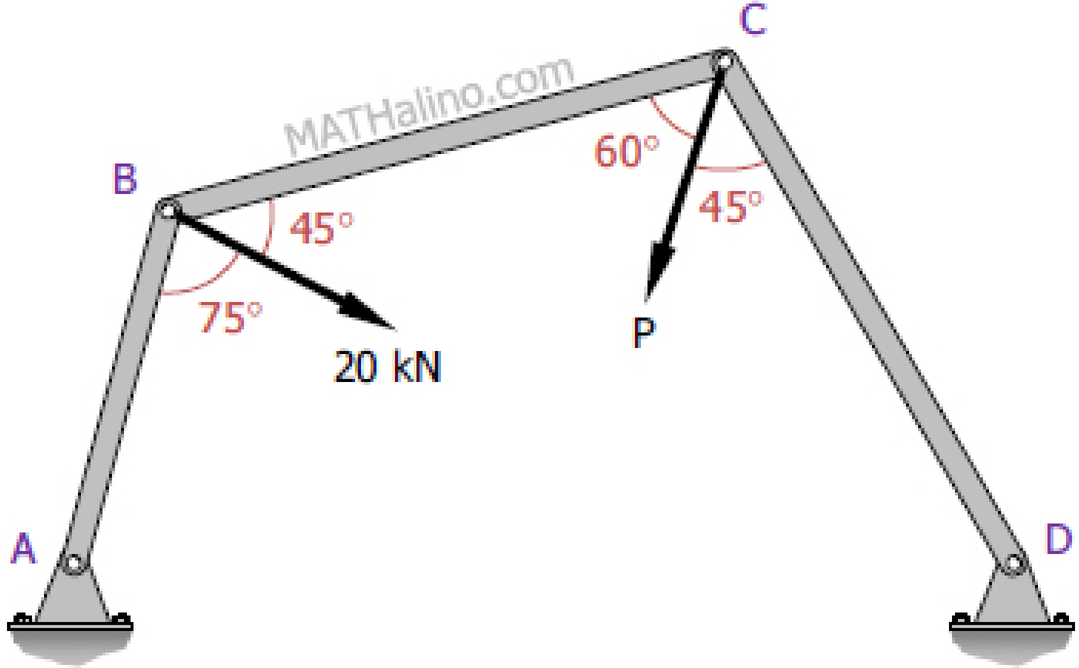



**Figure P-318**

Cords are loop around a small spacer separating two cylinders each weighing 400 lb and pass, as shown in Fig. P-319 over a frictionless pulleys to weights of 200 lb and 400 lb . Determine the angle θ and the normal pressure N between the cylinders and the smooth horizontal surface.

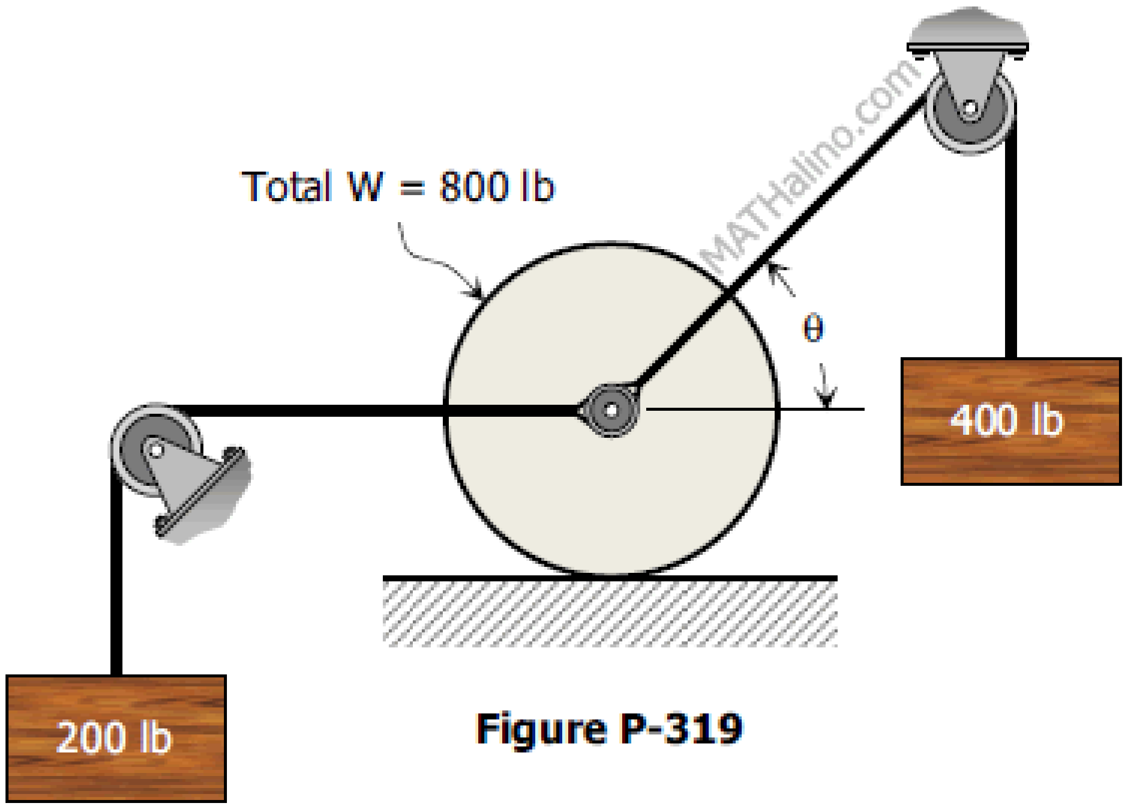



Figure P-319

A wheel of 10-in radius carries a load of 1000 lb, as shown in Fig. P-324. (a) Determine the horizontal force P applied at the center which is necessary to start the wheel over a 5-in. block. Also find the reaction at the block. (b) If the force P may be inclined at any angle with the horizontal, determine the minimum value of P to start the wheel over the block; the angle P makes with the horizontal; and the reaction at the block.

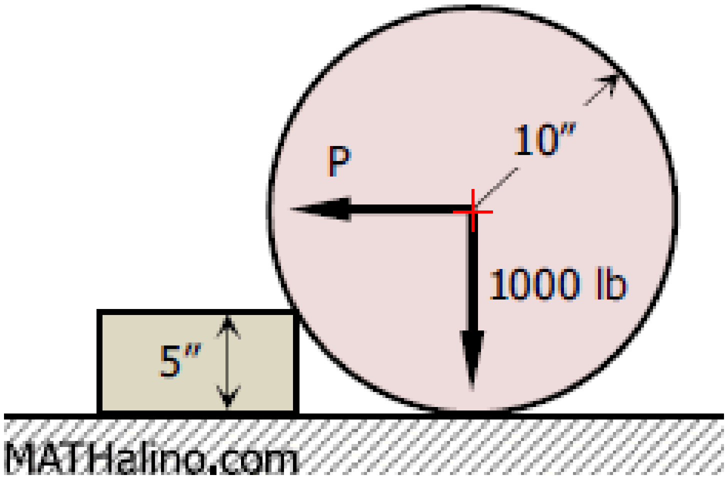



**Figure P-324**

Two cylinders A and B, weighing 100 lb and 200 lb respectively, are connected by a rigid rod curved parallel to the smooth cylindrical surface shown in Fig. P-329. Determine the angles α and β that define the position of equilibrium.

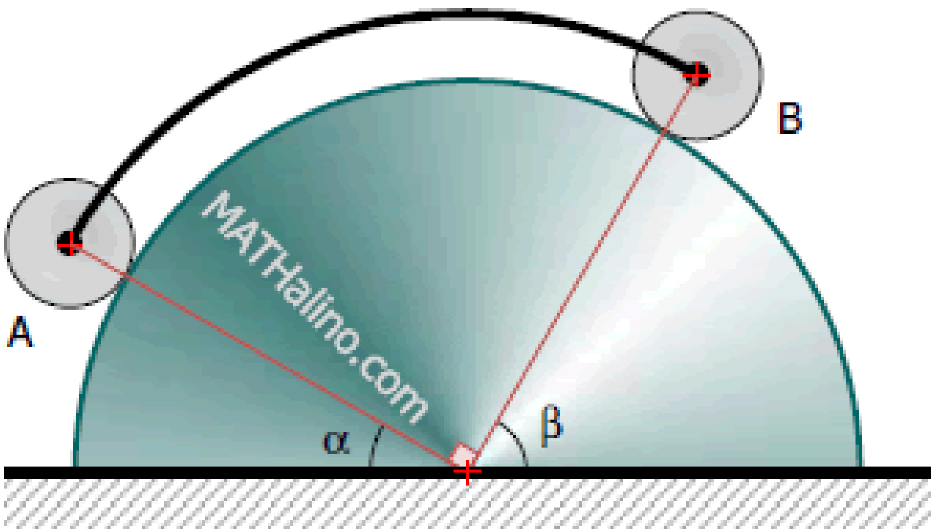



Figure P-329

Determine the reactions for the beam loaded as shown in Fig. P-334.

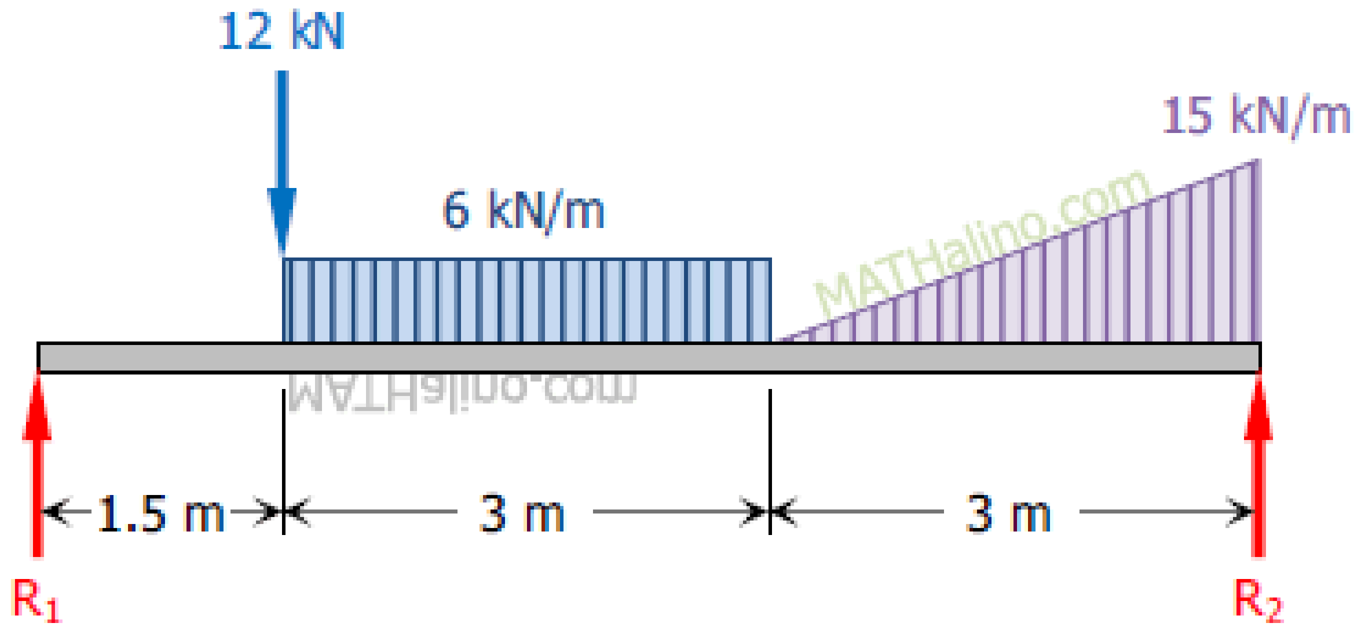



**Figure P-334**

The upper beam in Fig. P-337 is supported at D and a roller at C which separates the upper and lower beams. Determine the values of the reactions at A, B, C, and D. Neglect the weight of the beams.

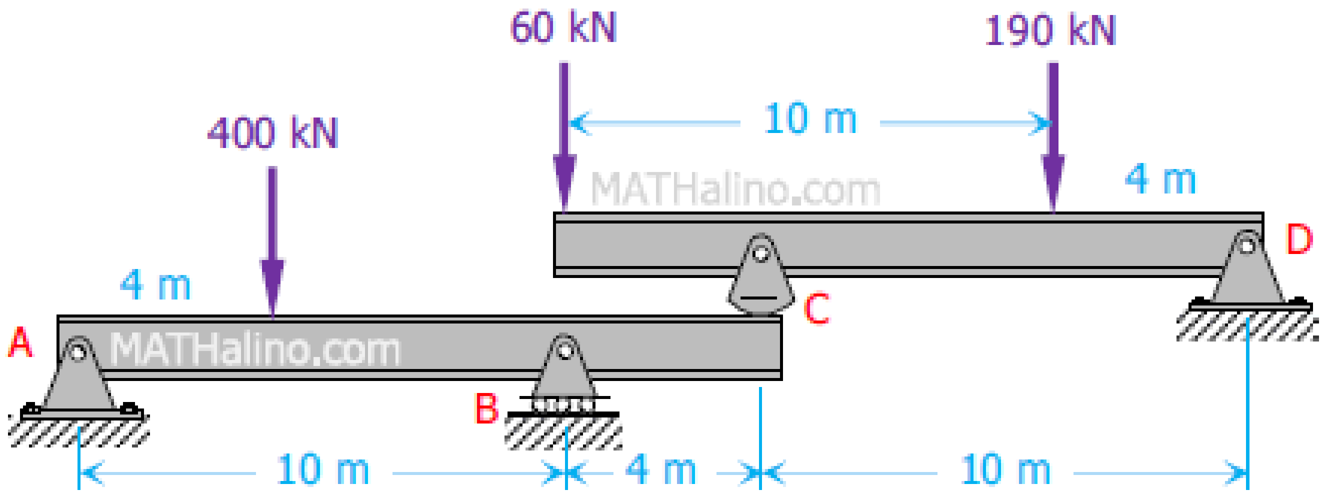



**Figure P-337**

The differential chain hoist shown in Fig. P-339 consists of two concentric pulleys rigidly fastened together. The pulleys form two sprockets for an endless chain looped over them in two loops. In one loop is mounted a movable pulley supporting a load W. Neglecting friction, determine the maximum load W that can just be raised by a pull P supplied as shown.

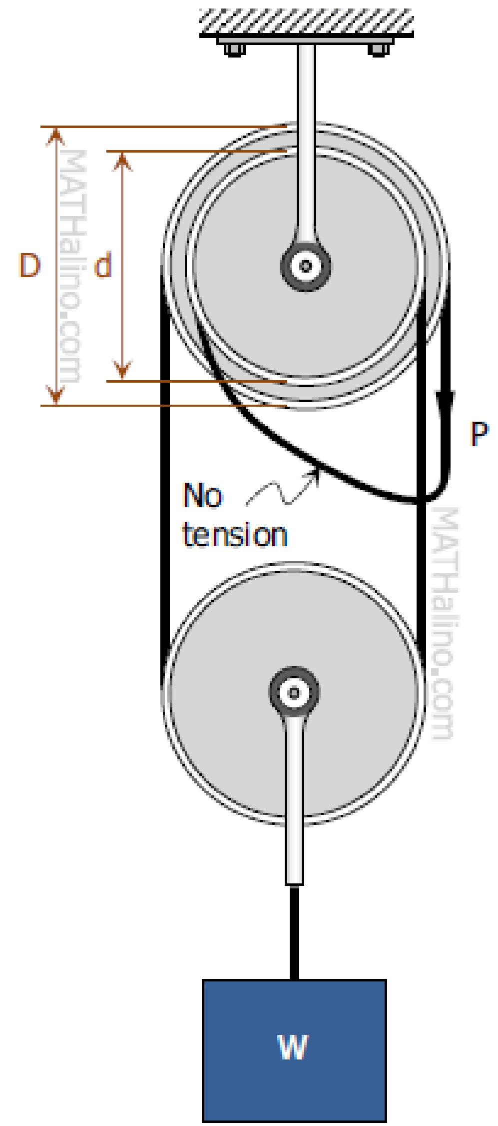



**Figure P-339**

The weight W of a traveling crane is 20 tons acting as shown in Fig. P-343. To prevent the crane from tipping to the right when carrying a load P of 20 tons, a counterweight Q is used. Determine the value and position of Q so that the crane will remain in equilibrium both when the maximum load P is applied and when the load P is removed.

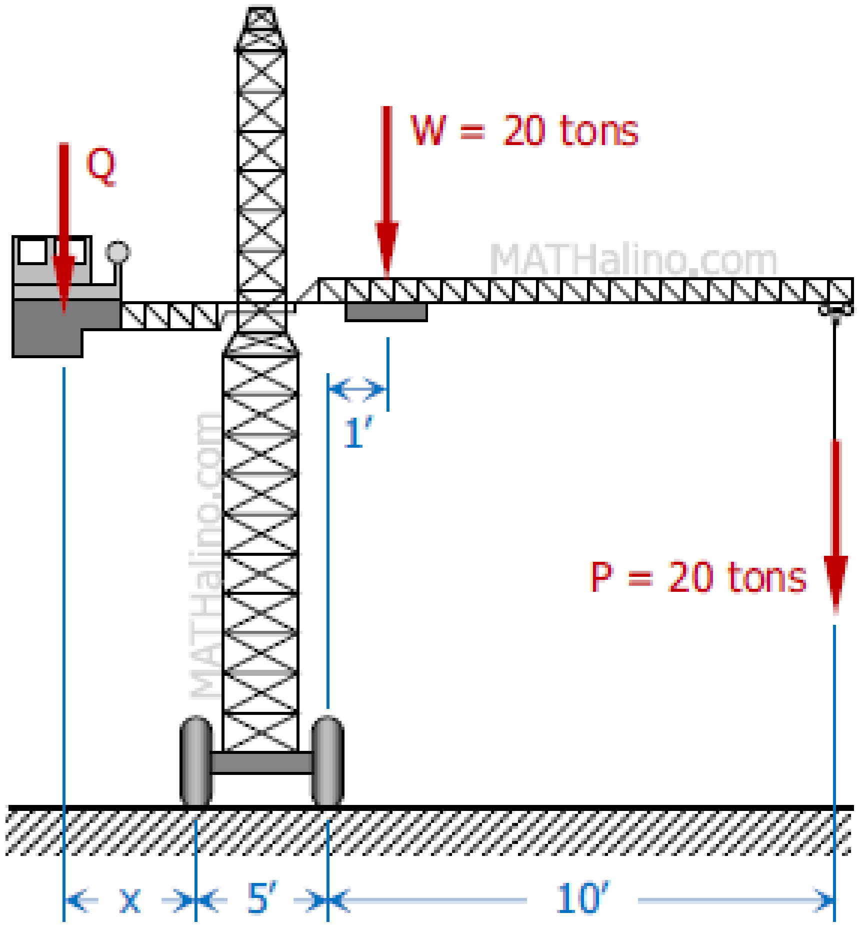



**Figure P-343**

The forces acting on a 1-m length of a dam are shown in Fig. P-353. The upward ground reaction varies uniformly from an intensity of $p_1$ kN/m to $p_2$ kN/m at B. Determine p1 and p2 and also the horizontal resistance to sliding.

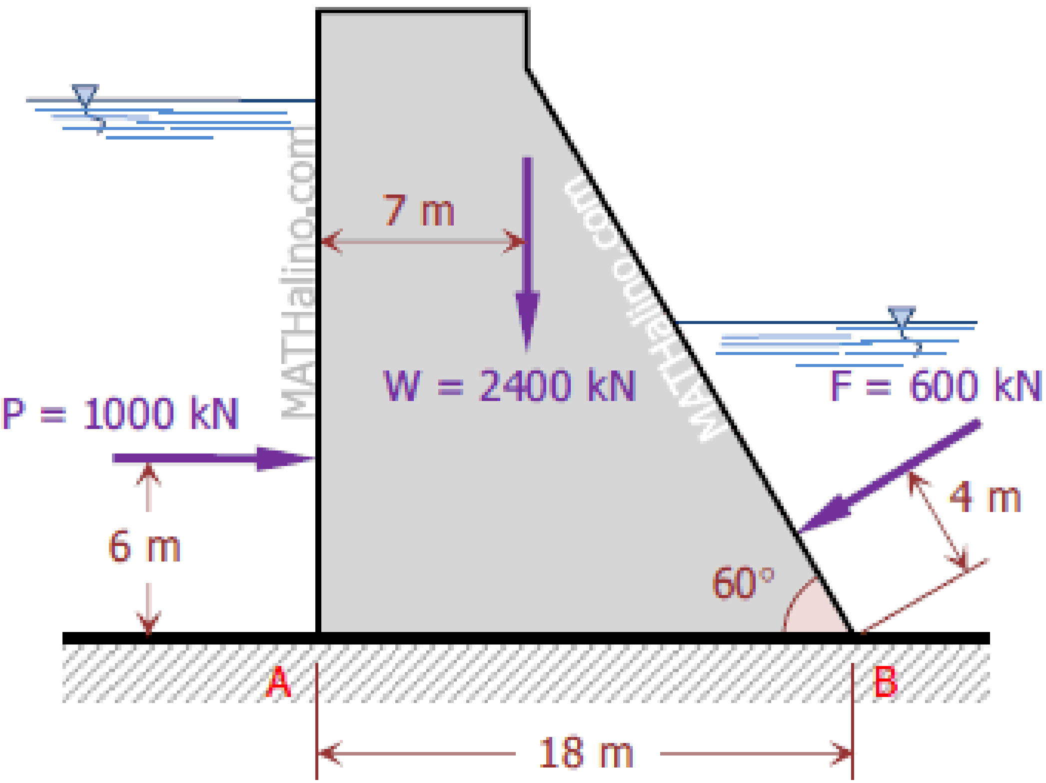



**Figure P-353**

The cantilever truss shown in Fig. P-356 is supported by a hinge at A and a strut BC. Determine the reactions at A and B.

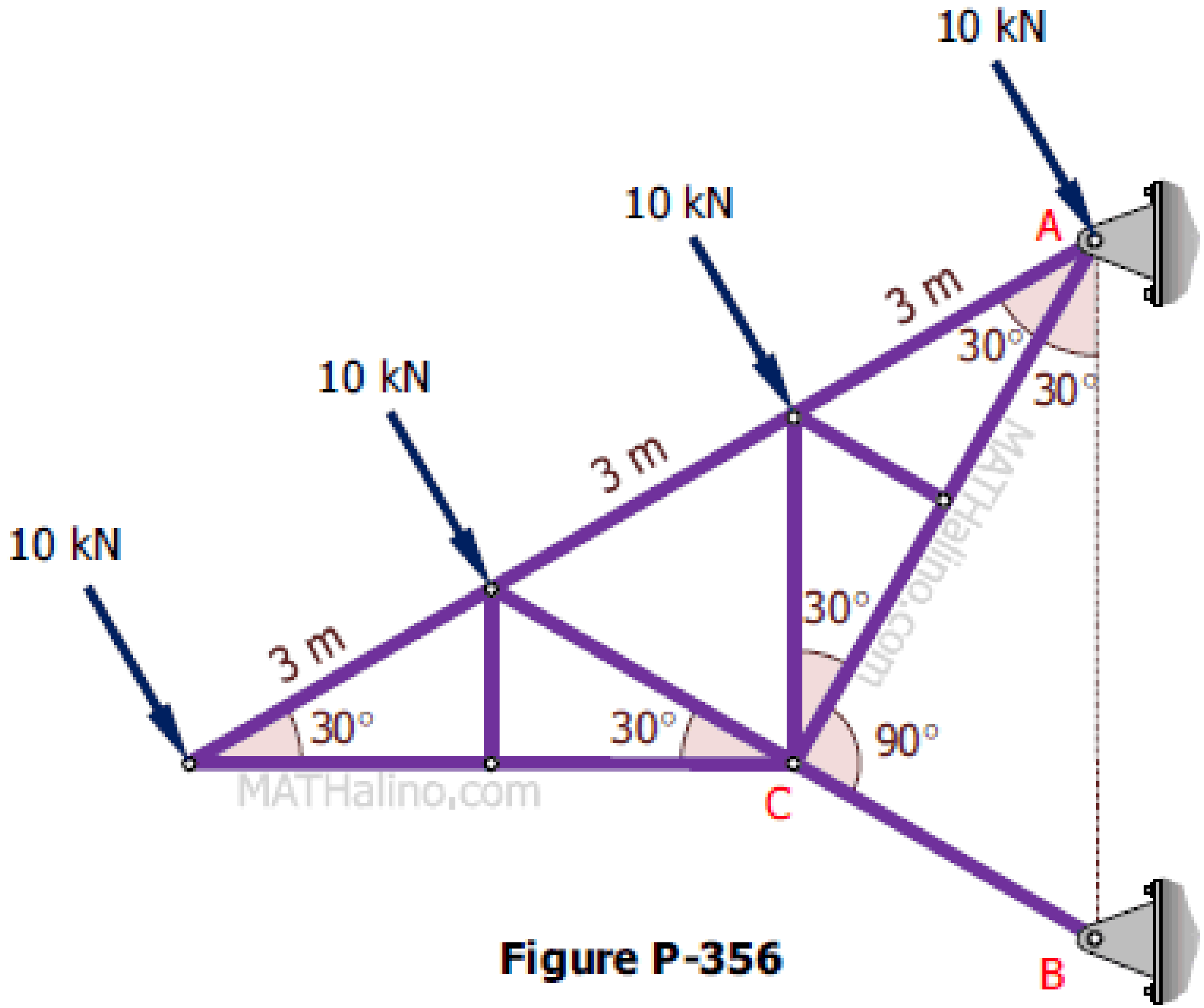



**Figure P-356**

The uniform rod in Fig. P-357 weighs 420 lb and has its center of gravity at G. Determine the tension in the cable and the reactions at the smooth surfaces at A and B.

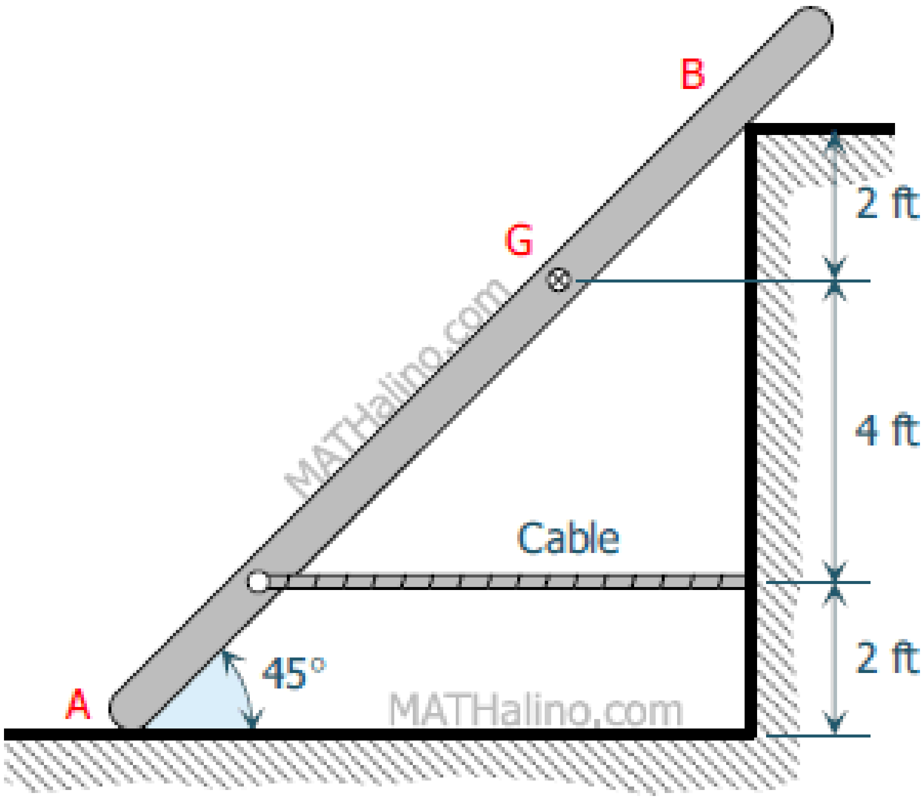



**Figure P-357**

A 4-m bar of negligible weight rests in a horizontal position on the smooth planes shown in Fig. P-359. Compute the distance x at which load T = 10 kN should be placed from point B to keep the bar horizontal.

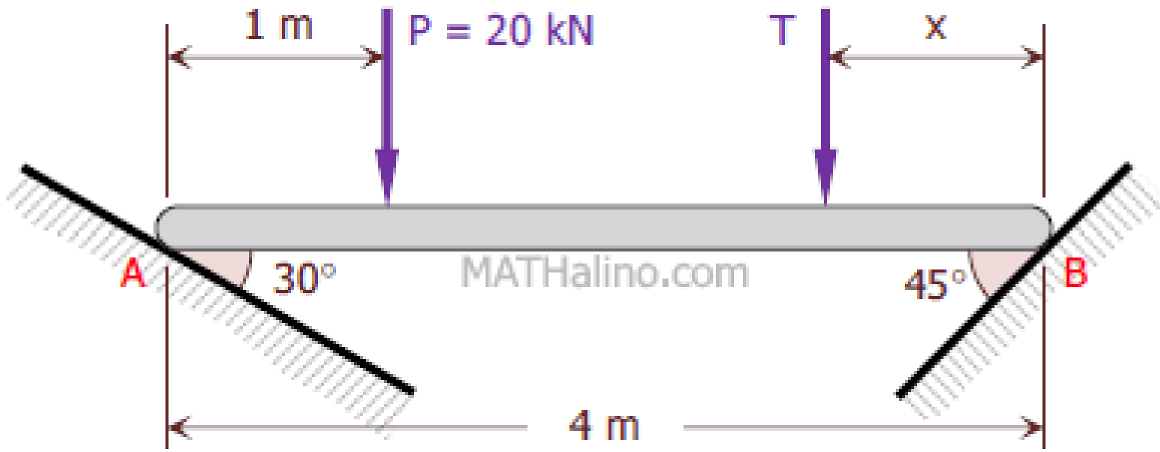



**Figure P-359, P-360, and P-361**

Referring to Problem 359, if T = 30 kN and x = 1 m, determine the angle θ at which the bar will be inclined to the horizontal when it is in a position of equilibrium.

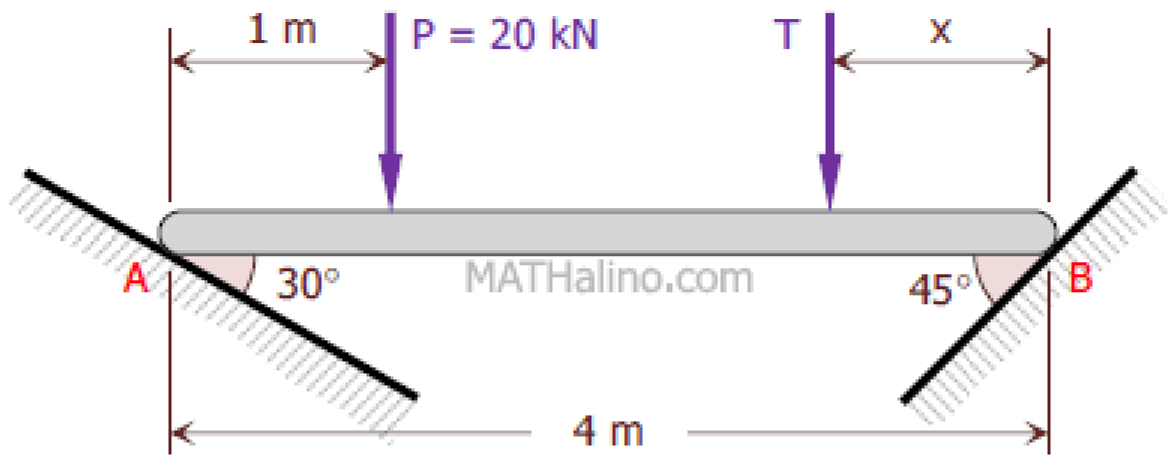



**Figure P-359, P-360, and P-361**

A homogeneous block of weight W rests upon the incline shown in Fig. P-512. If the coefficient of friction is 0.30, determine the greatest height h at which a force P parallel to the incline may be applied so that the block will slide up the incline without tipping over.

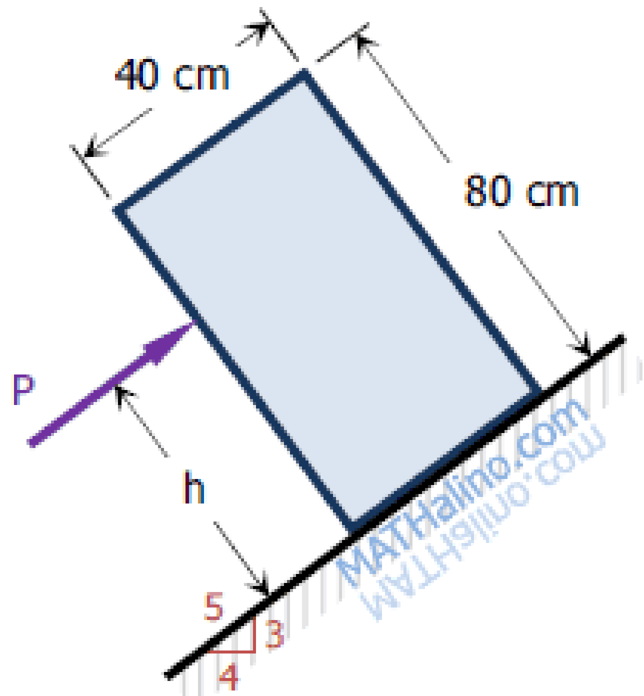



**Figure P-512 and P-513**

In Fig. P-519, two blocks are connected by a solid strut attached to each block with frictionless pins. If the coefficient of friction under each block is 0.25 and B weighs 2700 N, find the minimum weight of A to prevent motion.

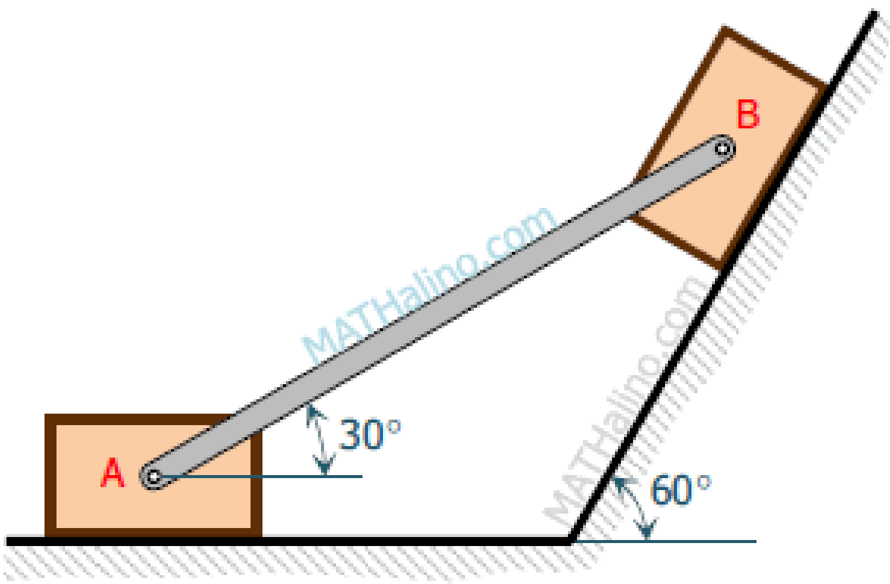



**Figure P-519, P-520, and P-521**

A force of 400 lb is applied to the pulley shown in Fig. P-523. The pulley is prevented from rotating by a force P applied to the end of the brake lever. If the coefficient of friction at the brake surface is 0.20, determine the value of P.

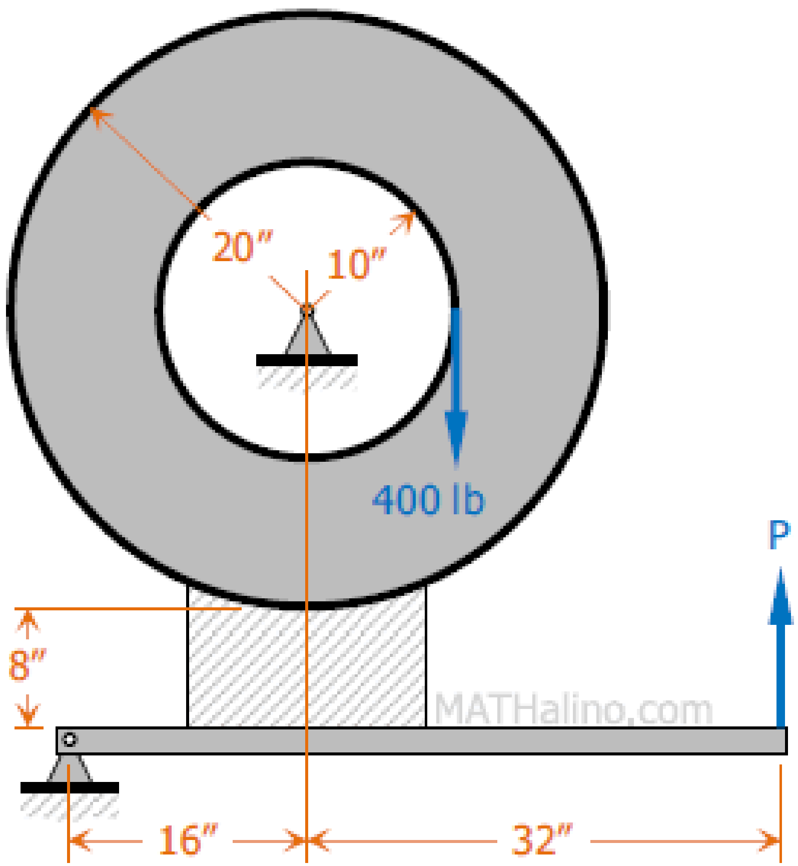



**Figure P-523**

A horizontal arm having a bushing of 20 mm long is slipped over a 20-mm diameter vertical rod, as shown in Fig. P-524. The coefficient of friction between the bushing and the rod is 0.20. Compute the minimum length L at which a weight W can be placed to prevent the arm from slipping down the rod. Neglect the weight of the arm.

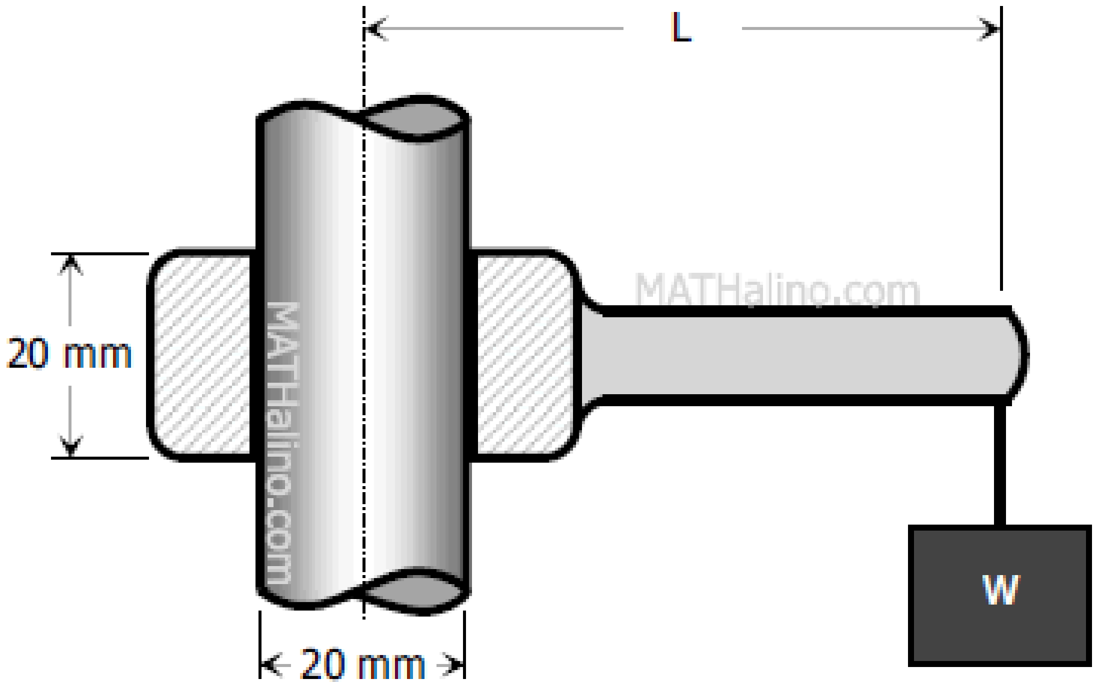



**Figure P-524**

A plank 10 ft long is placed in a horizontal position with its ends resting on two inclined planes, as shown in Fig. P-530. The angle of friction is 20°. Determine how close the load P can be placed to each end before slipping impends.

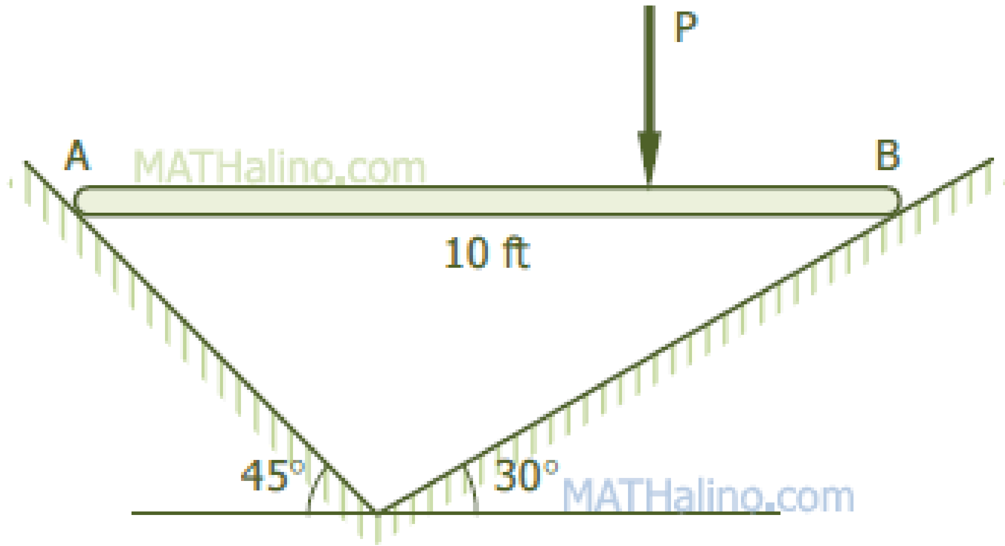



**Figure P-530**

A homogeneous cylinder 3 m in diameter and weighing 30 kN is resting on two inclined planes as shown in Fig. P-527. If the angle of friction is 15° for all contact surfaces, compute the magnitude of the couple required to start the cylinder rotating counterclockwise.

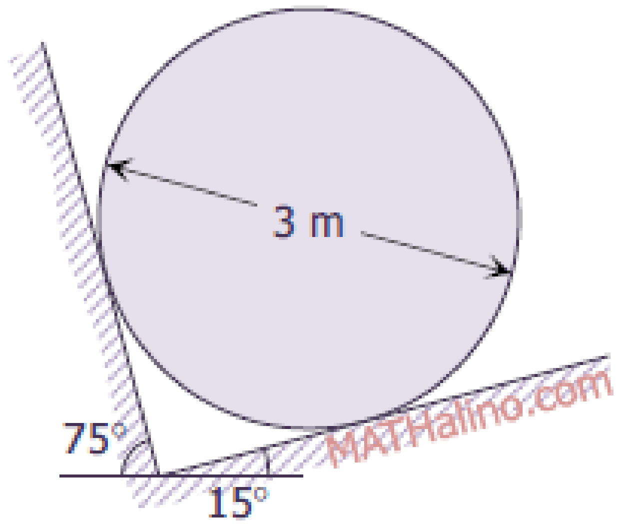



**Figure P-527 and P-528**

A uniform plank of weight W and total length 2L is placed as shown in Fig. P-531 with its ends in contact with the inclined planes. The angle of friction is 15°. Determine the maximum value of the angle a at which slipping impends.

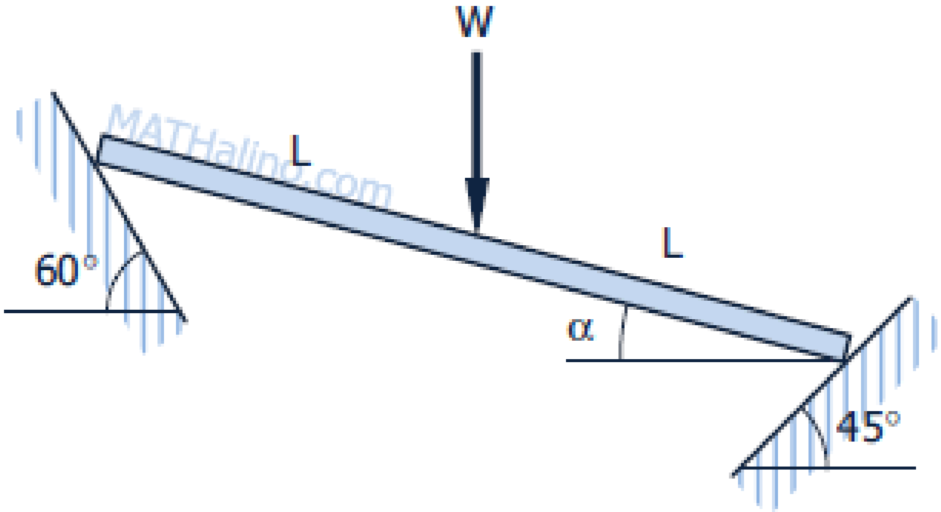



**Figure P-531**

In Fig. P-532, two blocks each weighing 1.5 kN are connected by a uniform horizontal bar which weighs 1.0 kN. If the angle of friction is 15° under each block, find P directed parallel to the 45° incline that will cause impending motion to the left.

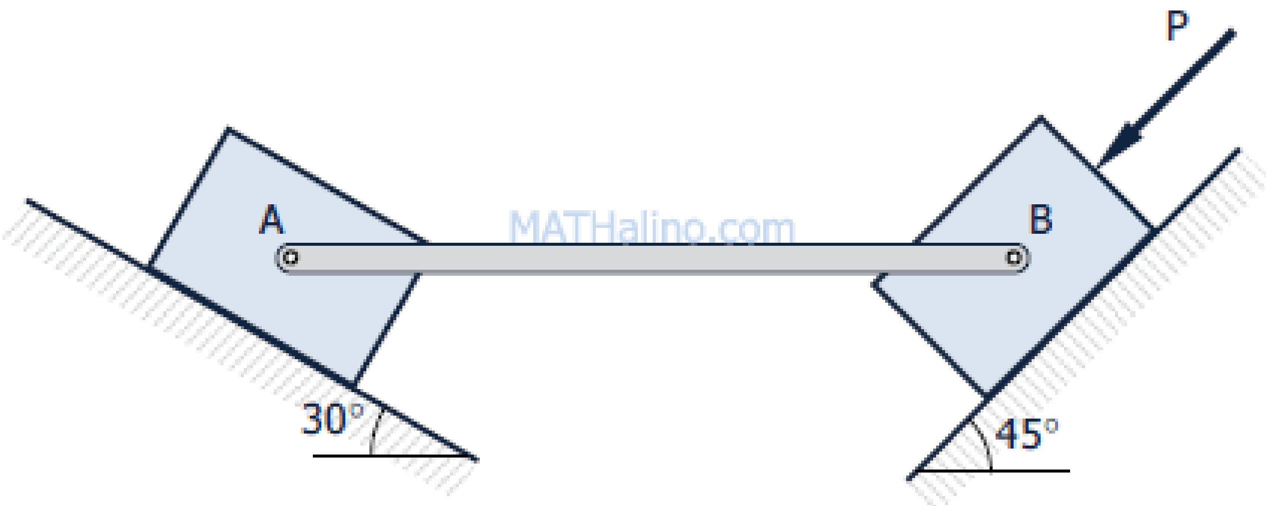



**Figure P-532**

Locate the centroid of the shaded area in Fig. P-722 created by cutting a semicircle of diameter r from a quarter circle of radius r.

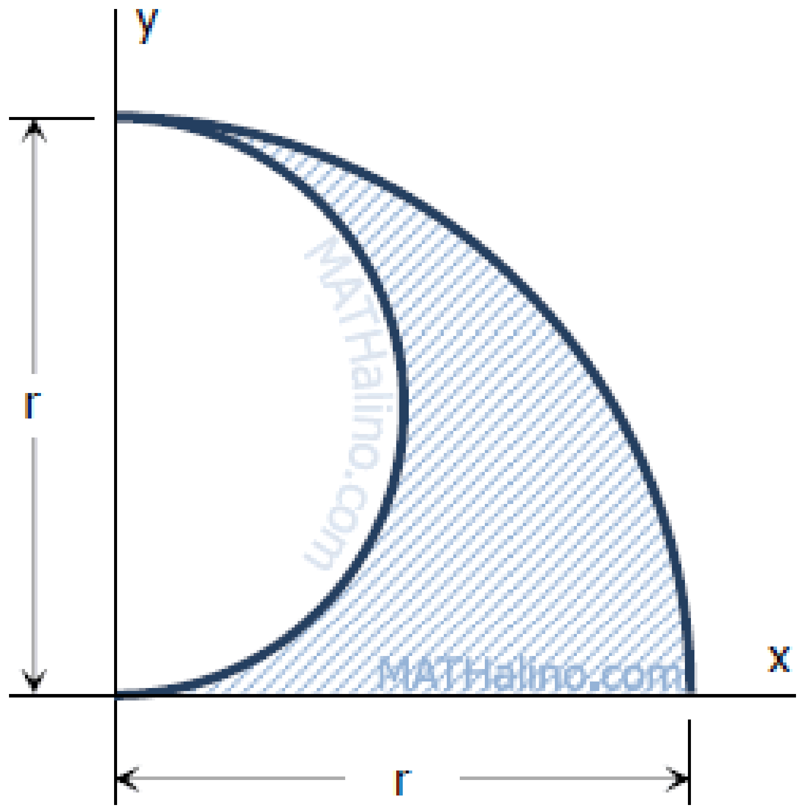



**Figure P-722**

Determine the moment of inertia of the area shown in Fig. P-819 with respect to its centroidal axes.

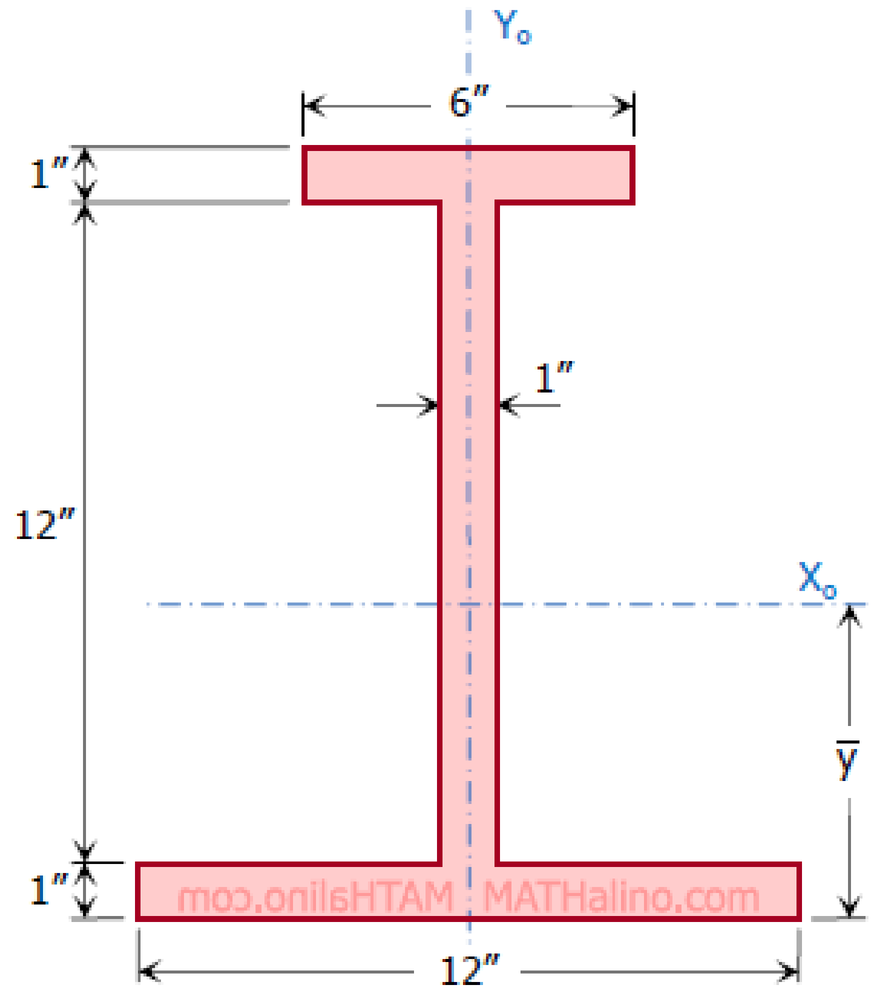



**Figure P-820**

Given the parabola $3x^2 + 40y - 4800 = 0$.

Part 1: What is the area bounded by the parabola and the X-axis?

A. 6 200 $unit^2$

B. 8 300 $unit^2$

C. 5 600 $unit^2$

D. 6 400 $unit^2$

Given the parabola $3x^2 + 40y - 4800 = 0$.

Part 2: What is the moment of inertia, about the X-axis, of the area bounded by the parabola and the X-axis?

A. 15 045 000 $unit^4$

B. 18 362 000 $unit^4$

C. 11 100 000 $unit^4$

D. 21 065 000 $unit^4$

Given the parabola $3x^2 + 40y - 4800 = 0$.

Part 3: What is the radius of gyration, about the X-axis, of the area bounded by the parabola and the X-axis?

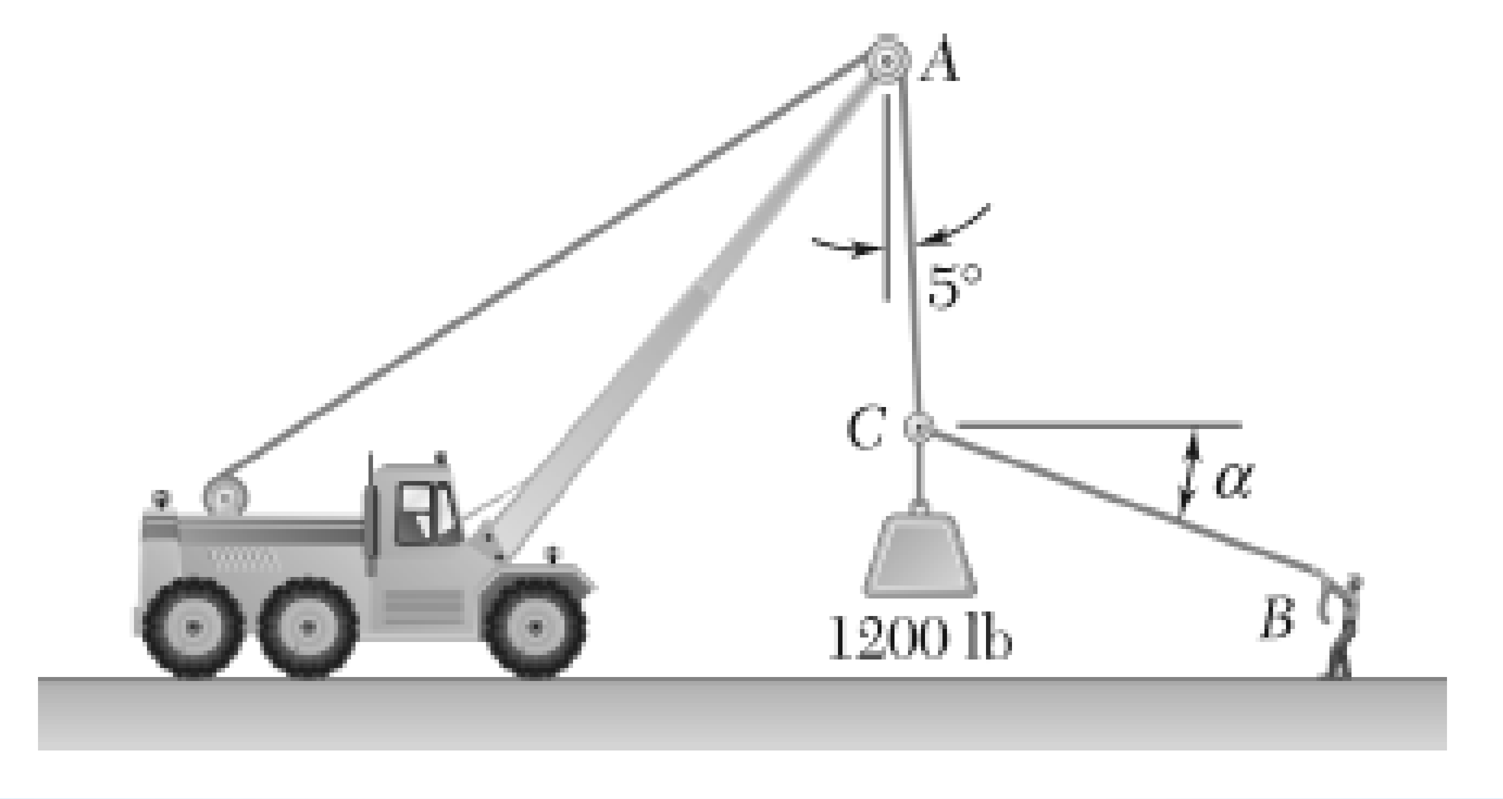
A. 57.4 units

B. 63.5 units

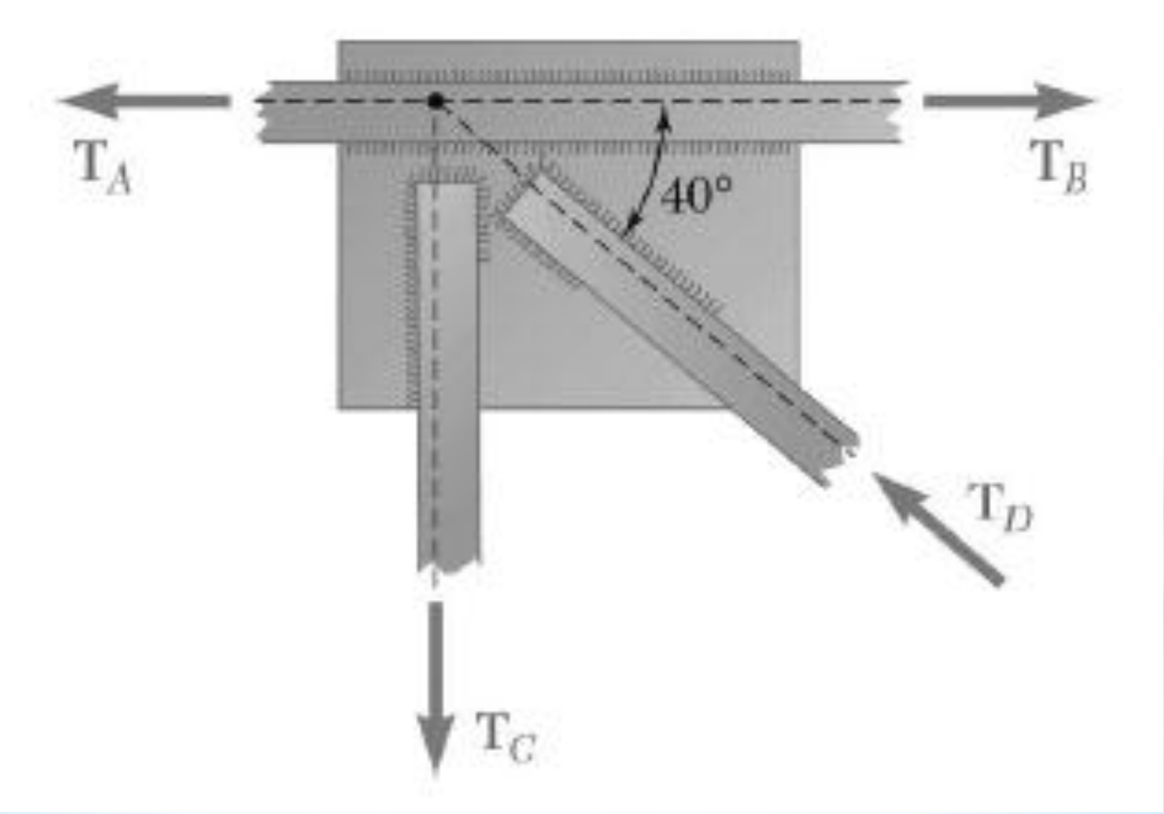
C. 47.5 units

D. 75.6 units

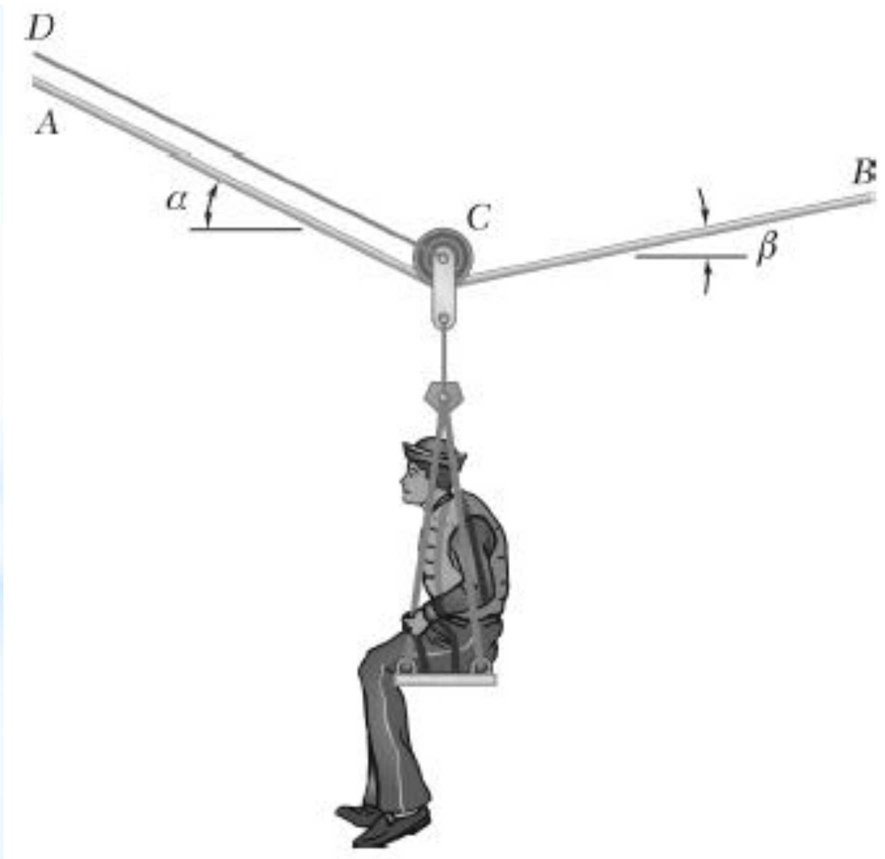
Knowing that $\alpha = 20°$, determine the tension (*a*) in cable *AC*, (*b*) in rope *BC*.



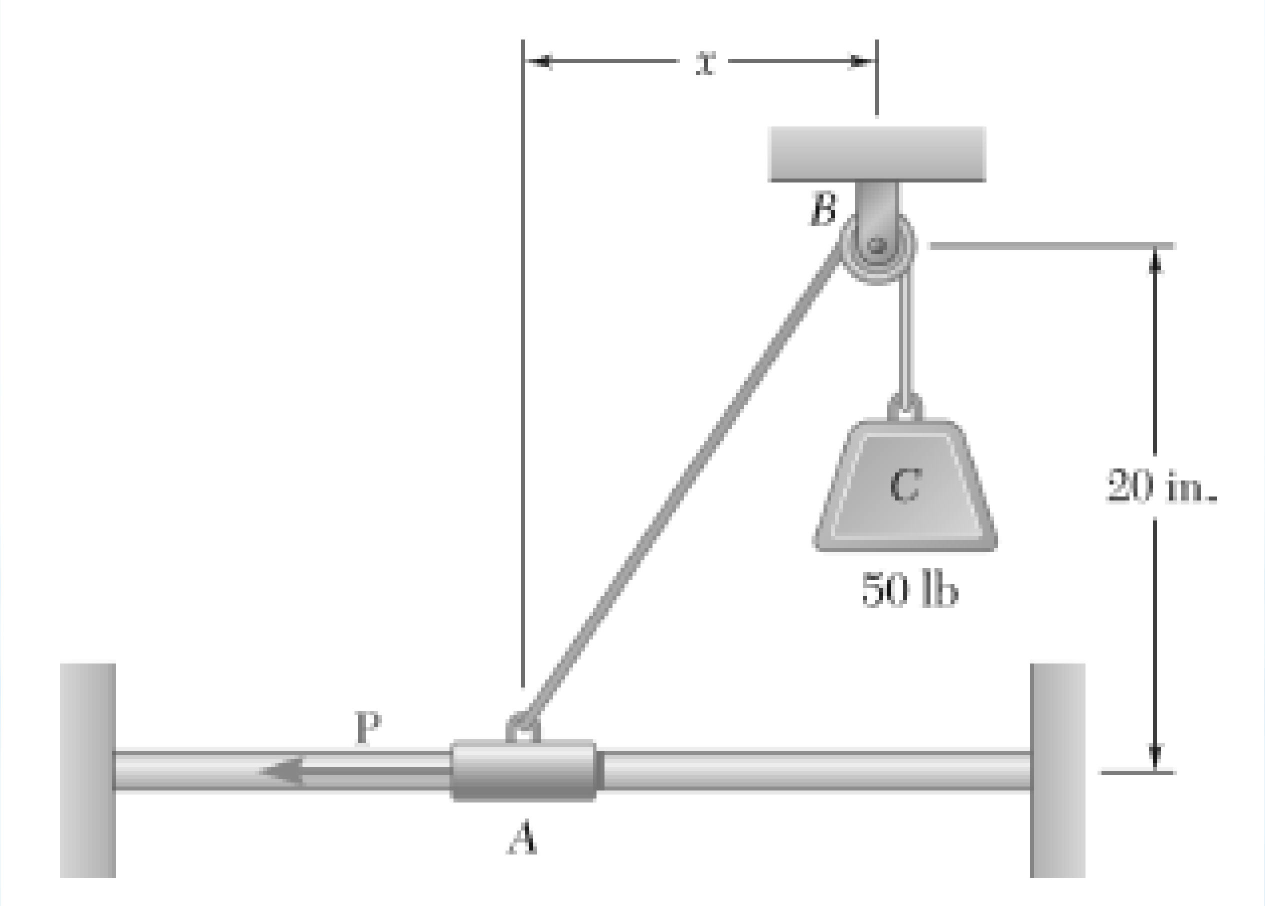
Two forces of magnitude $T_A = 8$ kips and $T_B = 15$ kips are applied as shown to a welded connection. Knowing that the connection is in equilibrium, determine the magnitudes of the forces $T_C$ and $T_D$.



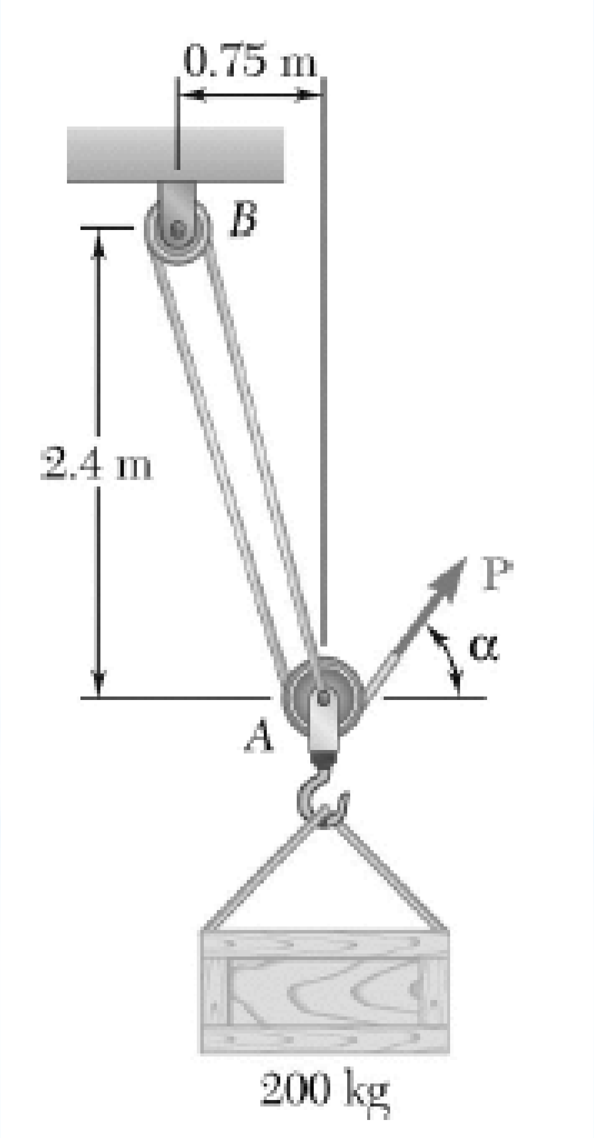
A sailor is being rescued using a boatswain's chair that is suspended from a pulley that can roll freely on the support cable $ACB$ and is pulled at a constant speed by cable $CD$. Knowing that $\alpha = 30°$ and $\beta = 10°$ and that the combined weight of the boatswain's chair and the sailor is 900 N, determine the tension (*a*) in the support cable $ACB$, (*b*) in the traction cable $CD$.



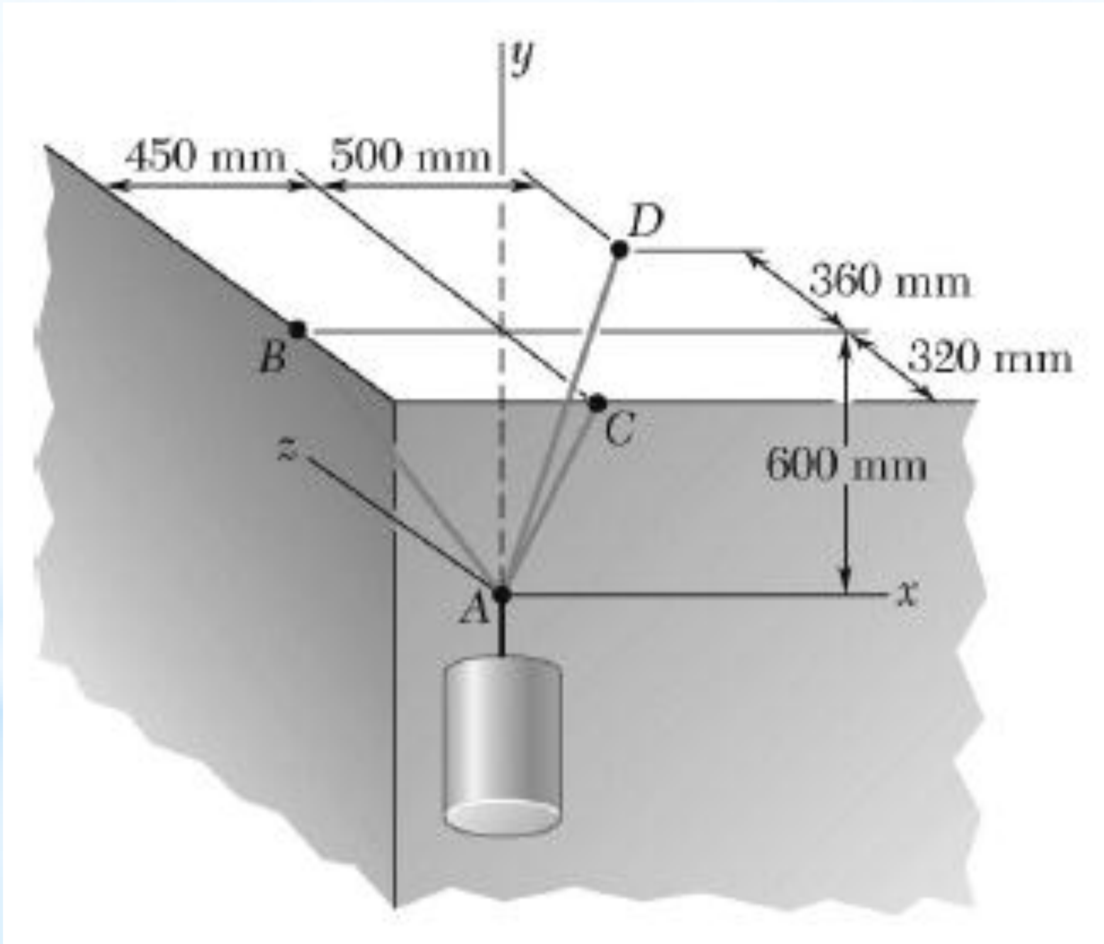
Collar $A$ is connected as shown to a 50-lb load and can slide on a frictionless horizontal rod. Determine the magnitude of the force **P** required to maintain the equilibrium of the collar when (*a*) $x = 4.5$ in., (*b*) $x = 15$ in.



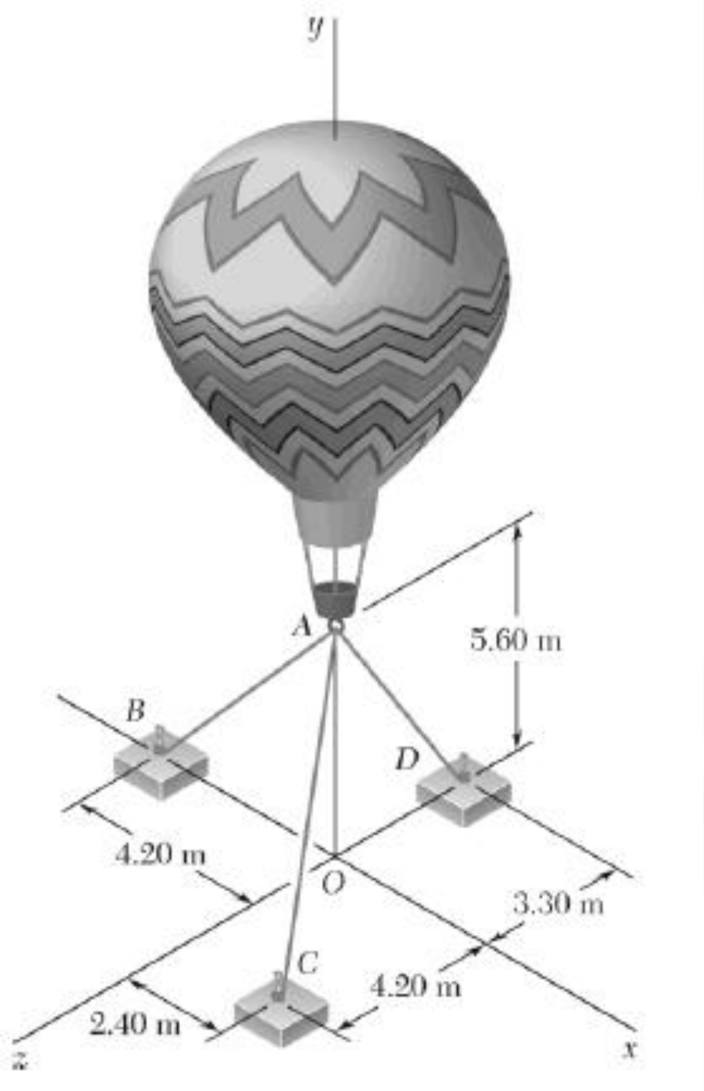
A 200-kg crate is to be supported by the rope-and-pulley arrangement shown. Determine the magnitude and direction of the force **P** that must be exerted on the free end of the rope to maintain equilibrium. (*Hint:* The tension in the rope is the same on each side of a simple pulley. This can be proved by the methods of Ch. 4.)



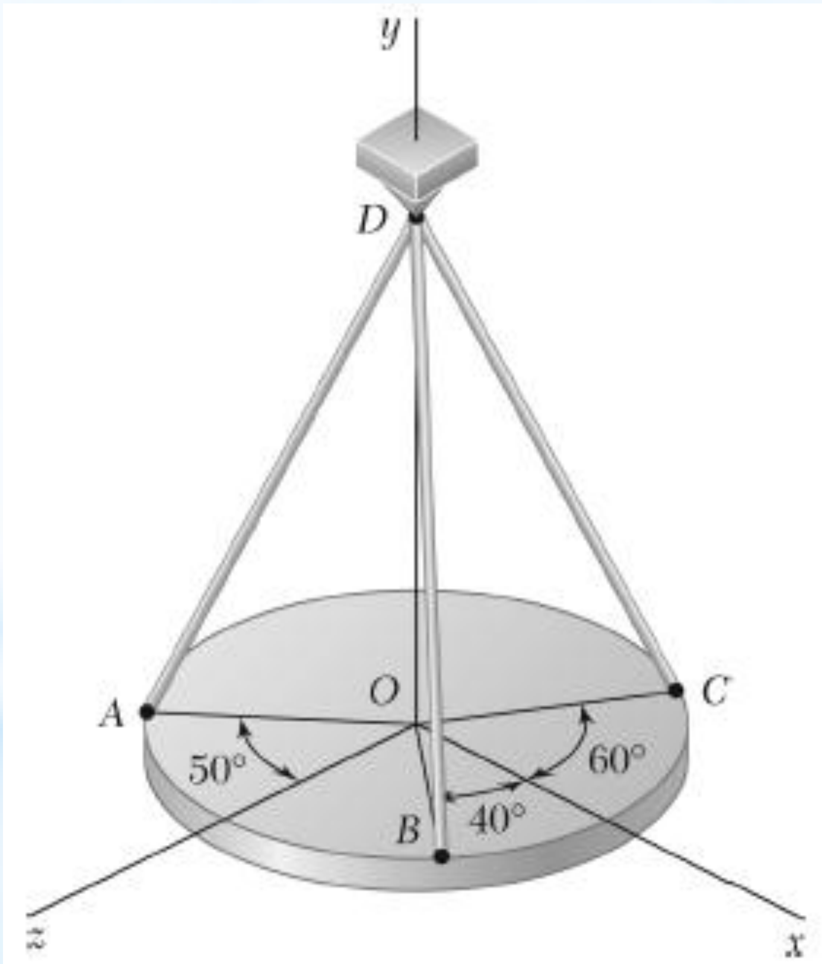
A container is supported by three cables that are attached to a ceiling as shown. Determine the weight $W$ of the container, knowing that the tension in cable $AD$ is 4.3 kN.



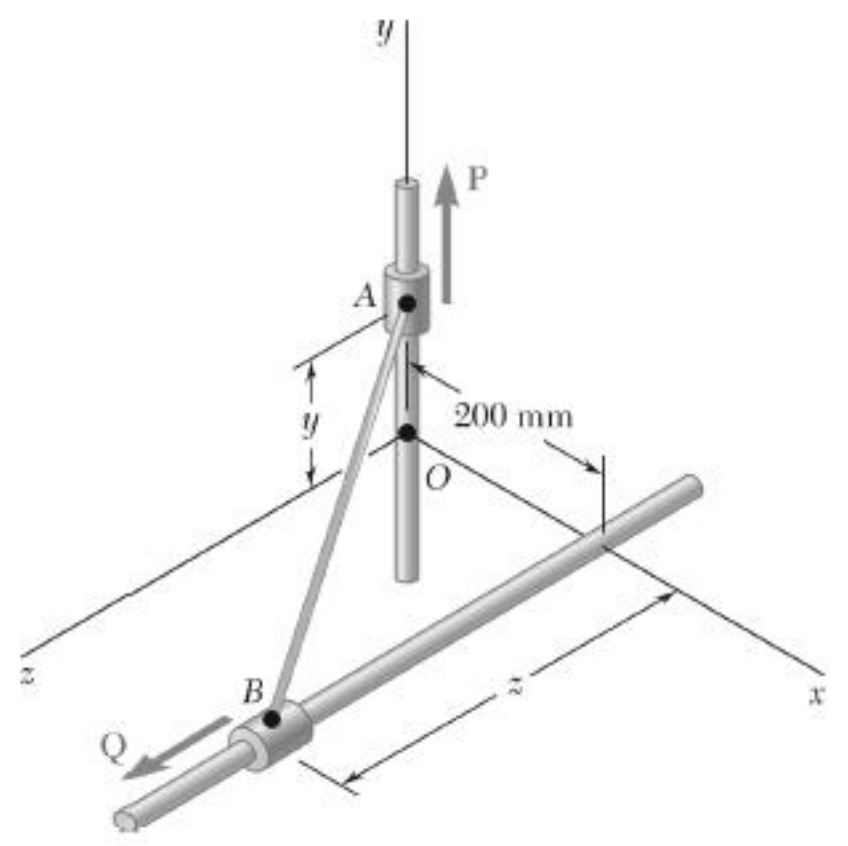
Three cables are used to tether a balloon as shown. Knowing that the balloon exerts an 800-N vertical force at $A$, determine the tension in each cable.



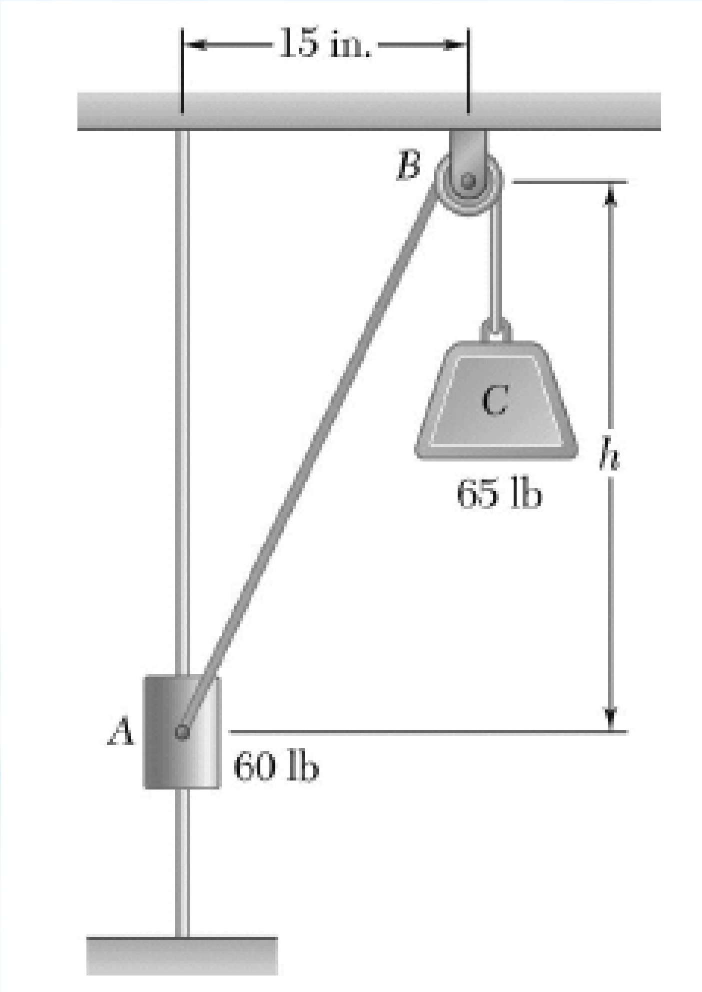
A horizontal circular plate weighing 60 lb is suspended as shown from three wires that are attached to a support at $D$ and form 30° angles with the vertical. Determine the tension in each wire.



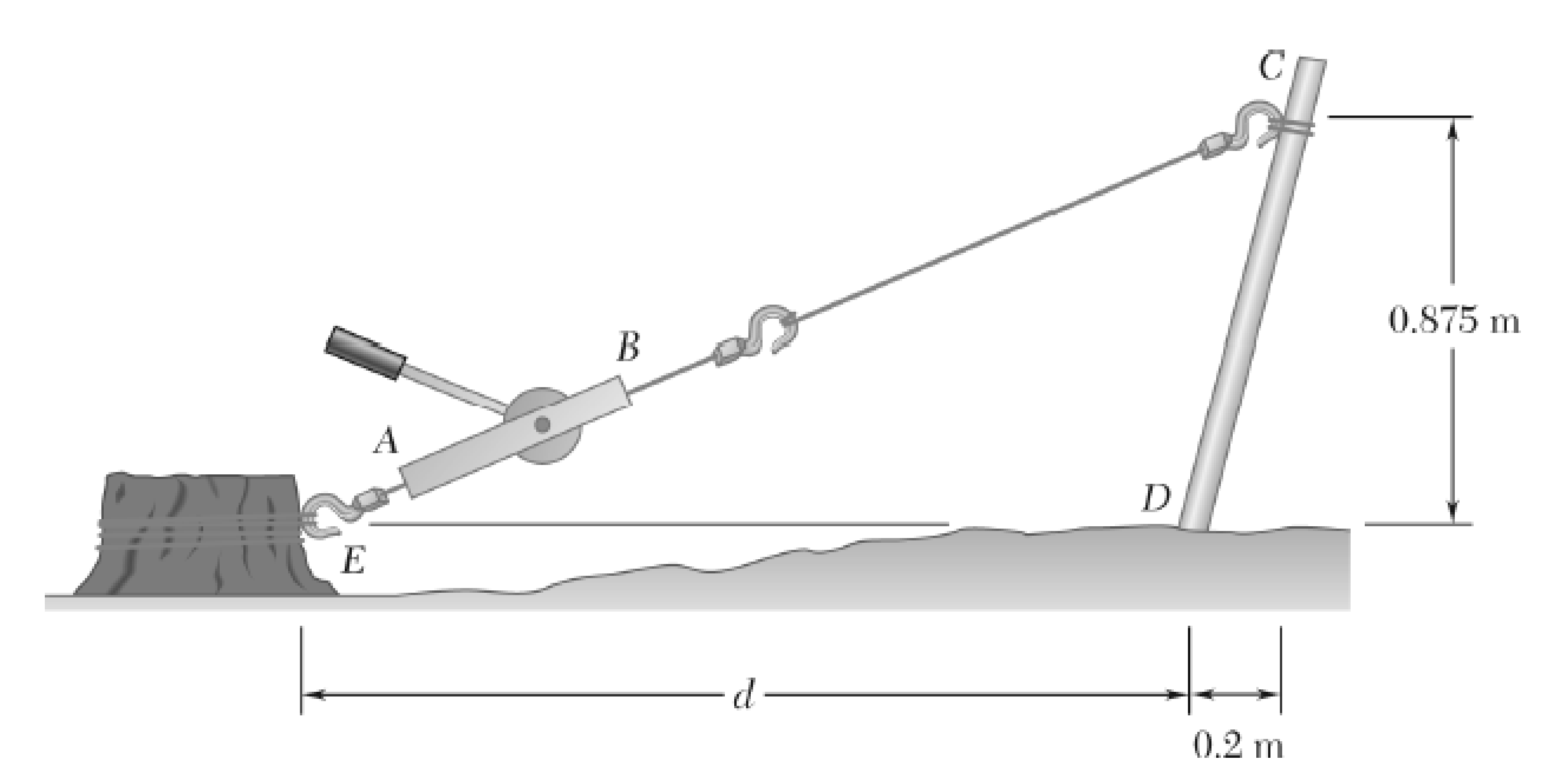
Collars $A$ and $B$ are connected by a 525-mm-long wire and can slide freely on frictionless rods. If a force $\mathbf{P} = (341\text{ N})\mathbf{j}$ is applied to collar $A$, determine (*a*) the tension in the wire when $y = 155$ mm, (*b*) the magnitude of the force $\mathbf{Q}$ required to maintain the equilibrium of the system.



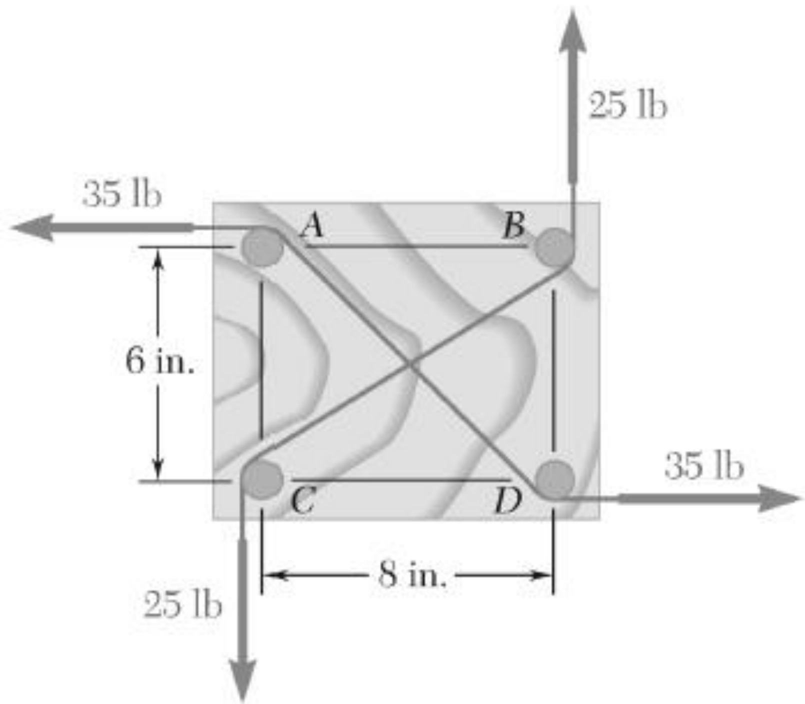
The 60-lb collar $A$ can slide on a frictionless vertical rod and is connected as shown to a 65-lb counterweight $C$. Draw the free-body diagram needed to determine the value of $h$ for which the system is in equilibrium.



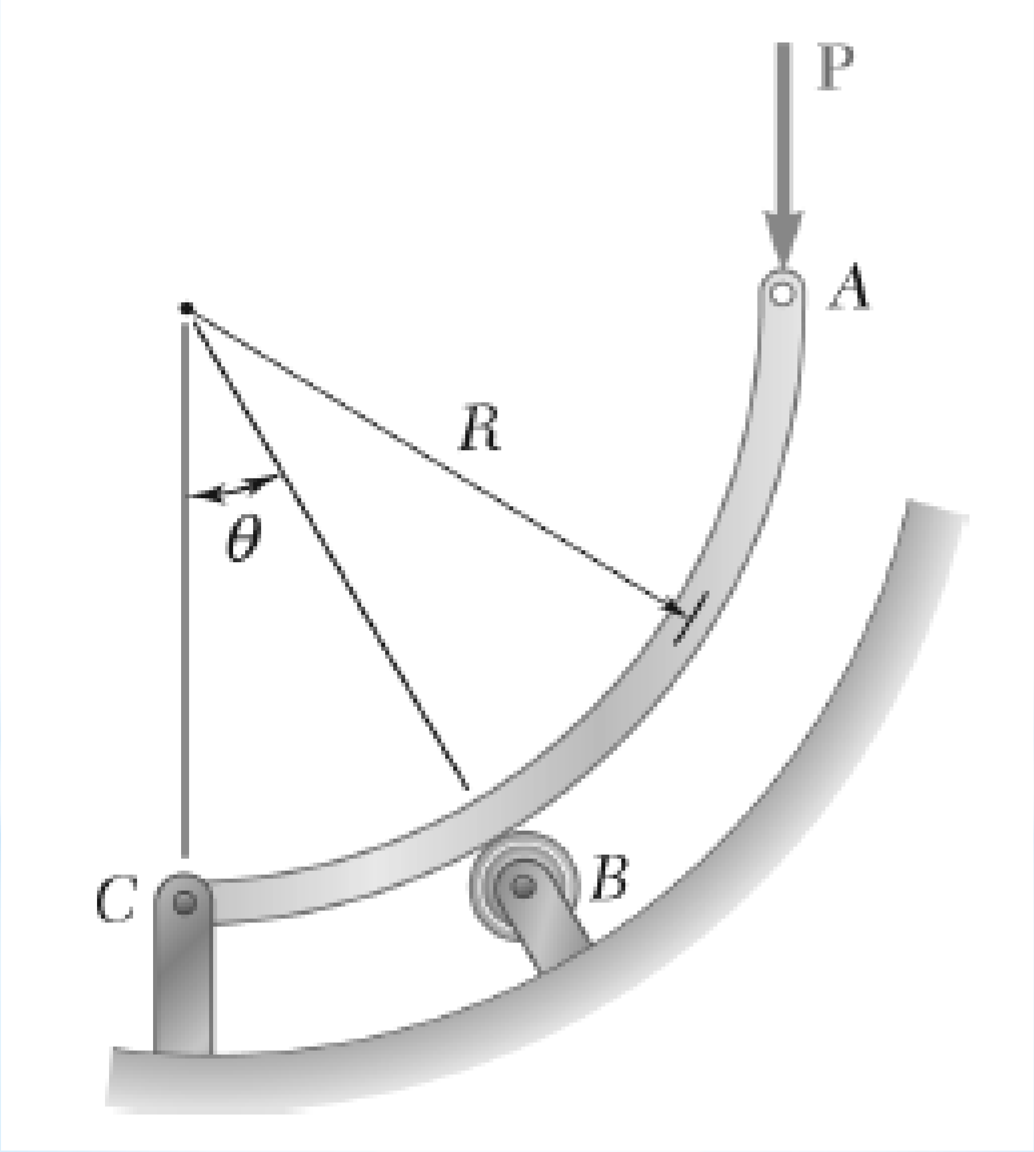
A winch puller *AB* is used to straighten a fence post. Knowing that the tension in cable *BC* is 1040 N and length *d* is 1.90 m, determine the moment about *D* of the force exerted by the cable at *C* by resolving that force into horizontal and vertical components applied (*a*) at Point *C*, (*b*) at Point *E*.



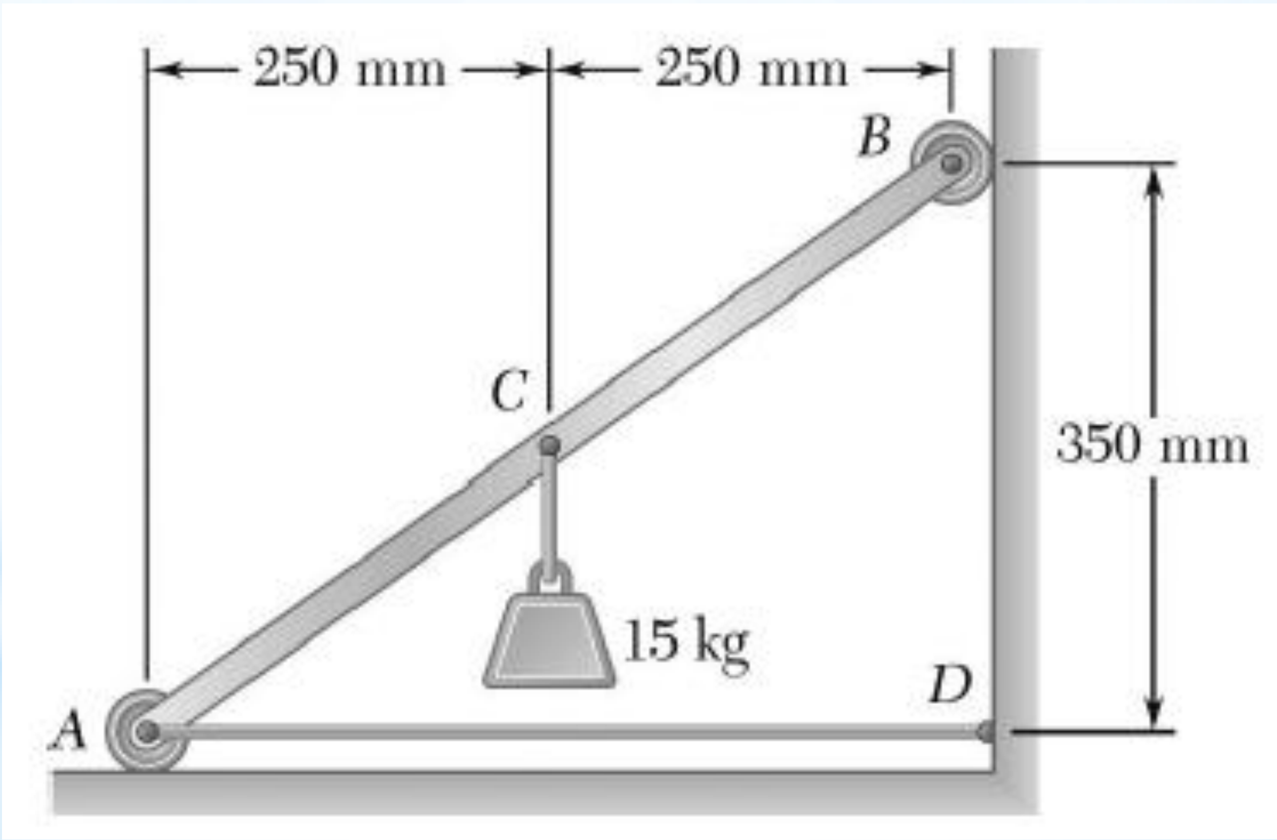
Four 1-in.-diameter pegs are attached to a board as shown. Two strings are passed around the pegs and pulled with the forces indicated. (*a*) Determine the resultant couple acting on the board. (*b*) If only one string is used, around which pegs should it pass and in what directions should it be pulled to create the same couple with the minimum tension in the string? (*c*) What is the value of that minimum tension?



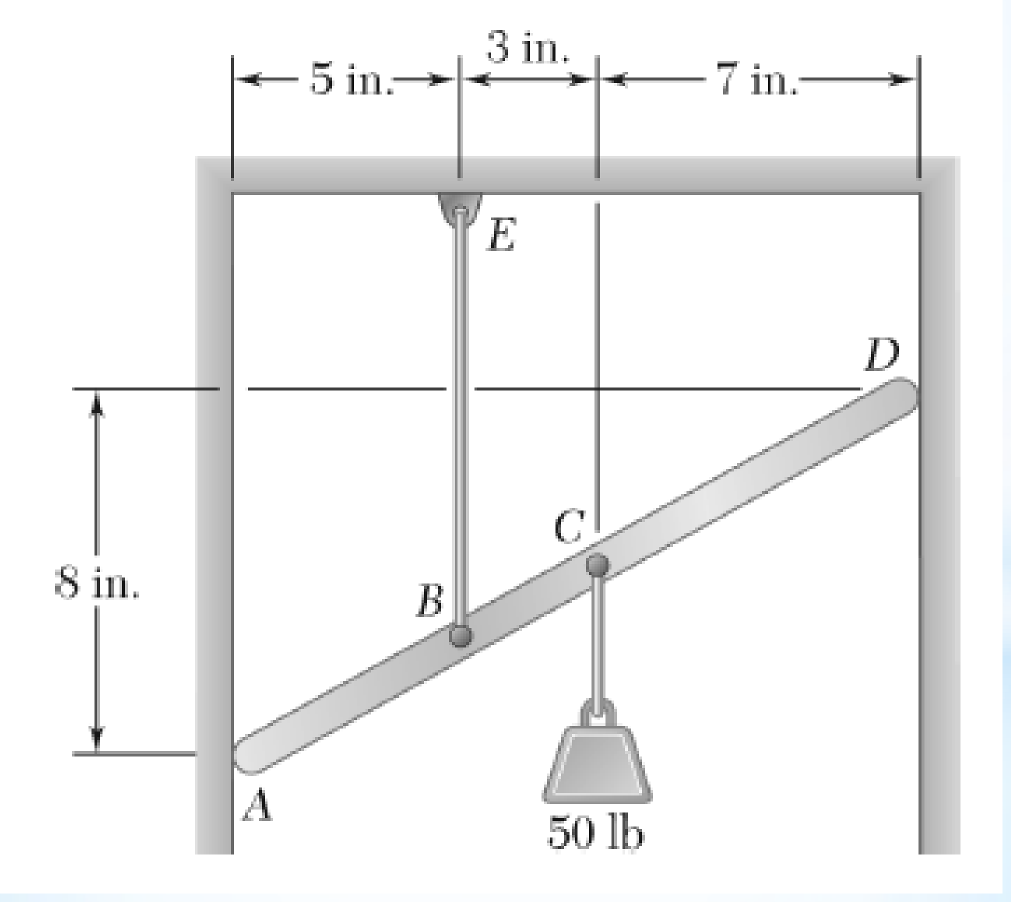
Rod $ABC$ is bent in the shape of an arc of circle of radius $R$. Knowing the $\theta = 30°$, determine the reaction (*a*) at $B$, (*b*) at $C$.



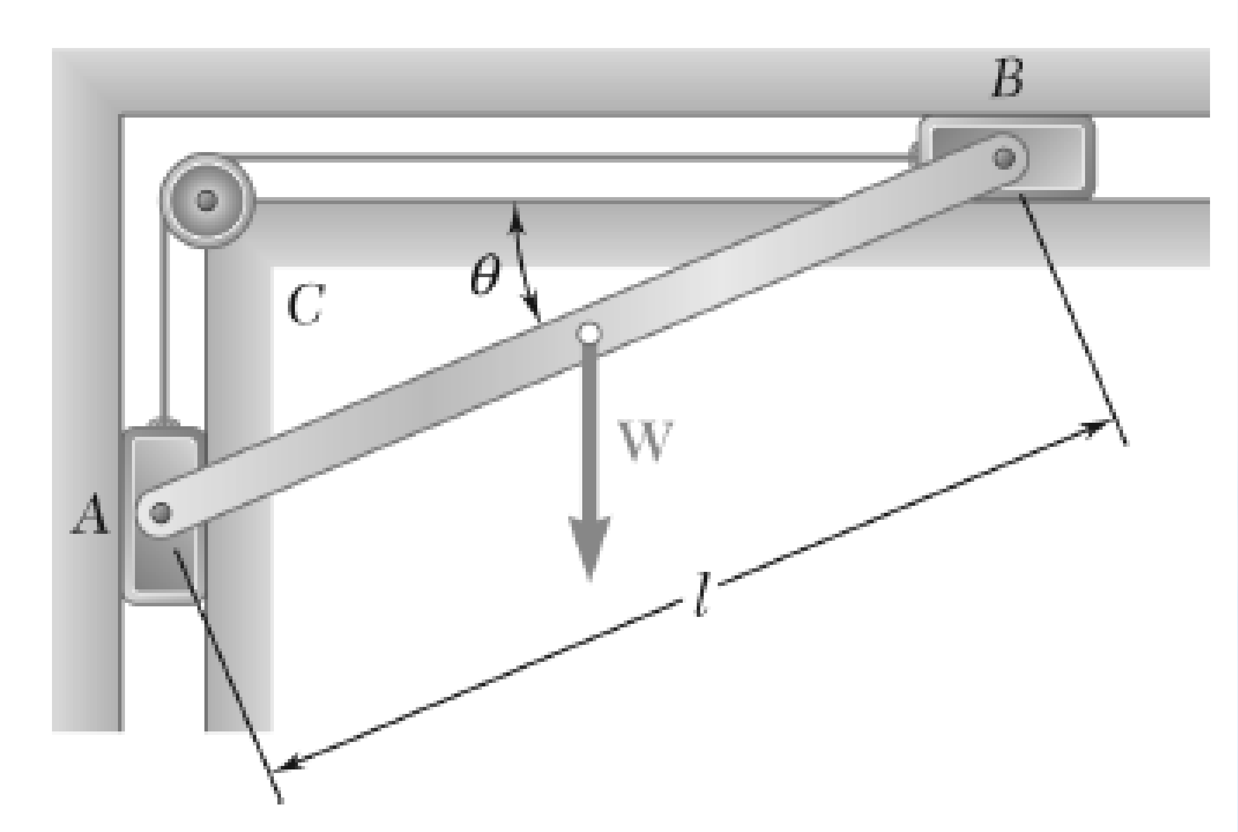
A light bar *AB* supports a 15-kg block at its midpoint *C*. Rollers at *A* and *B* rest against frictionless surfaces, and a horizontal cable *AD* is attached at *A*. Determine (*a*) the tension in cable *AD*, (*b*) the reactions at *A* and *B*.



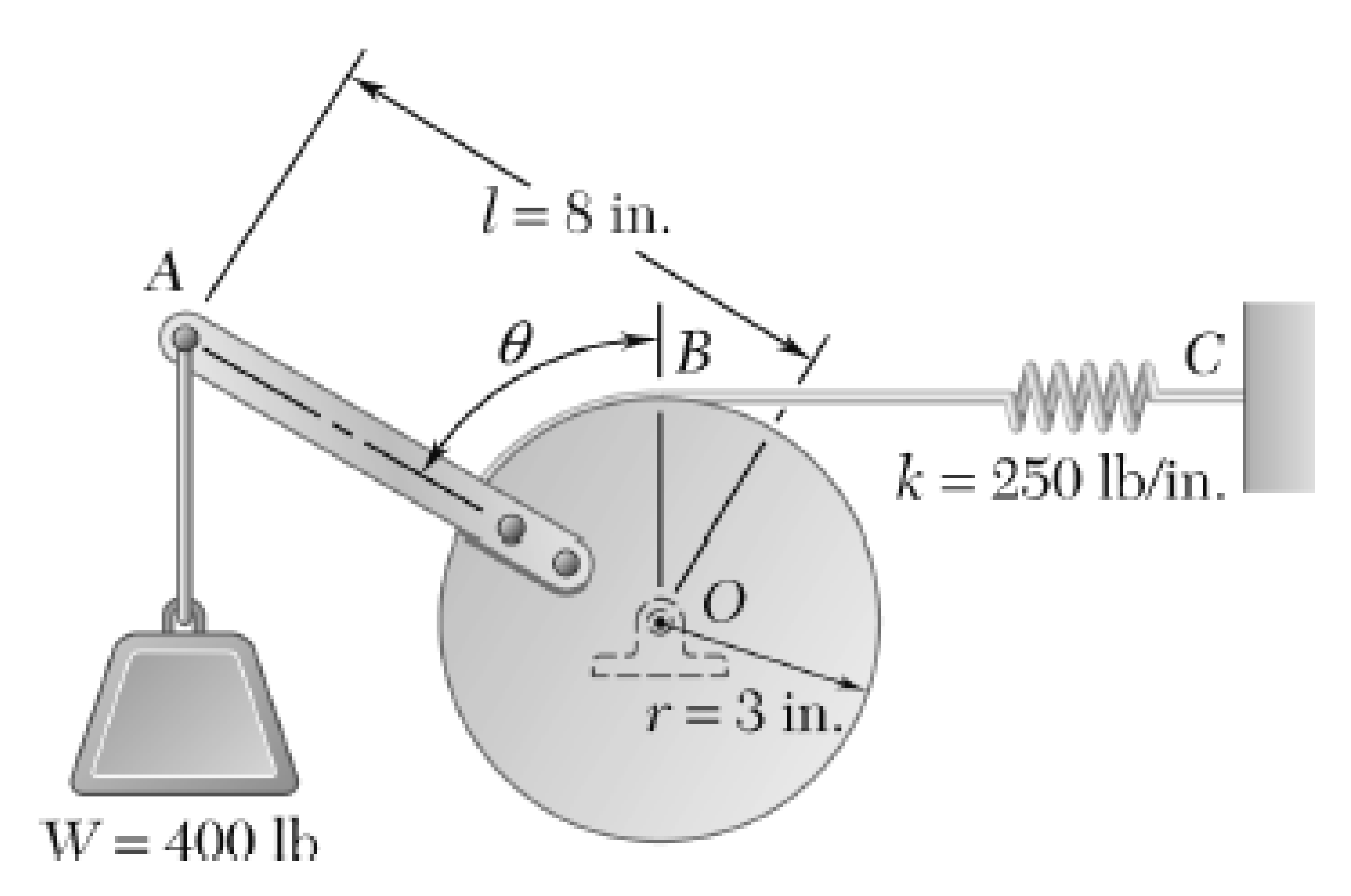
A light bar *AD* is suspended from a cable *BE* and supports a 50-lb block at *C*. The ends *A* and *D* of the bar are in contact with frictionless vertical walls. Determine the tension in cable *BE* and the reactions at *A* and *D*.



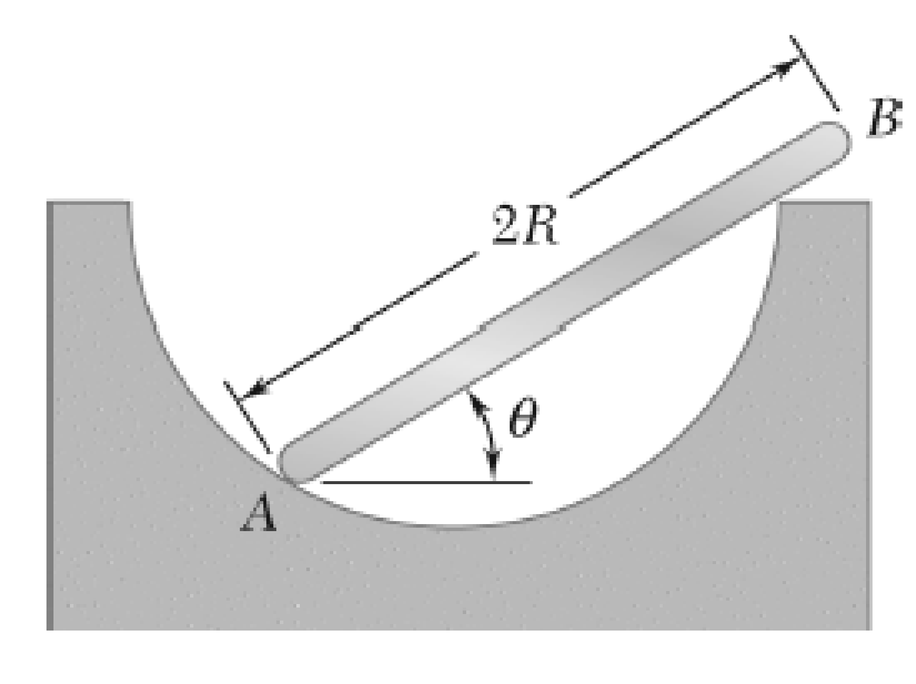
A slender rod $AB$, of weight $W$, is attached to blocks $A$ and $B$, which move freely in the guides shown. The blocks are connected by an elastic cord that passes over a pulley at $C$. ($a$) Express the tension in the cord in terms of $W$ and $\theta$. ($b$) Determine the value of $\theta$ for which the tension in the cord is equal to $3W$.



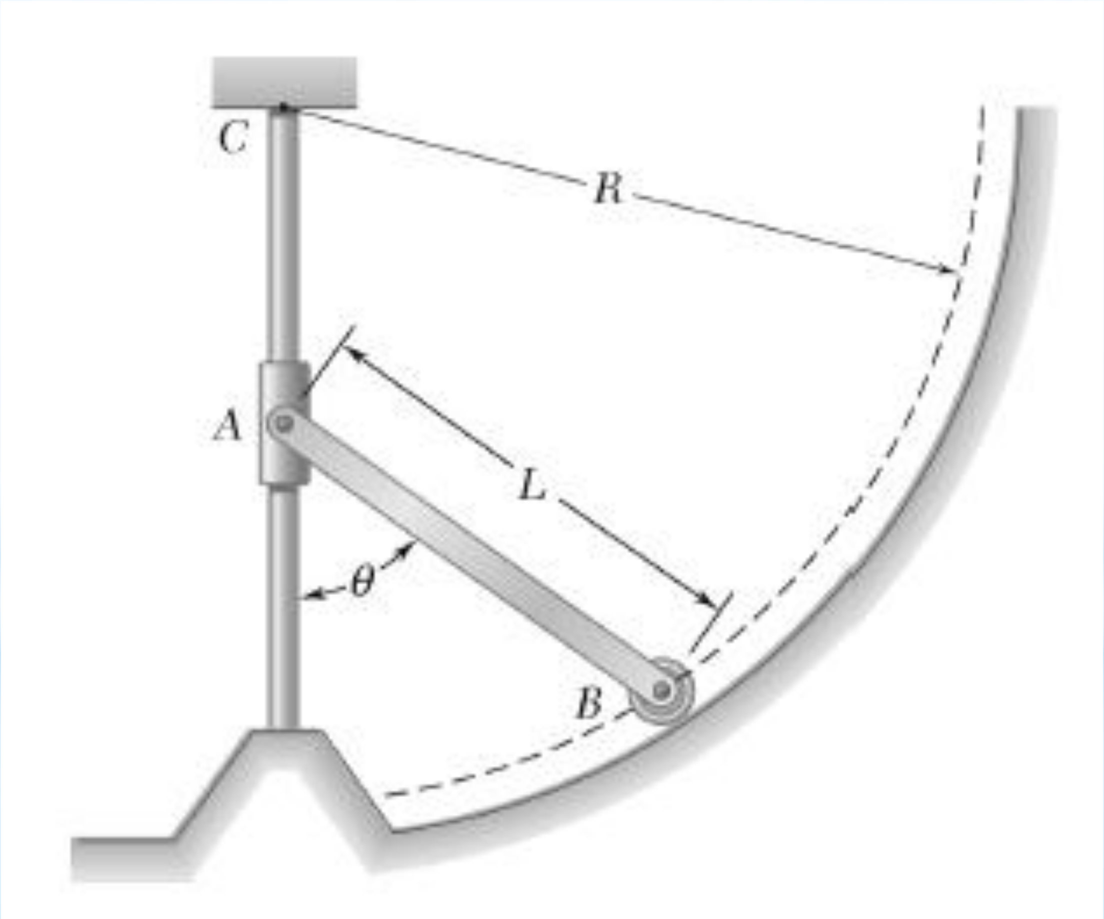
Solve Sample Problem 4.5, assuming that the spring is unstretched when $\theta = 90°$.



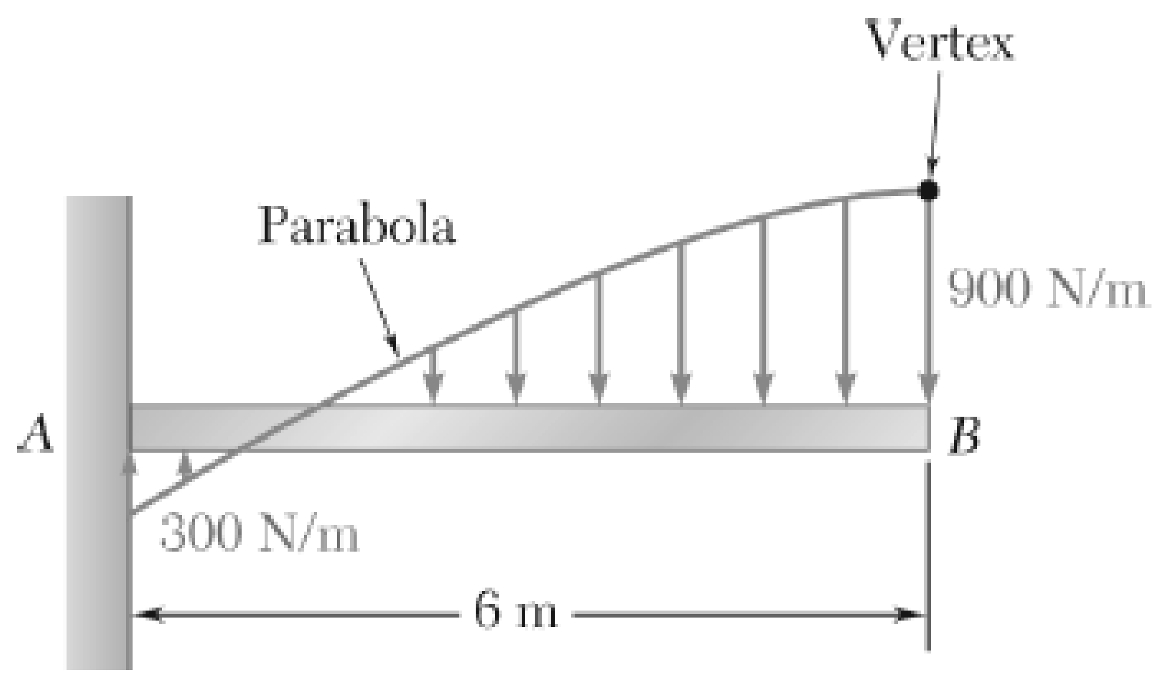
A uniform rod $AB$ of length $2R$ rests inside a hemispherical bowl of radius $R$ as shown. Neglecting friction, determine the angle $\theta$ corresponding to equilibrium.



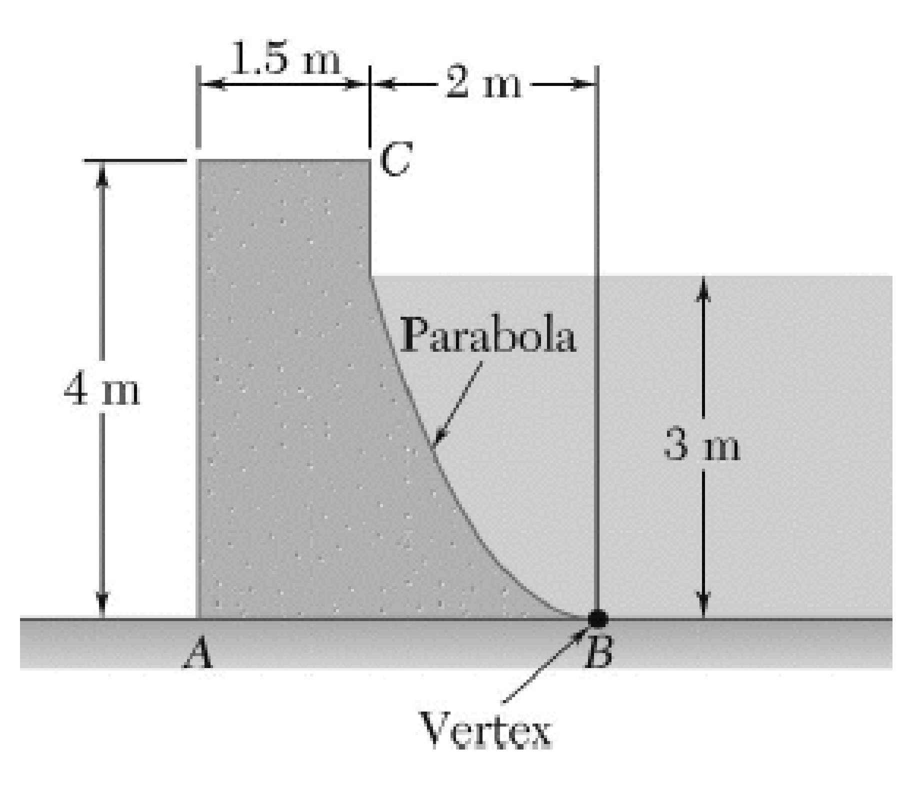
A slender rod of length $L$ and weight $W$ is attached to a collar at $A$ and is fitted with a small wheel at $B$. Knowing that the wheel rolls freely along a cylindrical surface of radius $R$, and neglecting friction, derive an equation in $\theta$, $L$, and $R$ that must be satisfied when the rod is in equilibrium.



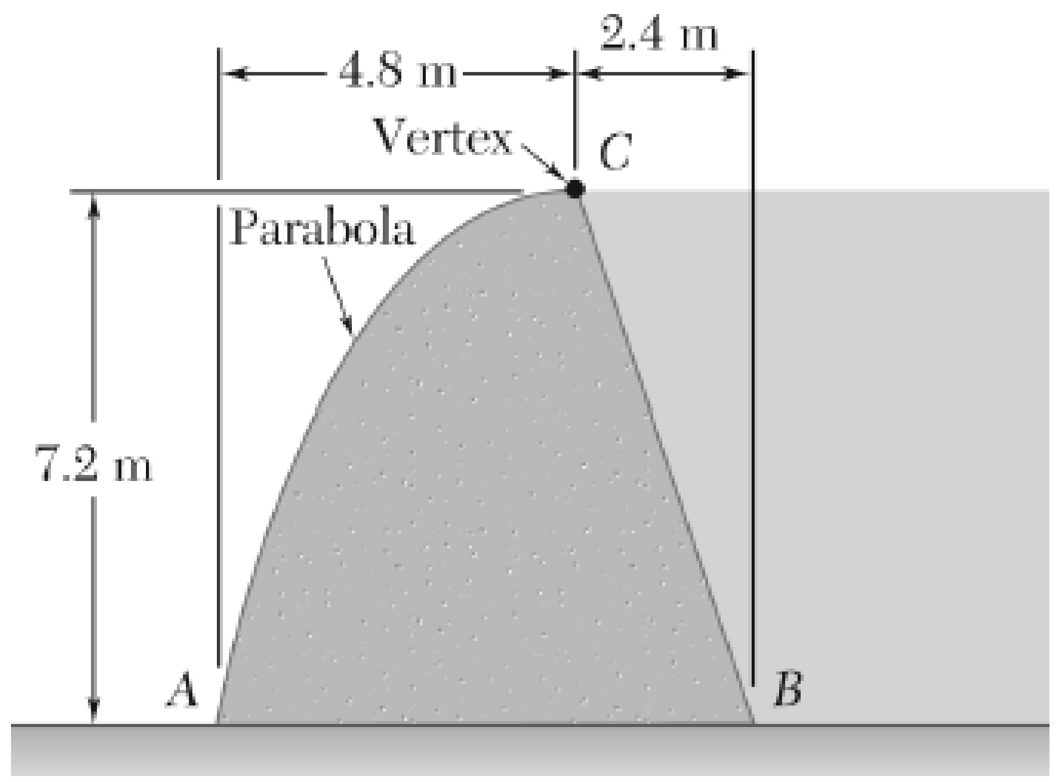
Determine the reactions at the beam supports for the given loading.



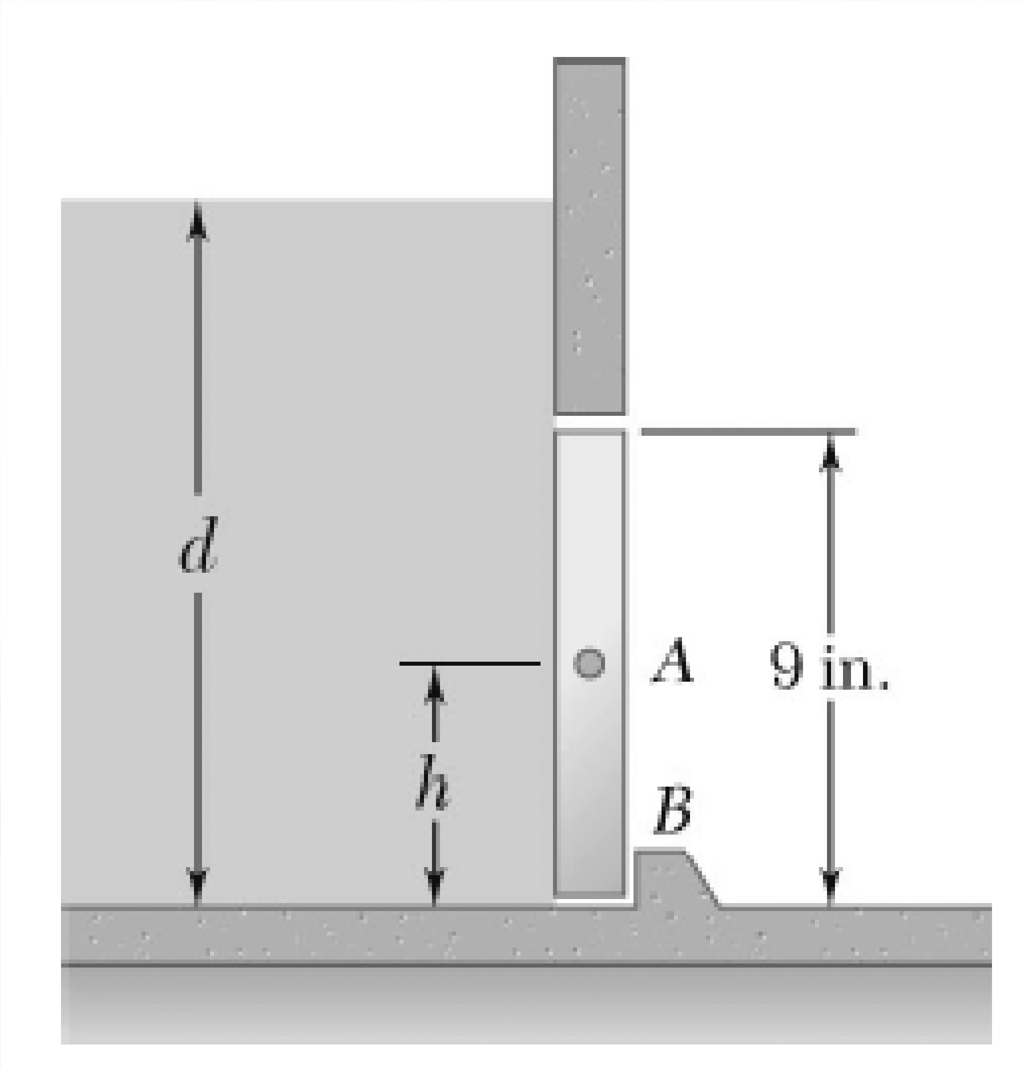
The cross section of a concrete dam is as shown. For a 1-m-wide dam section, determine (*a*) the resultant of the reaction forces exerted by the ground on the base *AB* of the dam, (*b*) the point of application of the resultant of part *a*, (*c*) the resultant of the pressure forces exerted by the water on the face *BC* of the dam.



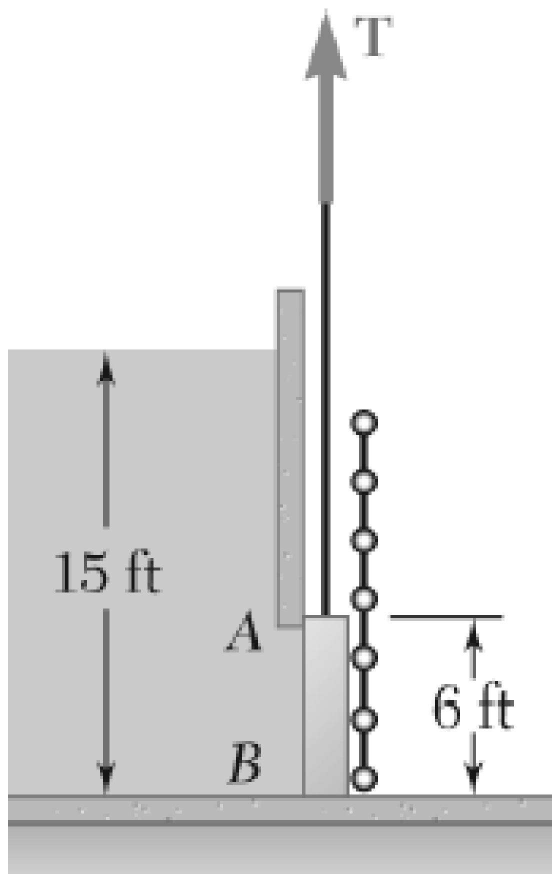
The cross section of a concrete dam is as shown. For a 1-m-wide dam section, determine (*a*) the resultant of the reaction forces exerted by the ground on the base *AB* of the dam, (*b*) the point of application of the resultant of part *a*, (*c*) the resultant of the pressure forces exerted by the water on the face *BC* of the dam.



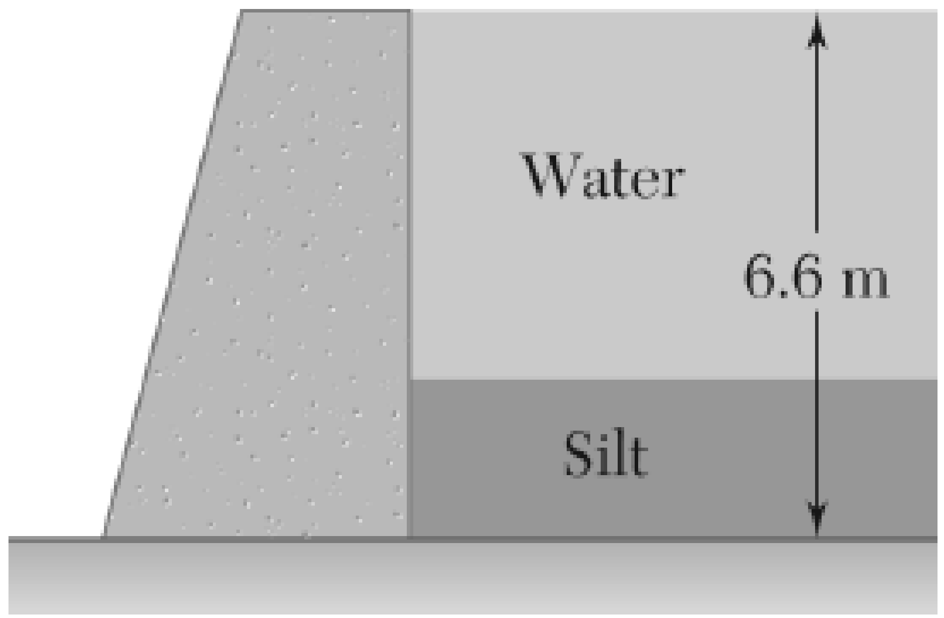
An automatic valve consists of a 9 × 9-in. square plate that is pivoted about a horizontal axis through $A$ located at a distance $h = 3.6$ in. above the lower edge. Determine the depth of water $d$ for which the valve will open.



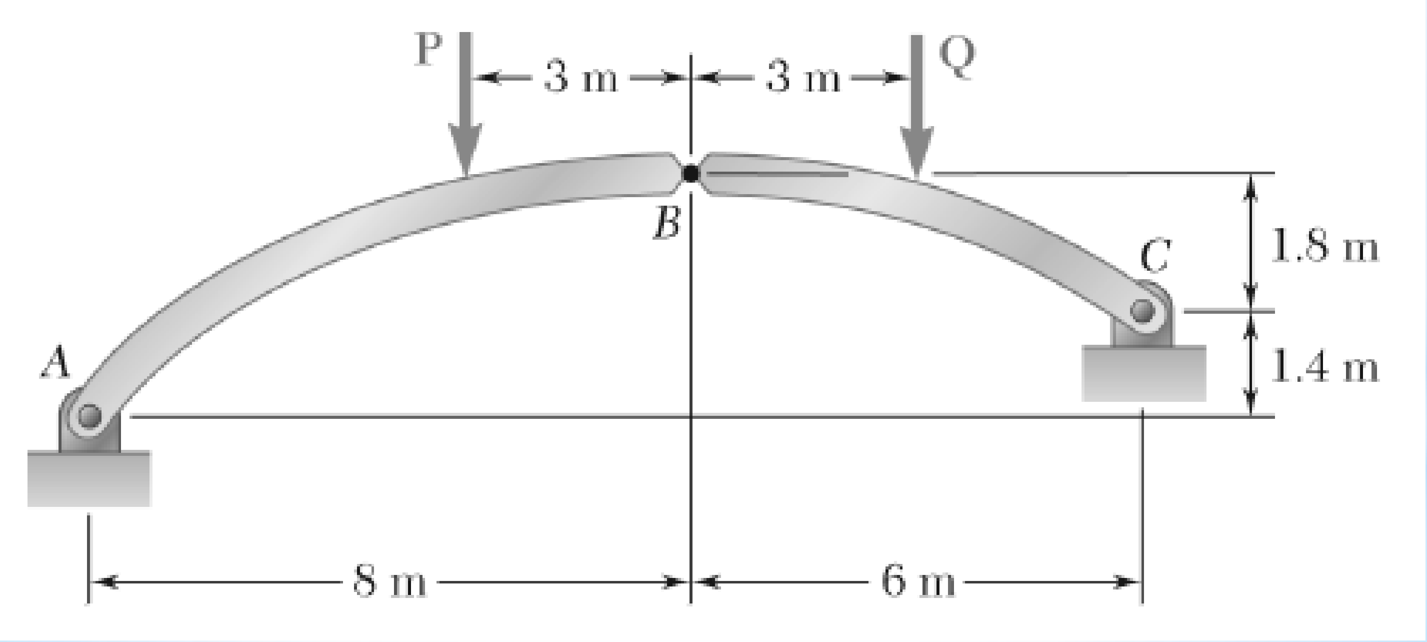
The friction force between a 6 × 6-ft square sluice gate *AB* and its guides is equal to 10 percent of the resultant of the pressure forces exerted by the water on the face of the gate. Determine the initial force needed to lift the gate if it weighs 1000 lb.



The base of a dam for a lake is designed to resist up to 120 percent of the horizontal force of the water. After construction, it is found that silt (that is equivalent to a liquid of density $\rho_s = 1.76 \times 10^3$ kg/m$^3$) is settling on the lake bottom at the rate of 12 mm/year. Considering a 1-m-wide section of dam, determine the number of years until the dam becomes unsafe.



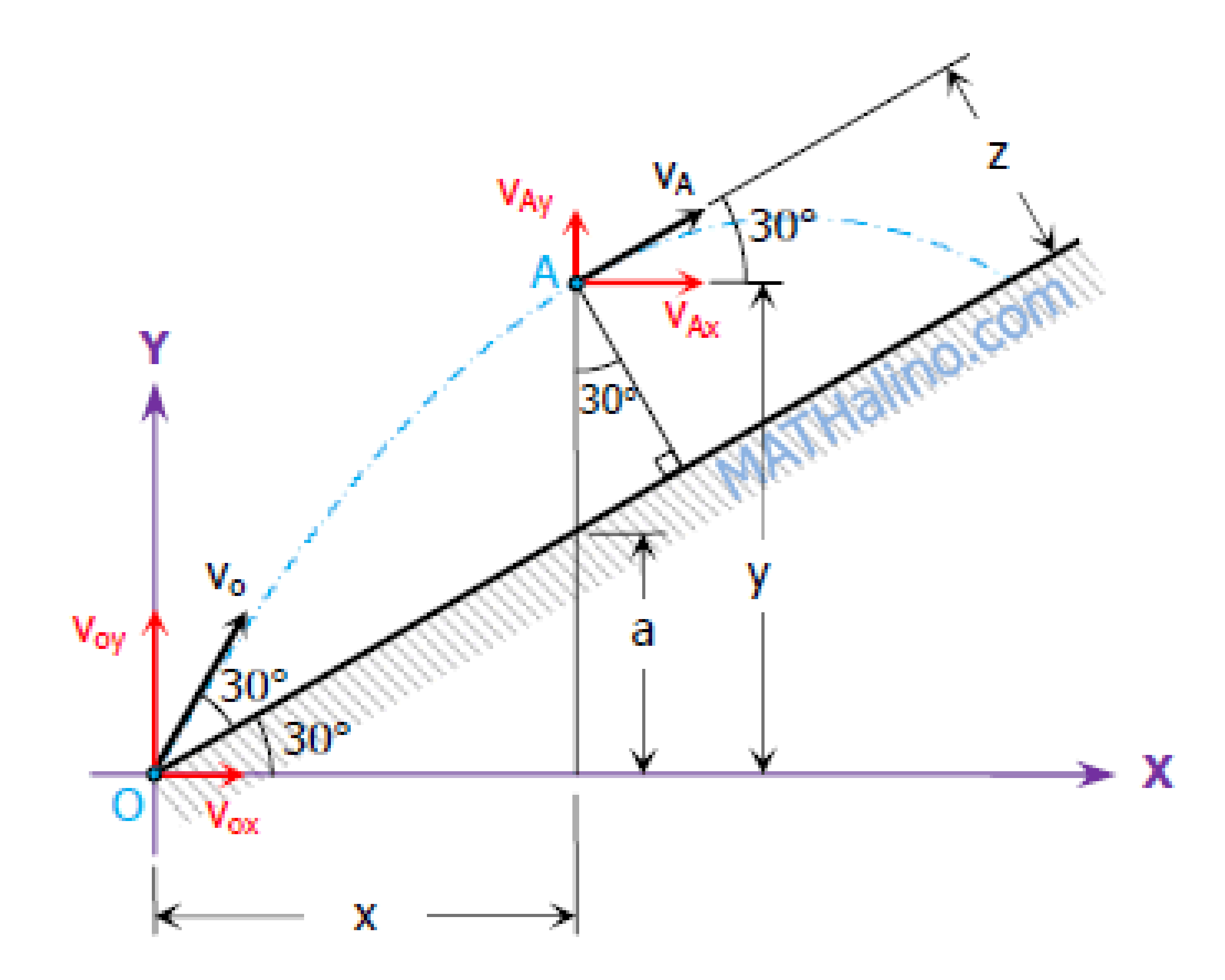
The axis of the three-hinge arch $ABC$ is a parabola with the vertex at $B$. Knowing that $P = 140$ kN and $Q = 112$ kN, determine (*a*) the components of the reaction at $A$, (*b*) the components of the force exerted at $B$ on segment $AB$.



# Engineering Mechanics

## Dynamics

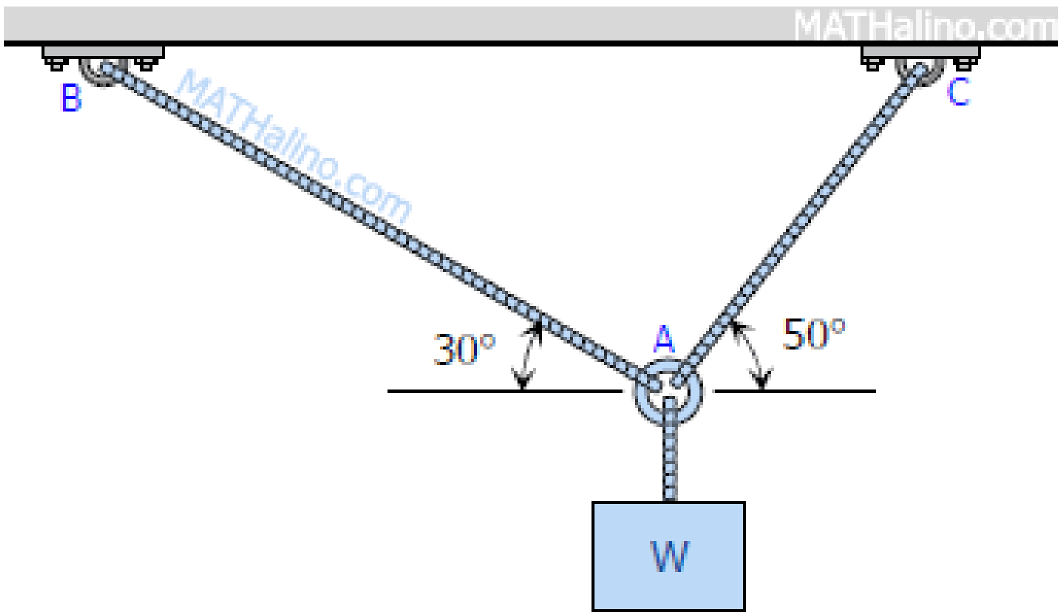
A projectile is fired up the inclined plane at an initial velocity of 15 m/s. The plane is making an angle of 30° from the horizontal. If the projectile was fired at 30° from the incline, compute the maximum height z measured perpendicular to the incline that is reached by the projectile. Neglect air resistance.



# Strength of Materials

Determine the largest weight W that can be supported by two wires shown in Fig. P-109. The stress in either wire is not to exceed 30 ksi. The cross-sectional areas of wires AB and AC are 0.4 in2 and 0.5 in2, respectively.

**Figure P-109**



The homogeneous bar ABCD shown in Fig. P-114 is supported by a cable that runs from A to B around the smooth peg at E, a vertical cable at C, and a smooth inclined surface at D. Determine the mass of the heaviest bar that can be supported if the stress in each cable is limited to 100 MPa. The area of the cable AB is 250 $mm^2$ and that of the cable at C is 300 $mm^2$.

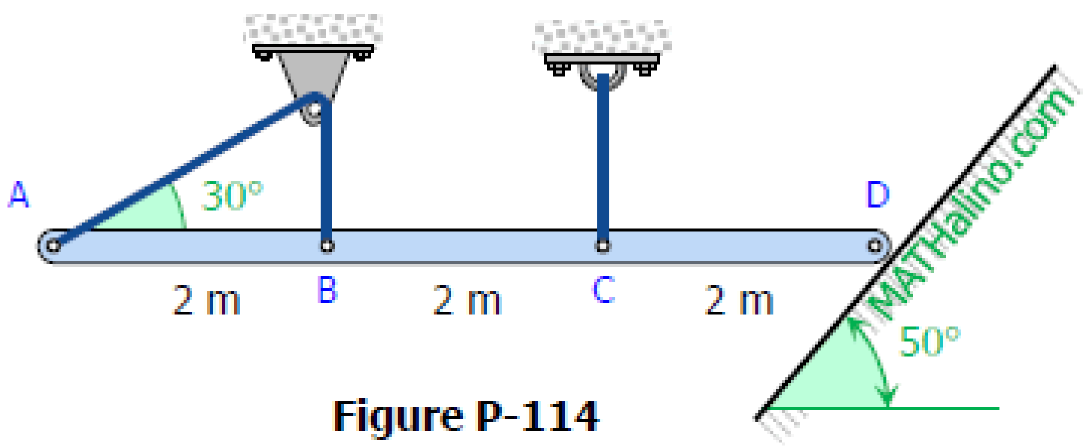



**Figure P-114**

A 200-mm-diameter pulley is prevented from rotating relative to 60-mm-diameter shaft by a 70-mm-long key, as shown in Fig. P-118. If a torque T = 2.2 kN·m is applied to the shaft, determine the width b if the allowable shearing stress in the key is 60 MPa.

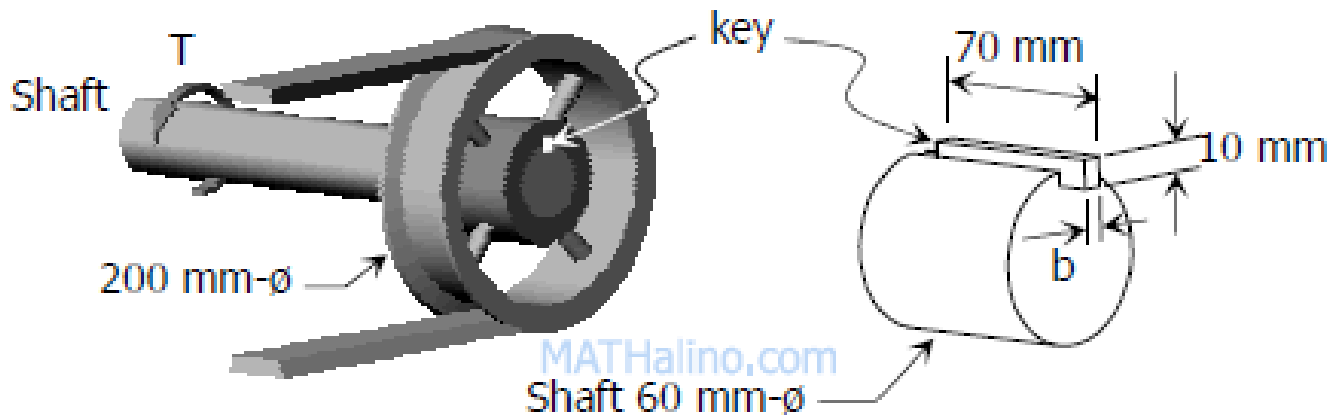



**Figure P-118**

A rectangular piece of wood, 50 mm by 100 mm in cross section, is used as a compression block shown in Fig. P-123. Determine the axial force P that can be safely applied to the block if the compressive stress in wood is limited to 20 $MN/m^2$ and the shearing stress parallel to the grain is limited to 5$MN/m^2$. The grain makes an angle of 20° with the horizontal, as shown. (Hint: Use the results in Problem 122.)

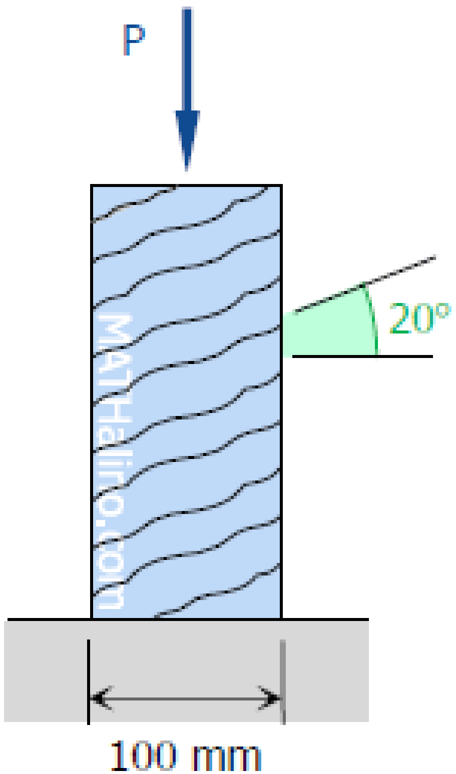



**Figure P-123**

In Fig. 1-12, assume that a 20-mm-diameter rivet joins the plates that are each 110 mm wide. The allowable stresses are 120 MPa for bearing in the plate material and 60 MPa for shearing of rivet. Determine (a) the minimum thickness of each plate; and (b) the largest average tensile stress in the plates.

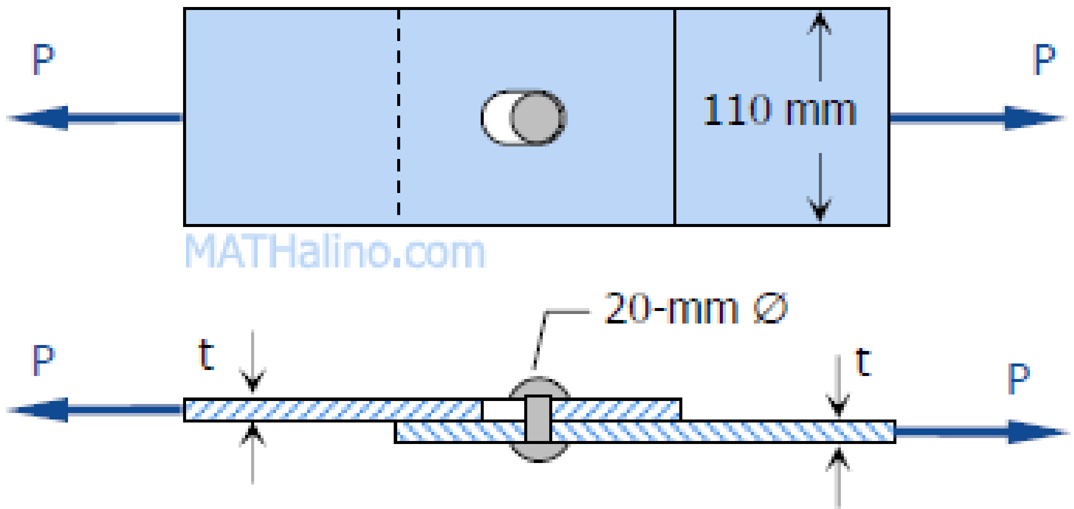



**Figure 1-12**

The strength of longitudinal joint in Fig. 1-17 is 33 kips/ft, whereas for the girth is 16 kips/ft. Calculate the maximum diameter of the cylinder tank if the internal pressure is 150 psi.

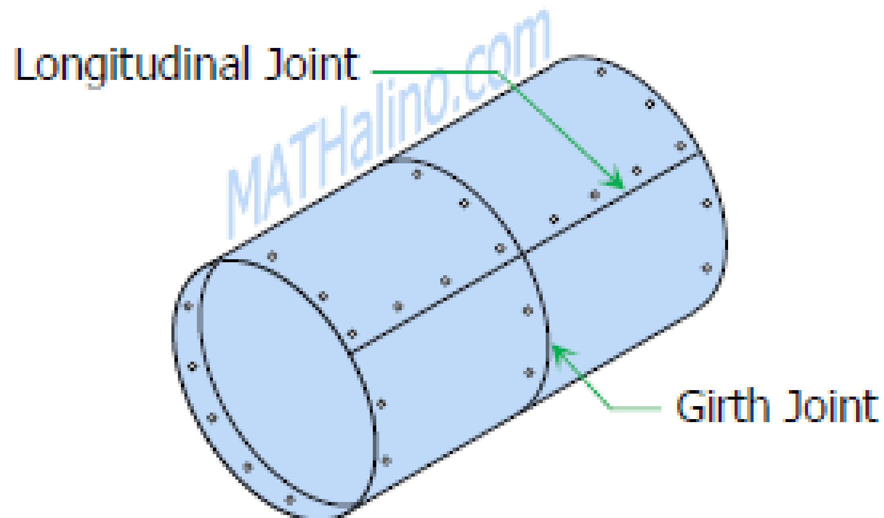



**Figure 1-17**

The tank shown in Fig. P-141 is fabricated from 1/8-in steel plate. Calculate the maximum longitudinal and circumferential stress caused by an internal pressure of 125 psi.

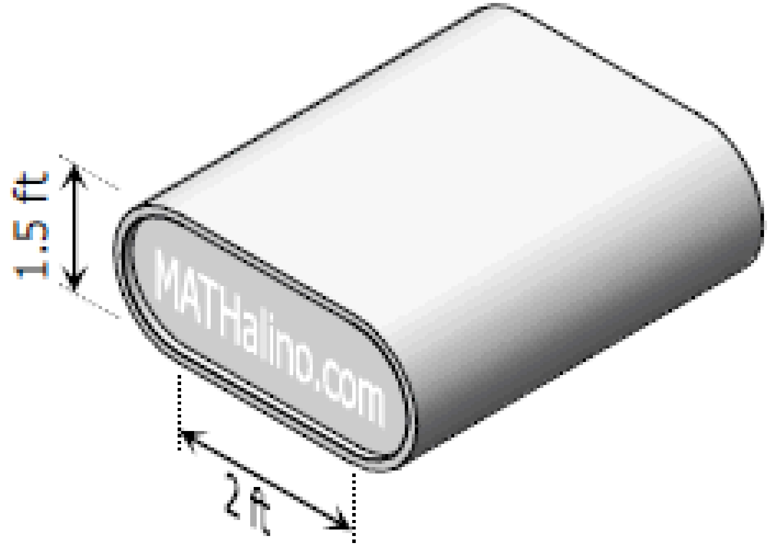
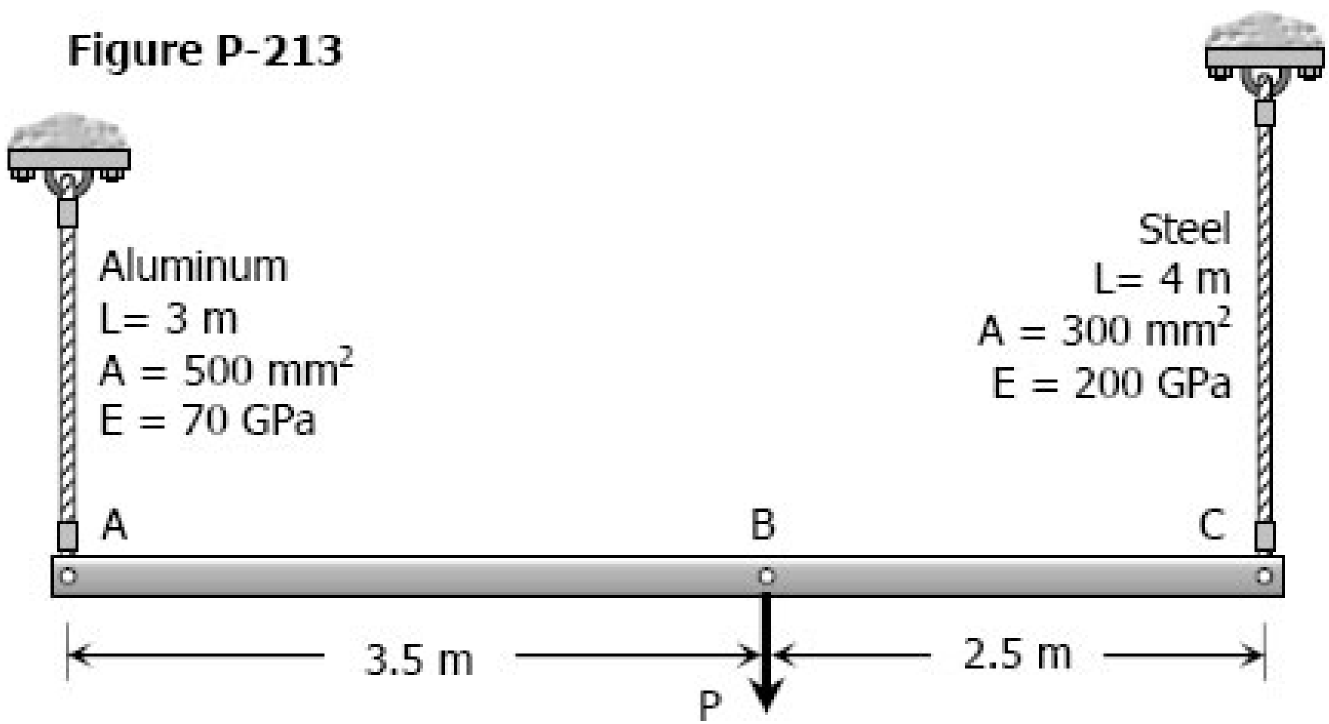


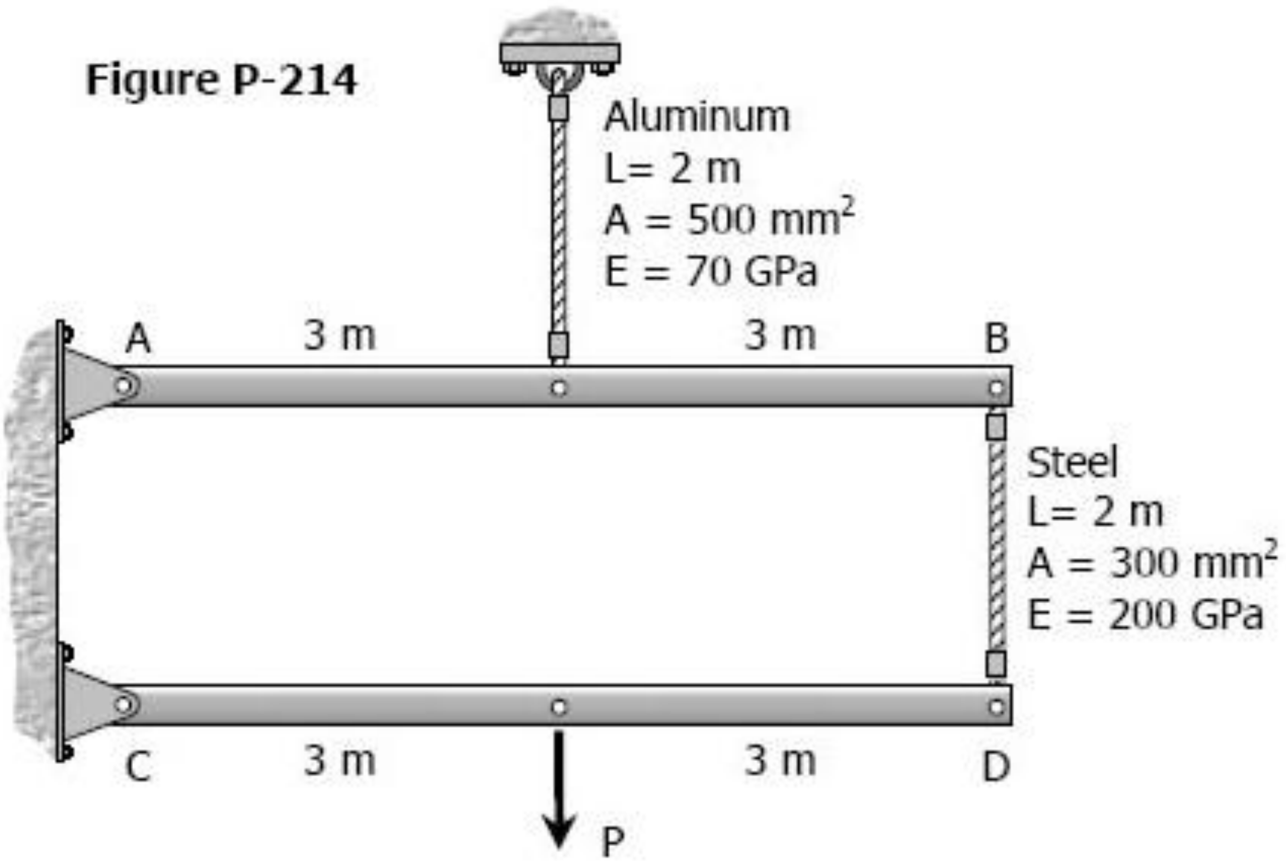

**Figure P-141**

The rigid bar AB, attached to two vertical rods as shown in Fig. P-213, is horizontal before the load P is applied. Determine the vertical movement of P if its magnitude is 50 kN.



The rigid bars AB and CD shown in Fig. P-214 are supported by pins at A and C and the two rods. Determine the maximum force P that can be applied as shown if its vertical movement is limited to 5 mm. Neglect the weights of all members.

**Figure P-214**



As shown in Fig. P-216, two aluminum rods AB and BC, hinged to rigid supports, are pinned together at B to carry a vertical load P = 6000 lb. If each rod has a cross-sectional area of 0.60 in.$^2$ and $E = 10 \times 10^6$ psi, compute the elongation of each rod and the horizontal and vertical displacements of point B. Assume $\alpha = 30°$ and $\theta = 30°$.

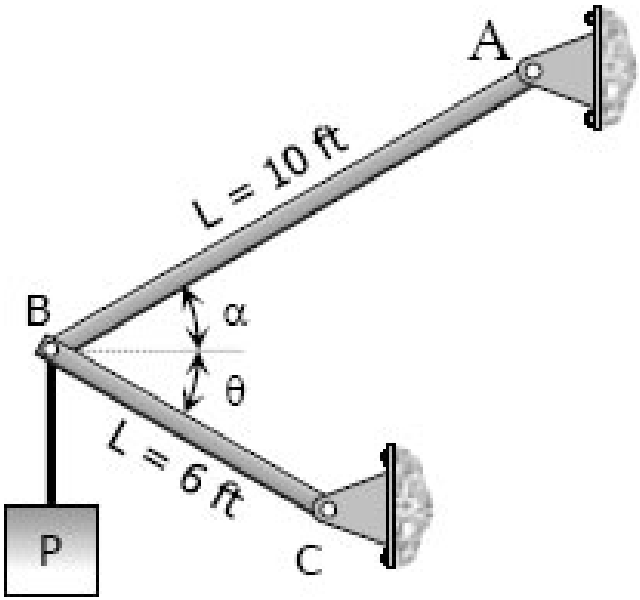



Figure P-216 and P-217

A rigid block of mass M is supported by three symmetrically spaced rods as shown in Fig. P-236. Each copper rod has an area of 900 $mm^2$; E = 120 GPa; and the allowable stress is 70 MPa. The steel rod has an area of 1200 $mm^2$; E = 200 GPa; and the allowable stress is 140 MPa. Determine the largest mass M which can be supported.

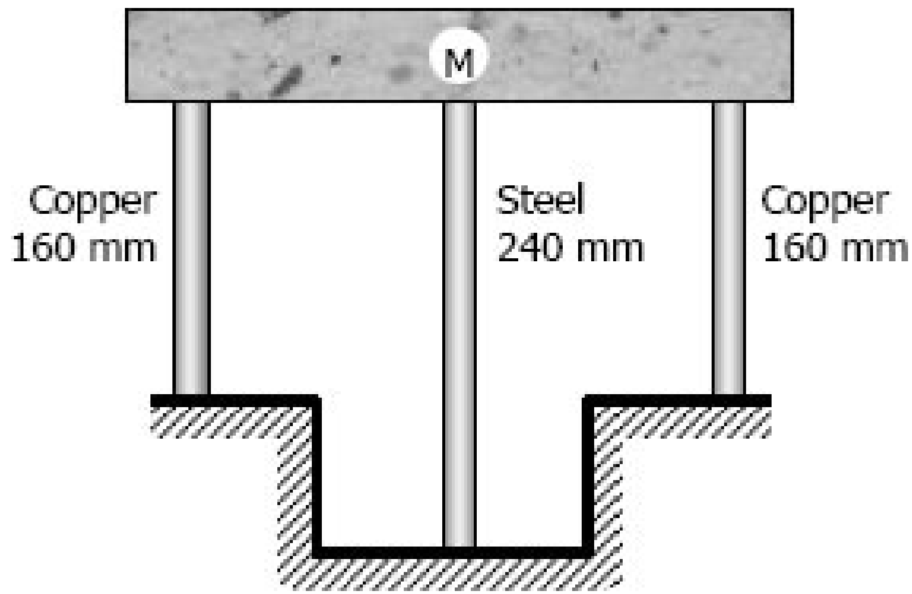
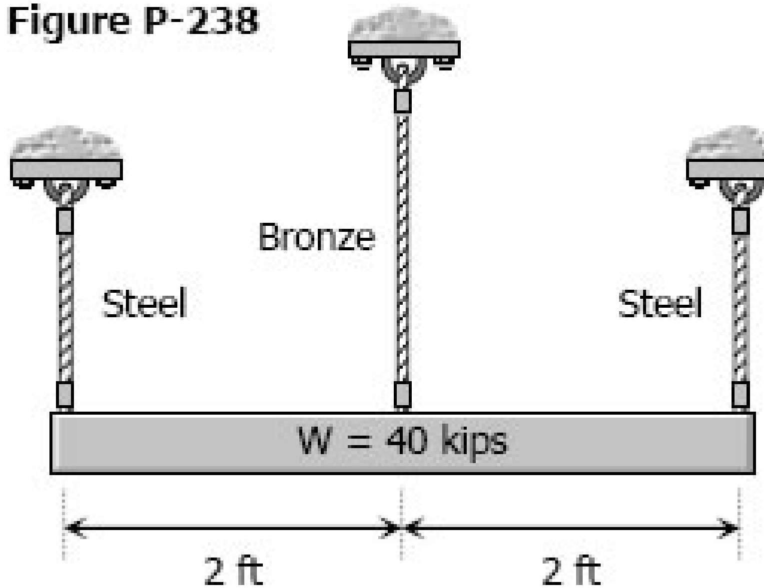



**Figure P-236 and P-237**

The lower ends of the three bars in Fig. P-238 are at the same level before the uniform rigid block weighing 40 kips is attached. Each steel bar has a length of 3 ft, and area of 1.0 in.$^2$, and E = 29 × 10$^6$ psi. For the bronze bar, the area is 1.5 in.$^2$ and E = 12 × 10$^6$ psi. Determine (a) the length of the bronze bar so that the load on each steel bar is twice the load on the bronze bar, and (b) the length of the bronze that will make the steel stress twice the bronze stress.

**Figure P-238**



The rigid platform in Fig. P-239 has negligible mass and rests on two steel bars, each 250.00 mm long. The center bar is aluminum and 249.90 mm long. Compute the stress in the aluminum bar after the center load P = 400 kN has been applied. For each steel bar, the area is 1200 mm2 and E = 200 GPa. For the aluminum bar, the area is 2400 mm2 and E = 70 GPa.

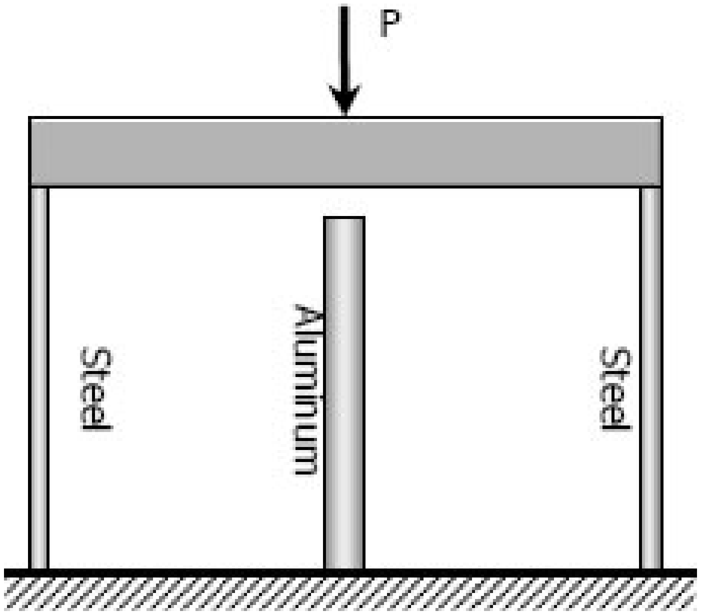



Figure P-239

As shown in Fig. P-241, three steel wires, each 0.05 in.2 in area, are used to lift a load W = 1500 lb. Their unstressed lengths are 74.98 ft, 74.99 ft, and 75.00 ft.

(a) What stress exists in the longest wire?

(b) Determine the stress in the shortest wire if W = 500 lb.


Figure P-241

The assembly in Fig. P-242 consists of a light rigid bar AB, pinned at O, that is attached to the steel and aluminum rods. In the position shown, bar AB is horizontal and there is a gap, Δ = 5 mm, between the lower end of the steel rod and its pin support at C. Compute the stress in the aluminum rod when the lower end of the steel rod is attached to its support.

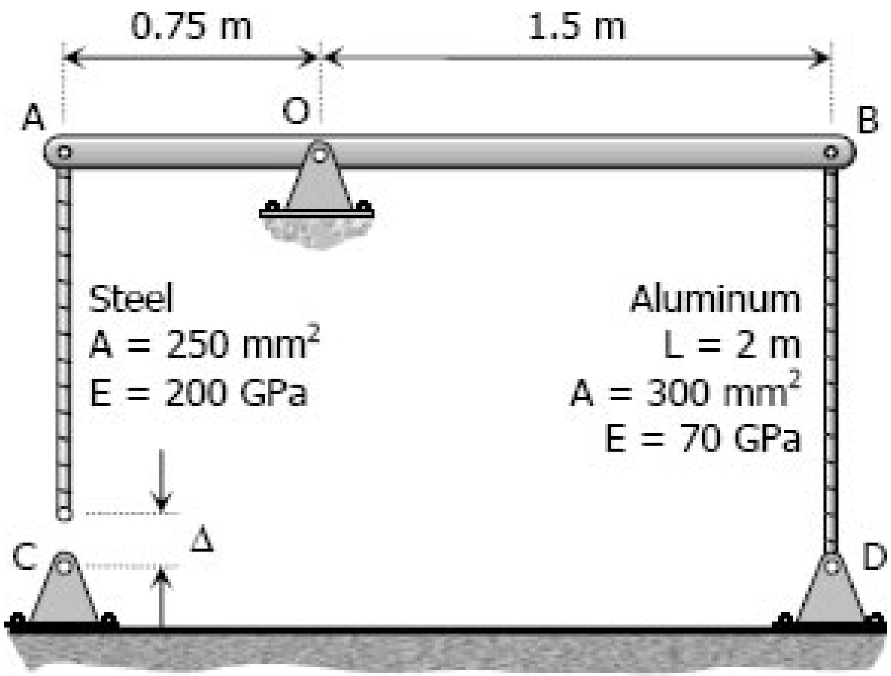



Figure P-242

A homogeneous rod of constant cross section is attached to unyielding supports. It carries an axial load P applied as shown in Fig. P-243. Prove that the reactions are given by $R_1 = Pb/L$ and $R_2 = Pa/L$.

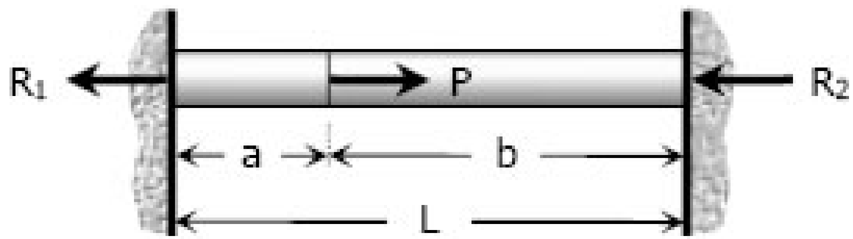



**Figure P-243**

In the assembly of the bronze tube and steel bolt shown in Fig. P-250, the pitch of the bolt thread is p = 1/32 in.; the cross-sectional area of the bronze tube is 1.5 in.$^2$ and of steel bolt is 3/4 in.$^2$ The nut is turned until there is a compressive stress of 4000 psi in the bronze tube. Find the stresses if the nut is given one additional turn. How many turns of the nut will reduce these stresses to zero? Use $E_{br} = 12 \times 10^6$ psi and $E_{st} = 29 \times 10^6$ psi.

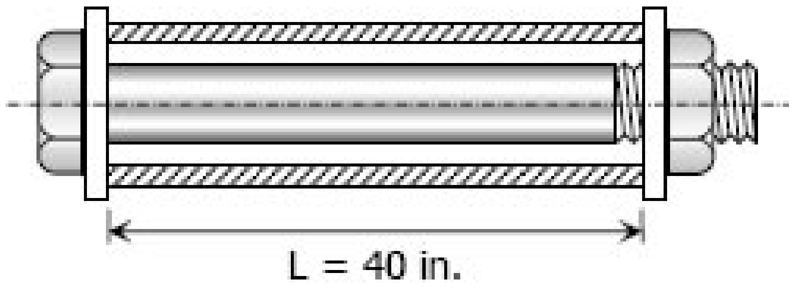



**Figure P-250**

Three rods, each of area 250 $mm^2$, jointly support a 7.5 kN load, as shown in Fig. P-256. Assuming that there was no slack or stress in the rods before the load was applied, find the stress in each rod. Use $E_{st}$ = 200 GPa and $E_{br}$ = 83 GPa.

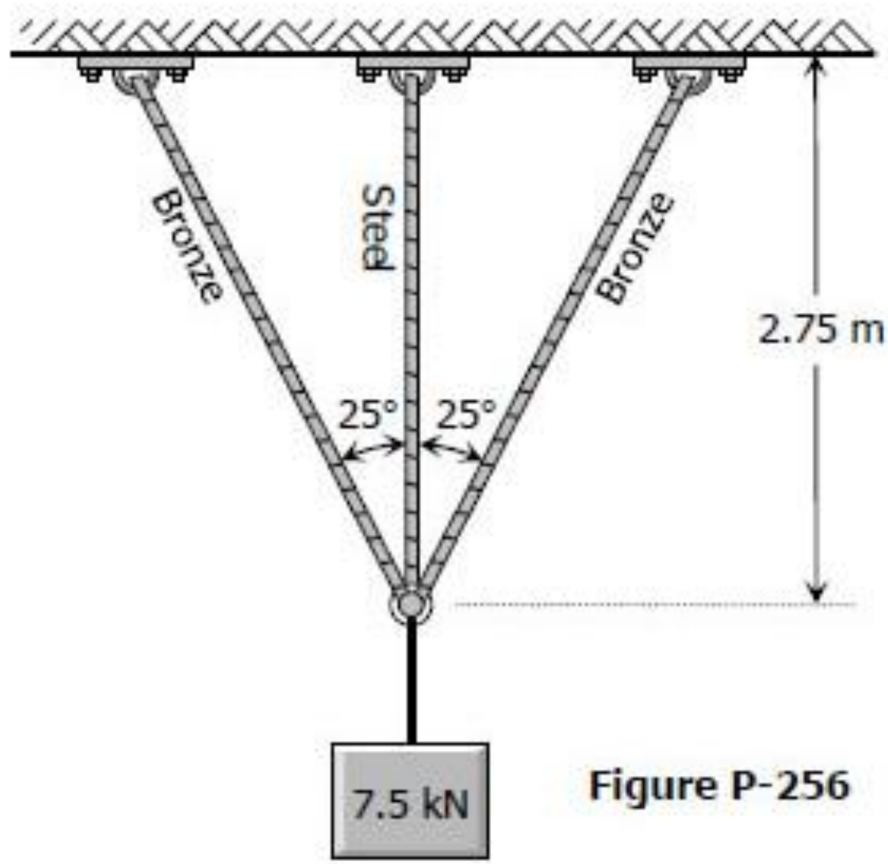



Figure P-256

A bronze bar 3 m long with a cross sectional area of 320 $mm^2$ is placed between two rigid walls as shown in Fig. P-265. At a temperature of -20°C, the gap Δ = 2.5 mm. Find the temperature at which the compressive stress in the bar will be 35 MPa. Use $\alpha = 18.0 \times 10^{-6}$ m/(m·°C) and E = 80 GPa.

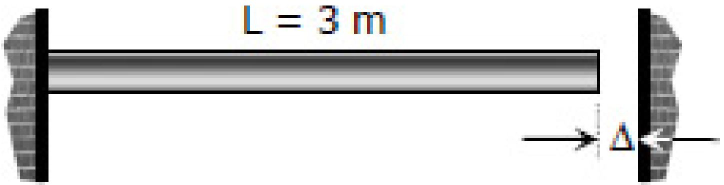
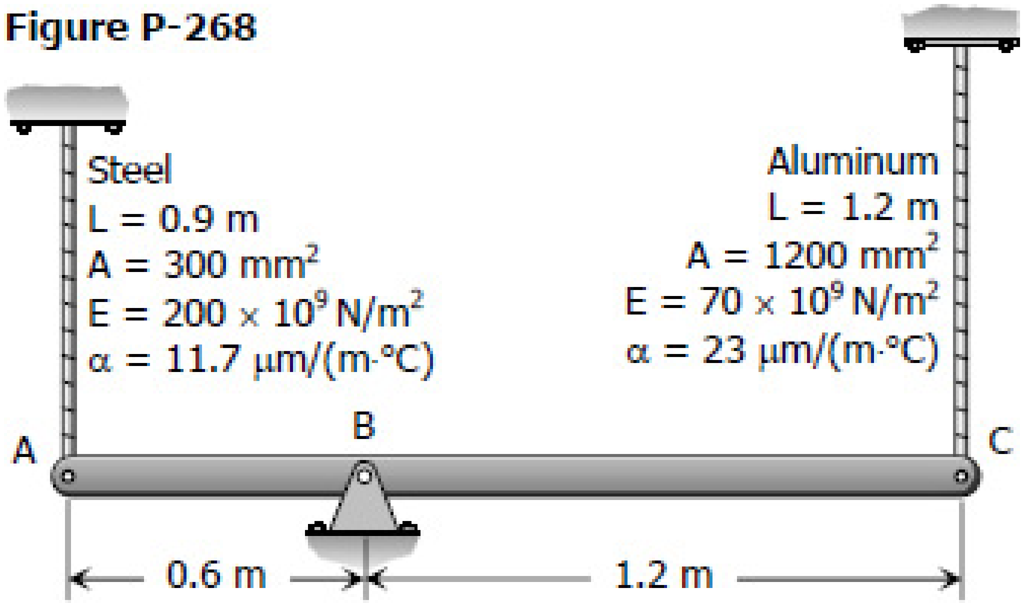



**Figure P-265**

The rigid bar ABC in Fig. P-268 is pinned at B and attached to the two vertical rods. Initially, the bar is horizontal and the vertical rods are stress-free. Determine the stress in the aluminum rod if the temperature of the steel rod is decreased by 40°C. Neglect the weight of bar ABC.

**Figure P-268**



As shown in Fig. P-269, there is a gap between the aluminum bar and the rigid slab that is supported by two copper bars. At 10°C, $\Delta = 0.18$ mm. Neglecting the mass of the slab, calculate the stress in each rod when the temperature in the assembly is increased to 95°C. For each copper bar, $A = 500\ mm^2$, $E = 120$ GPa, and $\alpha = 16.8\ \mu m/(m\cdot °C)$. For the aluminum bar, $A = 400\ mm^2$, $E = 70$ GPa, and $\alpha = 23.1\ \mu m/(m\cdot °C)$.

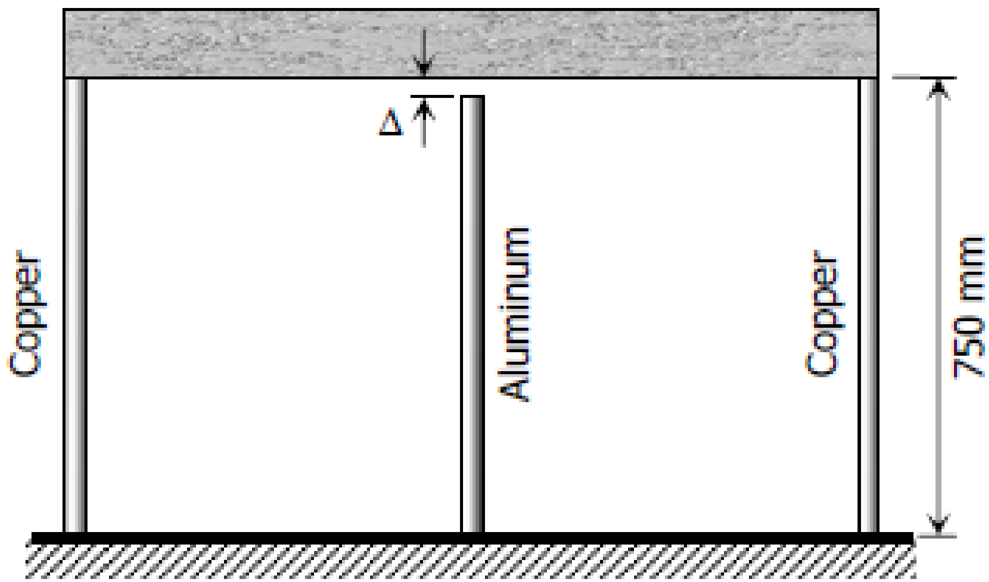



**Figure P-269**

Four steel bars jointly support a mass of 15 Mg as shown in Fig. P-276. Each bar has a cross-sectional area of 600 mm$^2$. Find the load carried by each bar after a temperature rise of 50°C. Assume $\alpha$ = 11.7 μm/(m·°C) and E = 200 GPa.

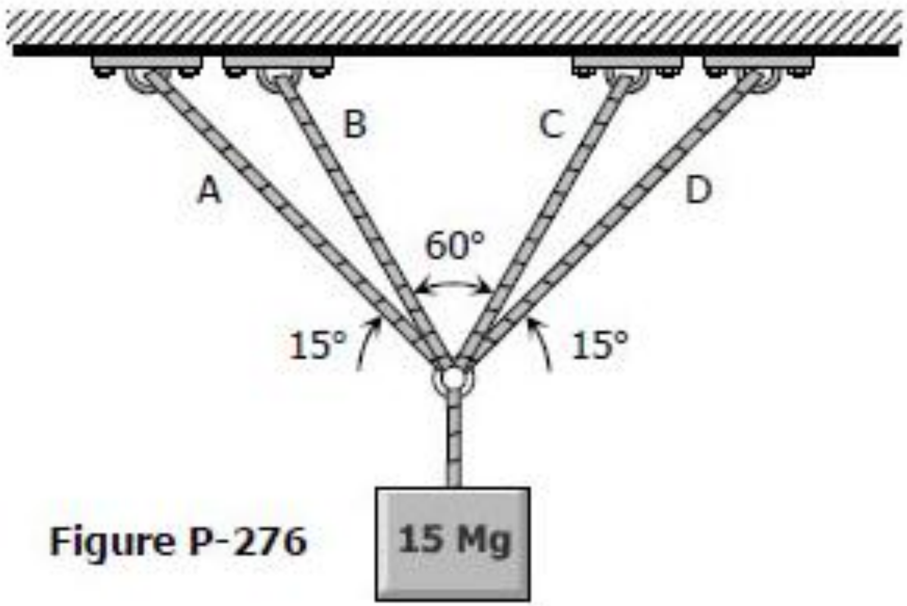



Figure P-276

# Axial, Shear & Moment